

# The $\Psi$ - $\Delta_{I \rightarrow G}$ Framework: $V$ -Structure Coupling and Structural Gaps under Finite Resolution

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## Abstract

This paper formulates the  $\Psi$ - $\Delta_{I \rightarrow G}$  framework as a  $V$ -conditioned diagnostic of structural gaps in multi-level systems. Its guiding claim is that finite descriptions do not report objective structure from an exterior position. A reportable object arises only through a declared coupling layer  $V$ , where objective organization meets finite access, admissible representation, compression, reconstruction, discrepancy, comparability, and resolution.

The package  $V$  therefore does not create the underlying structure, but it determines the form in which that structure becomes reportable. Changing  $V$  changes the reported structural object: it changes which distinctions are visible, which identifications are licensed, which residuals are measurable, and which comparisons are meaningful. Under a fixed  $V$  and finite reporting resolution  $\varepsilon$ , sufficiency, approximation, residual ambiguity, and undefinedness are package-locked outputs rather than viewpoint-free claims.

The framework separates two structural quantities. The intra-level structural tension readout  $\Psi(\mathcal{L}, A; V) \geq 0$  records the infimal residual tension within a declared perspective  $A$  under  $V$ . The cross-level diagnostic gap  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  records the infimal worst-case residual left when lower-level structure is compressed into an upper-level report and then reconstructed through the declared admissible lifting class. Thus  $\Psi$  and  $\Delta_{I \rightarrow G}$  are coupled but non-identical diagnostic objects:  $\Psi$  reads within-level tension, whereas  $\Delta_{I \rightarrow G}$  reads cross-level residuality under declared compression and reconstruction.

To describe persistence under declared evolution, refinement, or scale change, the paper introduces a minimal dynamical interface. Along a declared evolution parameter  $\gamma$ ,  $\Psi_{\text{run}}(\gamma)$  is treated as a running local readout, and an accumulated bookkeeping quantity  $\Delta_{\text{acc}}(\gamma)$  is defined by

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^{\gamma} \Psi_{\text{run}}(\gamma') d\gamma'.$$

When  $\Psi_{\text{run}}$  is locally integrable, the accumulated quantity is absolutely continuous and the running readout is its derivative almost everywhere along the declared axis. This interface formalizes a pattern abstracted from repeated cases in which local unresolved tension persists along a declared axis and contributes to an accumulated residual readout. It is presented as a definition-level reporting interface, not as an unconditional physical law.

The main contribution is therefore a minimal reportability framework: structural fracture is not treated as an informal slogan, but as a package-locked residual structure that can be statically diagnosed and dynamically tracked under declared resolution. The paper provides formal definitions, package cards, admissibility and comparability gates, resolution-aware reporting semantics, and explicit undefinedness diagnoses. It also inserts a fixed-target resolution-feasibility layer: when a certified fixed-target resolution refinement and typed null/separation query clauses are invoked, declaration of a threshold is distinguished from certified feasibility of that threshold, and the licensed completed packages are derived from one-sided error envelopes and the declared task margins. It should be read as the framework-level core for  $V$ -conditioned structural-gap reports. Detailed macro-closure mechanisms,

residual-complement licensing, closure-restoring extensions, and local fiber-activity diagnostics are treated in companion closure modules rather than built into the present framework as primitive assumptions.

## Contents

|          |                                                                                                         |           |
|----------|---------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction: <math>V</math>-Structure Coupling and Cross-Level Reportability</b>                    | <b>3</b>  |
| 1.1      | Motivating examples: calculus, Bayesian conditioning, and RG running . . . . .                          | 4         |
| 1.2      | Terminology and scope: levels, perspectives, and multi-level systems . . . . .                          | 6         |
| 1.3      | Structural fracture law as a working meta-constraint . . . . .                                          | 6         |
| 1.4      | From a static constraint to an operational diagnostic object . . . . .                                  | 6         |
| 1.5      | Declared regime and comparability layer (package discipline) . . . . .                                  | 6         |
| 1.6      | Positioning relative to existing frameworks . . . . .                                                   | 7         |
| 1.7      | Main objects, contributions, and what is proved . . . . .                                               | 7         |
| 1.8      | Roadmap . . . . .                                                                                       | 8         |
| <b>2</b> | <b>Conventions and Reader Guide</b>                                                                     | <b>8</b>  |
| 2.1      | Symbols and type discipline (system vs. declared package) . . . . .                                     | 8         |
| 2.2      | Equality and resolution-aware approximation . . . . .                                                   | 11        |
| 2.3      | Reader guide and dependency map . . . . .                                                               | 12        |
| <b>3</b> | <b>Framework Overview</b>                                                                               | <b>13</b> |
| 3.1      | Running minimal example: compression, hidden fibers, and future readout . . . .                         | 13        |
| 3.2      | Minimal notation quick reference (macro-consistent) . . . . .                                           | 14        |
| <b>4</b> | <b>Formal Objects: Regime, Perspectives, Packages, and Package-Locked Reporting Structures</b>          | <b>14</b> |
| 4.1      | Substrate regime and fine-grained domain . . . . .                                                      | 15        |
| 4.2      | Perspectives and cross-level interfaces . . . . .                                                       | 15        |
| 4.3      | Declared package $V$ as a coupling layer . . . . .                                                      | 16        |
| 4.4      | Resolution-controlled refinements and certified threshold feasibility . . . . .                         | 20        |
| 4.5      | Comparability across packages . . . . .                                                                 | 24        |
| <b>5</b> | <b>Core Quantities: <math>\Psi</math> and the Fixed-Slice Gap <math>\Delta_{I \rightarrow G}</math></b> | <b>24</b> |
| 5.1      | Intra-level structural tension readout $\Psi$ . . . . .                                                 | 25        |
| 5.2      | Inter-level structural gap $\Delta_{I \rightarrow G}$ . . . . .                                         | 25        |
| 5.3      | Geometric intuition: projection and fibers . . . . .                                                    | 26        |
| 5.4      | Optional H/M regime labels induced by $\Psi$ . . . . .                                                  | 30        |
| <b>6</b> | <b>Structural Fracture Law and Diagnostic Translation</b>                                               | <b>30</b> |
| 6.1      | Structural fracture law (canonical statement) . . . . .                                                 | 30        |
| 6.2      | Diagnostic translation into $\Psi$ - $\Delta_{I \rightarrow G}$ language . . . . .                      | 30        |
| 6.3      | Meta-gap $\Delta^*$ (cross-level non-identity postulate) . . . . .                                      | 31        |
| 6.4      | From static fracture to declared dynamics . . . . .                                                     | 31        |
| <b>7</b> | <b>Minimal Dynamical Interface</b>                                                                      | <b>32</b> |
| 7.1      | Declared evolution parameter and accumulated quantity . . . . .                                         | 32        |
| <b>8</b> | <b>Well-posed Reporting and Licensed Comparisons</b>                                                    | <b>34</b> |
| 8.1      | Package card template (fillable) . . . . .                                                              | 34        |
| 8.2      | What must be declared to make $\Psi$ and $\Delta_{I \rightarrow G}$ well-defined . . . . .              | 36        |
| 8.3      | Resolution robustness and locked adaptation . . . . .                                                   | 37        |

|           |                                                                                                          |           |
|-----------|----------------------------------------------------------------------------------------------------------|-----------|
| 8.4       | Partiality and licensed comparisons (when numerical reporting is undefined) . . .                        | 38        |
| 8.5       | Canonical audited example: Markov coarse-graining and fiber non-constancy . .                            | 40        |
| <b>9</b>  | <b>Discussion</b>                                                                                        | <b>41</b> |
| <b>10</b> | <b>Conclusion</b>                                                                                        | <b>41</b> |
| <b>A</b>  | <b>Diagnostic status tags and optional witnesses</b>                                                     | <b>42</b> |
| <b>B</b>  | <b>Extended probabilistic instantiation of <math>\Delta_{I \rightarrow G}</math></b>                     | <b>43</b> |
| B.1       | Minimax residual as a certificate-level structural limit . . . . .                                       | 43        |
| B.2       | Alignment with audit semantics: single-valued decidability and fiber non-constancy                       | 44        |
| B.3       | Probabilistic instantiation as reporting interface (TV/IPM) . . . . .                                    | 44        |
| B.4       | Bayesian intuition under declared evidence . . . . .                                                     | 45        |
| <b>C</b>  | <b>Extended <math>\Psi</math>-<math>\Delta_{I \rightarrow G}</math> Dynamics (short optional module)</b> | <b>46</b> |
| C.1       | Notation and $\Delta$ -variants . . . . .                                                                | 46        |
| C.2       | Optional proportional and conservation templates . . . . .                                               | 46        |
| <b>D</b>  | <b>Logarithmic Spiral Closure and Self-Similar Geometry</b>                                              | <b>47</b> |
| D.1       | Objects and scope of this appendix . . . . .                                                             | 47        |
| D.2       | Declared planar embedding (auxiliary) . . . . .                                                          | 47        |
| D.3       | Logarithmic-spiral closure as a local proportionality ansatz . . . . .                                   | 48        |
| D.4       | Equiangular property and a characterization . . . . .                                                    | 48        |
| D.5       | Self-similarity constraints and a functional characterization . . . . .                                  | 49        |
| D.6       | Discrete evolution index (hierarchy depth) version . . . . .                                             | 49        |
| D.7       | Reproducible reporting clauses for spiral closure . . . . .                                              | 50        |
| <b>E</b>  | <b>Resolution Feasibility and Control: Pairwise and Partition Semantics</b>                              | <b>50</b> |
| E.1       | Purpose and restricted scope . . . . .                                                                   | 50        |
| E.2       | Target discrepancy, interface family, and certification . . . . .                                        | 51        |
| E.3       | Threshold semantics and typed query clauses . . . . .                                                    | 52        |
| E.4       | Pairwise certified feasibility . . . . .                                                                 | 54        |
| E.5       | Partition semantics and chain separation . . . . .                                                       | 55        |
| E.6       | Numerical step size as a fixed-target instance . . . . .                                                 | 58        |
| E.7       | Certificate-relative infeasibility and localization . . . . .                                            | 59        |
| E.8       | Robustness, selection, and adaptive control . . . . .                                                    | 59        |
| E.9       | Reporting states and non-claims . . . . .                                                                | 61        |

# 1 Introduction: $V$ -Structure Coupling and Cross-Level Reportability

Across logic, physics, computation, learning, and coarse-grained modeling, a recurring difficulty appears: a compressed or upper-level description is often operationally useful and is then over-read as if it were structurally sufficient. The object may be an endpoint, a macro-label, a compressed state, a quotient variable, or a protocol-visible report. Such objects can be legitimate within a declared task, but their legitimacy does not by itself imply cross-level closure, reconstructive sufficiency, or ontological identity.

The problem is not compression itself. The problem is undeclared compression. A cross-level report becomes determinate only after one has specified what is visible, which objects are admissible, which comparisons are licensed, what reconstruction class is allowed, which discrepancy rule is used, and which residuals count as negligible at the declared reporting

resolution. Without these clauses, endpoint agreement, macro-label stability, approximate equality, and reconstructive adequacy can slide into one another without a licensed bridge.

This paper formulates that dependency through the viewpoint/package  $V$ .  $V$  is not metadata attached to a completed object; it is the declared coupling site at which objective organization becomes a typed report under finite access. Changing  $V$  changes the reported structural object because it changes visible distinctions, licensed identifications, residual readouts, and comparability.

Under this coupling layer, the framework separates two diagnostic quantities. The intra-level quantity  $\Psi$  records residual tension internal to a declared level and perspective. The cross-level quantity  $\Delta_{I \rightarrow G}$  records the infimal worst-case residual left by compression into an upper-level description followed by admissible reconstruction. These quantities are related but not interchangeable. A system may have low reportable  $\Psi$  at one declared level while still leaving a non-negligible  $\Delta_{I \rightarrow G}$  under cross-level transport; conversely, substantial within-level tension need not by itself imply a large cross-level residual.

Resolution is part of the semantics, not a numerical afterthought:  $\Delta_{I \rightarrow G} \approx_\varepsilon 0$  means that the declared residual lies below the threshold fixed by  $V$ . If the required clauses are absent, the output is not a guessed scalar but an undefined or non-licensed report.

Finally, the paper introduces a minimal dynamical interface. Along a declared evolution, refinement, hierarchy, or scale parameter  $\gamma$ , the running readout  $\Psi_{\text{run}}(\gamma)$  may be accumulated into a bookkeeping quantity  $\Delta_{\text{acc}}(\gamma)$ . This interface is introduced definitionally in order to express a pattern abstracted from repeated cases: unresolved local tension can persist along a declared axis and contribute to an accumulated residual readout. The interface is not asserted as a universal physical law and does not identify every local residual carrier with  $\Delta_{\text{acc}}$  by default.

The resulting pipeline is simple:  $V$  locks reportability;  $\Psi$  reads intra-level residual tension;  $\Delta_{I \rightarrow G}$  reads the cross-level compression–reconstruction residual; and  $\Delta_{\text{acc}}$  is introduced only when a declared evolution axis makes residual persistence reportable.

## 1.1 Motivating examples: calculus, Bayesian conditioning, and RG running

The claim that structure becomes reportable only through a declared coupling layer can sound more radical than it is. The point is not that objective structure is created by a description, nor that every structural statement is merely subjective. The point is that a finite report has to declare an interface through which structure is accessed, compressed, compared, and made negligible or non-negligible. Several standard scientific languages already exhibit this dependency. The following examples are not used as proofs of the framework. They calibrate the semantic point that a reportable object is not a viewpoint-free copy of structure, but a  $V$ -conditioned form in which structure becomes reportable.

**Calculus: whole structure through an infinitesimal interface.** In elementary calculus, a whole object is often re-expressed through local infinitesimal reports. For a differentiable function, one may write locally

$$f(x+h) - f(x) = f'(x)h + r(h), \quad \frac{r(h)}{h} \rightarrow 0,$$

and then pass to the differential expression

$$dy = f'(x)dx.$$

The differential  $dx$  should not be read simply as a discarded small quantity. It marks the local reporting interface through which the whole curve is re-expressed as an infinitesimal linear report. Likewise, an integral re-expresses a whole quantity by a declared decomposition into local pieces and a limiting summation protocol. Thus the passage between whole object, local

differential element, and reconstructed aggregate is licensed by coordinates, partition rules, limiting conventions, and negligibility clauses. Calculus therefore already displays a primitive form of  $V$ -structure coupling: the same structure is not reported in the same form when it is read as a whole curve, as a local tangent report, or as an accumulated integral. The equality notation is legitimate inside the declared calculus, but the semantic bridge is carried by the limiting interface, not by a viewpoint-free identity between all levels of description.

**Bayesian conditioning: event structure under declared evidence.** Bayesian conditioning gives a non-geometric example of the same pattern. The event  $A$  is not changed as a set-theoretic object when evidence  $B$  is declared, yet the report changes from  $P(A)$  to

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

when  $P(B) > 0$ . The conditional value is not attached to  $A$  alone; it is attached to  $A$  under a declared condition, background measure, and normalization domain. In this sense, the evidence event is not a verbal annotation appended after the probability has already been determined. It is part of the reporting interface that selects the relevant slice of the probability structure. This example is useful because it prevents  $V$  from being misread as merely a scale parameter. A package can include evidence, admissible information, likelihood/readout rules, and normalization clauses. When the report is distribution-valued, Bayesian conditioning may be viewed as a conditionalized update of unresolved  $V$ -fiber candidates after a declared evidence event. This reading introduces no new primitive; it is a special probabilistic instance of the broader claim that reported values are condition-dependent structural readouts.

**RG running: effective parameters under scale, scheme, and interface.** Renormalization-group running gives a physical version of the same dependency. A coupling written as  $g(\mu)$  is not a bare copy of objective structure. It is an effective coordinate of structure under a renormalization scale  $\mu$ , regulator and scheme conventions, threshold rules, and an observable interface. Changing  $\mu$  changes the reporting slice, and the parameter must flow so that physical reports remain comparable under the declared renormalization conditions:

$$\mu \frac{dg}{d\mu} = \beta(g).$$

The running of  $g(\mu)$  is therefore not best understood as a constant arbitrarily changing in itself. It is the compensatory flow of a  $V$ -conditioned coordinate when the scale component of  $V$  is varied. Residual scale dependence appears when the finite truncation, finite parameter family, or endpoint matching convention does not fully absorb the structure that remains visible along the declared scale window. This is the same discipline used in endpoint–path diagnostics: the strict or geometric residual is meaningful only after the viewpoint/package fixes what is observed, how scale is traversed, how discrepancies are measured, and what counts as negligible or reportable [18].

Together, these examples isolate the same semantic pattern in three different registers. Calculus shows whole/local re-expression through an infinitesimal limiting interface. Bayesian conditioning shows event/evidence re-expression through a conditional reporting interface. RG running shows physical-parameter re-expression through a scale–scheme–interface package. In all three cases, the reportable object is not obtained by simply copying structure into language. It is generated by the coupling between objective organization and declared access conditions. This is the role assigned here to  $V$ -structure coupling.

## 1.2 Terminology and scope: levels, perspectives, and multi-level systems

Throughout this paper, a *perspective*  $A$  means a declared representation scheme: it specifies (i) which variables or structural factors are treated as distinguishable, (ii) which background conditions are held fixed, and (iii) which aggregation/compression or identification rules are used to form the representation. A *level* is an abstraction layer induced by such a declared representation together with a declared resolution regime: different levels correspond to different representation/state spaces (e.g. micro/individual descriptions versus macro/global summaries) connected by generally non-invertible maps. A system is called *multi-level* when its analysis necessarily involves at least two such levels linked by an irreversible passage (coarse-graining, completion, identification, or any other genuinely cross-level gluing).

All operational diagnostic reports and numerical comparisons in this paper are made relative to a declared *package*  $V$ , which fixes (at minimum) the system boundary and reachable-set convention, the discrepancy conventions, the admissible lifting class, and the reporting threshold  $\varepsilon$  (and, when needed, explicit comparability/transport clauses). This package-relative stance is not stylistic: without explicit declarations, cross-domain comparisons of structural claims are not well-posed. More strongly,  $V$  is the declared coupling layer under which structure becomes reportable at all.

## 1.3 Structural fracture law as a working meta-constraint

The structural fracture law is used in this manuscript as a motivating meta-constraint, not as a theorem and not as a premise from which the formal objects are mechanically derived. The canonical statement is recorded in Sec. 6. At the level of the present introduction, its role is narrower: it names the recurring pressure that motivates the  $V$ -structure coupling framework.

Operationally, the law points to cases in which opposing contributions, feedback, and a single cross-level continuity demand make a package-free judgment unstable. The framework does not treat that instability as a slogan. It translates it into declared objects: intra-level residual tension is read through  $\Psi$ , cross-level compression-reconstruction residuals are read through  $\Delta_{I \rightarrow G}$ , and missing declaration clauses are reported as **undefined** rather than replaced by implicit comparisons. Thus the law functions as orientation; the reportable content is carried by the package-locked diagnostics and the resolution-aware reporting protocol.

## 1.4 From a static constraint to an operational diagnostic object

The paper proceeds by separating two distinct sources of obstruction:

- **Intra-level tension:** residual conflict that persists even when one stays within a fixed level and a fixed perspective (no cross-level gluing).
- **Inter-level residual:** residual mismatch generated by passing between levels (e.g. compressing to a global description and attempting to reconstruct individual structure), even when one allows an admissible class of reconstructions.

This separation is implemented by the  $\Psi$ - $\Delta_{I \rightarrow G}$  pair. The object is designed to be regime-relative: all statements are evaluated under a declared level specification, an admissible family of perspectives, and a declared resolution regime. This is the minimal requirement for controlled comparability across instantiations.

## 1.5 Declared regime and comparability layer (package discipline)

Every report in this framework is package-relative. The package  $V$  records the coupling conditions under which a structural quantity becomes reportable and comparable. The detailed semantics of resolution-aware equality, non-licensing outputs, and cross-package transport are fixed once and for all in Normative Rules 2.1–2.2.

## 1.6 Positioning relative to existing frameworks

The  $\Psi\text{-}\Delta_{I\rightarrow G}$  object is not introduced as a competing replacement for existing domain frameworks; it is a  $V$ -conditioned *reporting layer* that can be instantiated within them.

- **Renormalization/coarse-graining viewpoints.** The use of multi-level passages and admissible reconstructions is compatible with RG/EFT practice [7, 14, 15, 13], but the present manuscript treats endpoint matching and cross-level identification as *licensed only under a declared package* (interface, discrepancy, liftings, and resolution), with explicit undefined outputs when those clauses are absent.
- **Information-geometric and divergence-based diagnostics.** Many diagnostics are naturally expressed using relative entropy and its induced local geometry [2, 1, 18]. The framework keeps these tools admissible as package-level choices (e.g. as a discrepancy  $D$ ), while separating them from the *comparability layer* (transport, domain gates, resolution conventions).
- **Probabilistic instantiation as a reporting interface.** Appendix B records a distribution-valued reading in which the already-declared diagnostic  $\Delta_{I\rightarrow G}(\mathcal{L}; V)$  is translated into TV/IPM reporting readouts once the observation space, admissible lifting class, and discrepancy rule are locked. The instantiation introduces no new core object.
- **Transport and cross-report comparability.** When comparisons across packages are invoked, explicit transport clauses are required; when an optimal-transport style reconciliation is intended, the framework treats it as an explicit declared component rather than an implicit identification [12].

The distinguishing feature is therefore not the choice of discrepancy or the underlying domain model, but the insistence that: (i) reporting semantics is locked by a declared package, (ii) cross-package comparisons require declared transport, and (iii) under missing clauses the correct output is undefined rather than an unscoped surrogate.

## 1.7 Main objects, contributions, and what is proved

Fix a system  $\mathcal{L}$  and a declared package  $V$  with reporting threshold  $\varepsilon$ . Two structural quantities are introduced:

- **Intra-level tension  $\Psi$ .** Within a fixed perspective  $A$  at a fixed level specification,  $\Psi(\mathcal{L}, A; V)$  is the infimum of a declared nonnegative conflict functional over the declared reachable set.
- **Cross-level diagnostic gap  $\Delta_{I\rightarrow G}$ .**  $\Delta_{I\rightarrow G}(\mathcal{L}; V)$  is the minimax residual remaining after compressing to a coarse description and reconstructing via an admissible lifting class under the declared discrepancy rule.

Beyond definitions and reporting discipline, the paper establishes strictly operational non-vanishing mechanisms at the package-relative level: under metric or symmetric-pseudometric discrepancy rules, whenever a macro representation non-injectively identifies interface-distinguishable states within a single macro fiber (i.e., induces fiber non-constancy), the minimax residual  $\Delta_{I\rightarrow G}(\mathcal{L}; V)$  admits an explicit positive lower bound (Theorem 5.1 and Corollary 5.1). This supports witness-based “not resolution-negligible” reports at finite  $\varepsilon$  without invoking any ontological equality claim.

**Separation of roles (definitions vs. proved statements vs. protocol).** To support long-term study and controlled reuse, the manuscript keeps four roles explicit:

- **Defined objects:**  $\Psi(\mathcal{L}, A; V)$ ,  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ , the accumulated bookkeeping quantity  $\Delta_{\text{acc}}(\gamma)$  (Sec. 7), and the comparability/undefinedness semantics (Sec. 8).
- **Adopted semantic axioms/postulates:** resolution-aware cross-level “=” is read as  $\approx_\epsilon$  in reporting (Normative Rule 2.1); and genuine cross-level relations carry a non-vanishing meta-gap schema  $\Delta_R^* > 0$  (Sec. 6.3).
- **Proved operational claims:** when the required package clauses are declared, certain non-negligibility reports (e.g. positive lower bounds for  $\Delta_{I \rightarrow G}$  under fiber non-constancy of the coarse representation relative to the interface content) are supported by explicit witnesses (Theorem 5.1 and Corollary 5.1).
- **Protocol clauses:** package cards, admissibility/comparability gating, resolution-feasibility checks when a certified mechanism family is invoked, and explicit undefined outputs are treated as part of the reporting semantics, not as mathematical conclusions (Sec. 8).

The framework also provides: (i) a reproducibility-oriented reporting protocol (package cards, admissibility/comparability gating, and explicit undefinedness diagnoses, optionally tagged in Appendix A); (ii) a fixed-target resolution-feasibility core, formulated as typed refinements of the existing package clauses, with a full pairwise/partition module in Appendix E; (iii) a minimal dynamical interface in which accumulated residual readouts are tracked along a declared evolution coordinate  $\gamma$ ; and (iv) optional probabilistic, dynamical, and geometric-closure modules that remain subordinate to the same package and reporting protocol. Detailed macro-closure, residual-complement licensing, closure-restoring extensions, and local fiber-activity diagnostics are treated as companion closure modules rather than as primitive assumptions of the present framework [17, 16].

**Minimal guiding example.** Suppose a compression map identifies two lower-level states into the same upper-level report. If every admissible future readout remains constant on that fiber, the compression is sufficient under the declared package. If an audited readout varies within the fiber, the endpoint report is not reconstructively sufficient under that package. The framework records this either as a non-negligible  $\Delta_{I \rightarrow G}$ , when the discrepancy and lifting clauses are declared, or as **undefined** when the required clauses are missing.

## 1.8 Roadmap

Section 2 fixes the type discipline and the resolution-aware semantics used throughout. Sections 3–5 introduce the package-relative objects, the fixed-target threshold-feasibility layer, and the core diagnostics. Section 7 introduces the minimal dynamical interface and clarifies its definition-level status. Section 8 formalizes well-posedness, licensed comparisons, package cards, resolution robustness, locked adaptation, and undefined outputs. Appendices record optional status-tag, probabilistic, dynamical, geometric-closure, and full resolution-feasibility modules. Any illustrative material retained in the appendices is optional and is not used by the core claims. None of these materials changes the core  $V$ -structure coupling semantics.

## 2 Conventions and Reader Guide

### 2.1 Symbols and type discipline (system vs. declared package)

This paper distinguishes *target-system objects* from *analysis/reporting objects*. The distinction is semantic (what is being studied vs. how it is reported) and is enforced to avoid silent shifts of meaning across sections.



**Target system (object-level).** A target system is denoted by  $\mathcal{L}$  (static or dynamical, depending on the chapter). Its state space is denoted by  $\Omega$ , with a declared reachable set  $\Omega_{\text{reach}} \subseteq \Omega$  (once the boundary/reachability convention in the declared package is fixed). When a task restricts attention to a regime relevant for a report, we use  $\Omega_{\text{reach}}^{\text{rel}} \subseteq \Omega_{\text{reach}}$  (task-relevant reachable subset).

**Declared package  $V$  (reporting-level).** A declared package  $V$  is an *audit bundle* that fixes the operational conventions under which  $\mathcal{L}$  is analyzed and reported. At minimum, a package declaration pins down:

- the system boundary and the reachable set convention (what is counted as a state in  $\Omega_{\text{reach}}$ );
- the level specification and admissible perspective family (what representations are allowed);
- the discrepancy structure used for reporting (micro discrepancy and, when needed, induced macro discrepancy);
- the admissible lifting / reconstruction class (what “best reconstruction” means);
- the reporting threshold  $\varepsilon$  (the operational meaning of “resolution-negligible”);
- the comparability clauses: transport rules (when defined), domain gating, and optional diagnostic tags.

Accordingly, numerical statements about  $\Psi$  or  $\Delta_{I \rightarrow G}$  are always package-relative, even when  $V$  is suppressed typographically.

**Type discipline (what may depend on what).**

- $\mathcal{L}, \Omega, \Omega_{\text{reach}}, \Omega_{\text{reach}}^{\text{rel}}$  are target-system objects (what is being studied).
- $A$  (perspective),  $E$  (conflict functional),  $D$  (declared discrepancy),  $\mathcal{U}$  (admissible liftings), and  $\varepsilon$  (threshold) are package-level objects (how the study is operationalized).
- $\Psi$  is an intra-level structural quantity *for*  $(\mathcal{L}, A)$  but is *reported under*  $V$ .
- $\Delta_{I \rightarrow G}$  is an inter-level residual defined only after  $V$  declares the relevant compression/reconstruction interface and admissible liftings; cross-package comparison requires an explicit transport declaration (otherwise the correct output is “undefined” with a diagnostic).

**Partial semantics (well-posedness vs. undefinedness).** All reporting-level quantities in this paper are treated as *partial* objects whose meanings are locked only on the domain of declared packages satisfying the relevant clauses. If required clauses are missing (under-specification), if an invoked comparison lacks declared transport/comparability, or if a referenced domain condition is not satisfied, then the corresponding numerical or logical statement is **undefined** relative to the declared regime and must be reported as such. For compactness, a report may attach a short diagnostic status tag (Appendix A) together with a minimal witness list, but the core semantic point is the same: “undefined” is an output state, not an error to be silently repaired by implicit conventions.

**Normative Rule 2.1 (Resolution-aware equality and non-licensing outputs).**

- **Cross-level equality.** Whenever an identification *glues distinct levels* (limits, completions, quotients/identifications, coarse-graining followed by reconstruction), any “=” is read in reporting through the declared symmetric resolution-indistinguishability predicate:  $X = Y$  in such a context is reported as  $X \approx_\varepsilon Y$  under the declared package  $V$ .
- **Ontological non-identity (meta-gap).** For genuine cross-level relations, ontological non-identity is recorded by the meta-gap axiom  $\Delta_R^* > 0$  (Def. 6.1) for the invoked genuine cross-level relation  $R$ ; accordingly, one may not silently replace a cross-level mismatch by a zero report  $\Delta_{I \rightarrow G} \approx_\varepsilon 0$  without satisfying the licensing conditions under the declared package  $V$ .
- **Reporting convention.** The statement  $\Delta_{I \rightarrow G} \approx_\varepsilon 0$  is *only* a reporting convention, read operationally as “ $\Delta_{I \rightarrow G} < \varepsilon$  under the declared clauses of  $V$ ”.
- **Refinement behavior.** If a declared or certified structural lower bound  $\ell_{I \rightarrow G} \leq \Delta_{I \rightarrow G}$  is available and satisfies  $\ell_{I \rightarrow G} \geq \varepsilon$ , then the convention  $\Delta_{I \rightarrow G} \approx_\varepsilon 0$  is unavailable: one must report  $\Delta_{I \rightarrow G} \not\approx_\varepsilon 0$  (or, when the needed clauses are missing, **undefined**) rather than forcing an equality claim.
- **Undefinedness / non-licensing.** Any numerical statement or cross-report comparison is licensed only on the domain where the required package clauses (interface, admissibility, and domain gating) are declared and any invoked certificate-relative feasibility gate is satisfied; cross-package comparison additionally requires an explicit transport law (Normative Rule 2.2). Otherwise, the correct output state is **undefined** (optionally with a short diagnostic status tag and a minimal witness list; Appendix A).

**Normative Rule 2.2 (Cross-package comparability and explicit transport).**

- **no implicit transport (Normative Rule 2.2).** Numerical reports produced under distinct declared packages  $V$  and  $V'$  are *not* treated as directly comparable scalars by default (even if they share a symbol name).
- **Explicit transport law.** A cross-package comparison (equality/inequality, limits across packages, or reusing a numerical threshold) is licensed only if a transport/identification rule  $\tau_{V \rightarrow V'}$  is explicitly declared together with its domain of validity and compatibility with the relevant representation maps and interface clauses.
- **Non-comparability output.** If  $\tau_{V \rightarrow V'}$  is not declared (or does not cover the reported objects), cross-package comparison statements are **undefined**; an optional diagnostic tag is **non-comparable**.
- **Resolution alignment.** When a transport is declared, comparison is made only after aligning reporting thresholds (or declaring an alignment rule). Otherwise the appropriate output is **undefined** (optional tag **RES / resolution-mismatch**).

**Notation collisions (avoidance rules).**

- Within the framework’s diagnostic notation, the symbol  $\Delta$  is reserved for cross-level residuals, and the bare symbol  $\Delta$  is not used as an alias for any specific gap object;

we always keep subscripts/markers (e.g.  $\Delta_{I \rightarrow G}$ ,  $\Delta_{\text{acc}}(\gamma)$ ,  $\Delta^*$ ) to prevent cross-module collisions.

- Generic scalar placeholders (dummy variables, coordinates, function arguments) are written as  $x, y, \dots$ , not  $\Delta$ .
- The symbol  $A$  is reserved for *perspectives* (Def. 4.4). Matrix operators used in local linearizations are written as  $J$  when local linear models are declared.
- The symbol  $V$  is reserved for declared packages (audit bundles). The package-locked reporting structure is denoted by  $\mathcal{V}$  (Def. 4.11).
- The symbol  $R$  is reserved for genuine cross-level relations (Sec. 6.3). Substrate constraint regimes are written as  $\mathfrak{R}$  (Def. 4.1) to avoid collision.
- Level-respecting re-expression maps are written as  $T_{A_2 \leftarrow A_1}$  (Def. 4.6).

## 2.2 Equality and resolution-aware approximation

This paper uses “=” as an academic writing convention, but its *reporting semantics* depends on whether a statement stays within a single declared calculus/module or crosses a genuine level boundary.

**(i) Within a declared calculus/module (single-level inference).** When a derivation is explicitly confined to a declared calculus/module (an H-calculus segment in the terminology used later), “=” is read as *equality under the rules of that declared module*: an inference-level identification licensed by the adopted rules and admissibility constraints of the module. This reading is local: it does not by itself assert that a cross-level identification has been made gap-free.

**Bare approximation vs. resolution-aware reporting indistinguishability.** Within a declared module, the symbol “ $\approx$ ” (without  $\varepsilon$ ) denotes a local analytic/discretization approximation (e.g. a local linearization  $v_{k+1} \approx Jv_k$ ). It does not carry the reporting semantics of  $\approx_\varepsilon$ . Only  $\approx_\varepsilon$  denotes the declared symmetric resolution-indistinguishability predicate under a package; it becomes an equivalence relation only when transitivity is separately established or imposed by declared closure.

**(ii) Cross-level identities (limits, completions, quotients, coarse-graining).** Whenever “=” arises from a construction that *glues distinct levels* (e.g. limiting, completion, quotienting/identification, coarse-graining followed by reconstruction), the default reading is *resolution-aware symmetric indistinguishability under the declared package*:

$$X = Y \quad (\text{in cross-level context}) \quad \rightsquigarrow \quad X \approx_\varepsilon Y,$$

where  $\approx_\varepsilon$  denotes the declared symmetric indistinguishability predicate at reporting threshold  $\varepsilon$  under  $V$ . The predicate need not be transitive; a resolution-equivalence relation arises only after transitivity is proved or a declared closure generates equivalence classes.

**Normative semantics.** All later uses of cross-level equality, resolution-aware approximation, and non-licensing/undefinedness inherit the single authoritative semantics fixed by Normative Rule 2.1. Declaration of a threshold fixes the meaning of the symbol; when a certified fixed-target resolution refinement is invoked, a separate certificate-relative feasibility check determines whether the corresponding ordinary completed package instance is licensed for the declared query clauses (Sec. 4.4).

**Two uses of  $\Delta$  (fixed-slice vs. accumulated).** To prevent a common confusion:

- $\Delta_{I \rightarrow G}$  denotes a fixed-slice inter-level residual (compression  $\rightarrow$  admissible reconstruction/lifting residual).
- $\Delta_{\text{acc}}(\gamma)$  (when used) denotes an accumulated scalar along a declared evolution coordinate  $\gamma$ . Its meaning is package-relative and requires that the package declare the carrier/averaging conventions used to define the accumulation. It is not interchangeable with the fixed-slice diagnostic  $\Delta_{I \rightarrow G}$ .

Throughout the manuscript we do not use the bare symbol  $\Delta$  as an alias for any specific gap object ( $\Delta_{I \rightarrow G}$ ,  $\Delta_{\text{acc}}(\gamma)$ , or  $\Delta^*$ ); we always keep subscripts/markers. Generic scalar placeholders (dummy variables, coordinates, function arguments) are written as  $x, y, \dots$ , not  $\Delta$ .

### 2.3 Reader guide and dependency map

This manuscript has a slim core and optional branches. The core consists of the  $V$ -structure coupling semantics, the definitions of  $\Psi$  and  $\Delta_{I \rightarrow G}$ , the resolution-aware reporting rules, the fixed-target threshold-feasibility interface, and the minimal dynamical interface. Optional branches are invoked only when their additional objects are declared: probabilistic interfaces for TV/IPM readouts, the full pairwise/partition resolution module, extended dynamical readouts, and geometric-closure modules for model-specific mechanisms. If the required branch declarations are absent, the branch-specific claim is simply not licensed.

### 3 Framework Overview

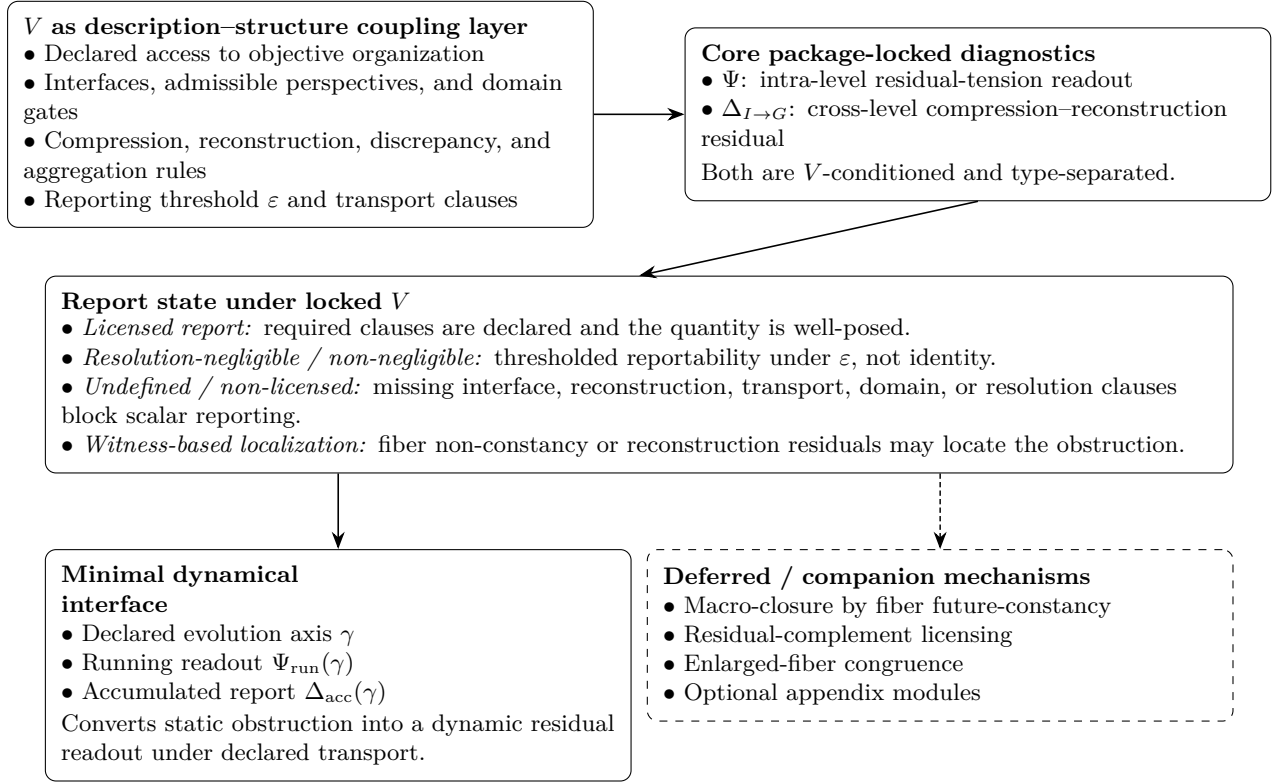


Figure 1: Overview of the  $V$ -structure coupling framework. The core line is: declared coupling layer  $V \rightarrow$  package-locked diagnostics  $(\Psi, \Delta_{I \rightarrow G}) \rightarrow$  report states under finite resolution. Once a declared evolution axis is supplied, the minimal dynamical interface converts the static diagnostic into a dynamic residual readout. Companion closure mechanisms and optional appendix modules require additional declarations and are not used by the core claims.

**Dependency map.** Figure 1 is a dependency map: reportable diagnostics appear only after  $V$  is declared; dynamical and closure statements require additional declared structure.

#### 3.1 Running minimal example: compression, hidden fibers, and future read-out

The following finite-state example is used only to display the type structure of the framework. Let

$$X = \{a, b, c, d\}, \quad \pi(a) = \pi(b) = \pi(c) = 0, \quad \pi(d) = 1,$$

and declare an evolution

$$T(a) = c, \quad T(b) = d, \quad T(c) = c, \quad T(d) = d.$$

Although  $a$  and  $b$  share the same current macro-report, their one-step macro-readouts separate:

$$\pi(Ta) = 0, \quad \pi(Tb) = 1.$$

Thus endpoint agreement does not by itself license future-readout sufficiency. To make a numerical claim one must declare the relevant package clauses: current interface, future readout, discrepancy, admissible reconstruction, and threshold  $\varepsilon$ . With those clauses, the within-fiber variation can witness a non-negligible  $\Delta_{I \rightarrow G}$ ; without them, the sufficiency claim is undefined rather than numerical.

### 3.2 Minimal notation quick reference (macro-consistent)

For quick reference, we list only the symbols that are most likely to be misread on first contact. All meanings are *package-relative* (i.e. evaluated and reported under a declared package  $V$  and its threshold  $\varepsilon$ ), unless explicitly stated otherwise.

Table 1: Minimal notation quick reference (restricted to the high-confusion set; consistent with the current macro layer).

| Symbol                               | Meaning (as used in this paper)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{L}$                        | Target system object (static or dynamical, depending on context).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| $\Omega$                             | Full state space of $\mathcal{L}$ (target-system object).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\Omega_{\text{reach}}$              | Reachable set (base) declared under the package; subset of $\Omega$ on which reporting is defined.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $\Omega_{\text{reach}}^{\text{rel}}$ | Task-relevant subset of $\Omega_{\text{reach}}$ (regime restriction used by a specific report/diagnostic).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| $A$                                  | Perspective / representation specification (declares distinguishable variables, fixed background, and aggregation/compression rules). Reported statements are always relative to an admissible family of such $A$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| $\Psi$                               | Intra-level structural tension readout (package-reported): infimal irreducible tension within a fixed perspective (via the declared conflict functional and reachable convention).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $\Delta_{I \rightarrow G}$           | Fixed-slice inter-level structural gap (package-reported): infimal residual under declared compression $\rightarrow$ admissible reconstruction/lifting. Throughout, we write $\Delta_{I \rightarrow G}$ explicitly; the bare symbol $\Delta$ is avoided as a standalone object and is not used as an alias for any specific object. Ontological non-identity for genuine cross-level relations is recorded by the meta-gap postulate $\Delta_R^* > 0$ (for any genuine cross-level relation $R$ ); operationally one may report $\Delta_{I \rightarrow G} \approx_\varepsilon 0$ only as a resolution convention (i.e. $\Delta_{I \rightarrow G} < \varepsilon$ under $V$ ). |
| $\Delta_{\text{acc}}(\gamma)$        | Accumulated residual scalar along a declared evolution coordinate $\gamma$ . This is distinct from the fixed-slice $\Delta_{I \rightarrow G}$ and is meaningful only after the package declares the accumulation carrier and averaging conventions.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| $\varepsilon$                        | Reporting threshold (resolution cutoff) declared as part of $V$ . Statements like “ $\Delta_{I \rightarrow G} \approx_\varepsilon 0$ ” are read operationally as “ $\Delta_{I \rightarrow G} < \varepsilon$ under the declared package.”                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| $\approx_\varepsilon$                | Resolution-aware symmetric indistinguishability predicate: “indistinguishable at threshold $\varepsilon$ under $V$ ” (default reporting predicate for cross-level “=”). It is not called an equivalence relation unless transitivity is separately licensed.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\Delta^*$                           | Meta-gap schema. For any genuine cross-level relation $R$ , $\Delta_R^* > 0$ . Not a tunable tolerance and not package-relative; it prevents cross-level “=” statements from being read as ontological identities (Sec. 6.3).                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## 4 Formal Objects: Regime, Perspectives, Packages, and Package-Locked Reporting Structures

This section fixes the primitives used throughout the manuscript. The central convention is:

*A reportable analysis context is specified by a target system equipped with a declared package; the package locks the reporting structure used for evaluation and comparison.*

Accordingly, we distinguish a *substrate regime* (which supplies a fine-grained domain on which representations act) from a *declared package* (which locks what is reportable/comparable

and at which resolution). The resulting package-locked reporting structure is denoted by  $\mathcal{V}$  and is determined by the pair  $(\mathcal{L}, V)$  (equivalently, the substrate regime together with the locked clauses of  $V$ ) (Def. 4.11).

#### 4.1 Substrate regime and fine-grained domain

**Definition 4.1** (Constraint regime). A *constraint regime* is a collection  $\mathfrak{R}$  of local constraint rules. Each rule  $r \in \mathfrak{R}$  involves only finitely many degrees of freedom and is viewed as a relation/predicate on those degrees of freedom.

**Definition 4.2** (Induced support and fine-grained state space). Let  $\text{Var}(\mathfrak{R})$  denote the set of degrees of freedom that appear in at least one rule in  $\mathfrak{R}$  (the support induced by the regime). A *fine-grained state* is an assignment of values to all degrees of freedom in  $\text{Var}(\mathfrak{R})$ . The set of all such assignments forms the fine-grained state space  $\Omega$ .

*Remark 4.1* (No global satisfiability assumption). No assumption is made that  $\mathfrak{R}$  is jointly satisfiable, nor that the regime occupies only states satisfying all constraints. Partial violations are permitted and treated as structurally relevant.

**Definition 4.3** (Reachability). When admissibility/dynamics are declared, the actually attainable fine-grained states form a reachable set

$$\Omega_{\text{reach}} \subseteq \Omega.$$

Unless stated otherwise, worst-case (supremum) statements are taken over  $\Omega_{\text{reach}}$ .

#### 4.2 Perspectives and cross-level interfaces

**Definition 4.4** (Perspective and representation map). Given a regime  $\mathfrak{R}$  with fine-grained state space  $\Omega$ , a *perspective* is specified as

$$A = (F, \text{Enc}),$$

where  $F$  is the family of structural factors made visible and  $\text{Enc}$  is the encoding rule implementing the chosen description. Each perspective  $A$  induces a representation space  $\Omega_A$  and a map

$$\Phi_A : \Omega \rightarrow \Omega_A.$$

**Definition 4.5** (Level specification). A *level* is specified operationally by a *level specification*

$$\Lambda := (\Pi_\Lambda, \mathcal{A}_\Lambda, \varepsilon),$$

where:

- $\Pi_\Lambda$  is a fixed level partition determining what is treated as “individual” versus “global” at that stage (i.e. which degrees of freedom are retained or suppressed by level-respecting encodings);
- $\mathcal{A}_\Lambda$  is a declared admissible family of level-respecting perspectives (re-expressions that do not change  $\Pi_\Lambda$ );
- $\varepsilon > 0$  is a declared resolution threshold used for reporting indistinguishability (Def. 4.10).

*Remark 4.2* (“Whole” versus “individual” is boundary-induced). Within this formalization, “whole” and “individual” are not intrinsic labels of an object; they are induced by the declared level specification (partition and admissible re-expressions) and by the boundary choices encoded in the package.

**Definition 4.6** (Level-respecting re-expression). For two perspectives  $A_1, A_2$  within the same admissible family  $\mathcal{A}_\Lambda$ , a *level-respecting re-expression* is an invertible map

$$T_{A_2 \leftarrow A_1} : \Omega_{A_1} \rightarrow \Omega_{A_2}$$

such that for all  $S \in \Omega$ ,

$$\Phi_{A_2}(S) = T_{A_2 \leftarrow A_1}(\Phi_{A_1}(S)).$$

When such a  $T_{A_2 \leftarrow A_1}$  exists,  $A_1$  and  $A_2$  are treated as equivalent encodings at the same declared level (up to reparameterization).

**Definition 4.7** (Designated cross-level pair). We fix two designated perspectives, interpreted as an individual-level and a global-level description:

$$A_I, A_G, \quad \Phi_I : \Omega \rightarrow \Omega_I, \Phi_G : \Omega \rightarrow \Omega_G.$$

Here  $\Omega_I$  and  $\Omega_G$  are the respective representation spaces induced by  $A_I$  and  $A_G$ .

**Definition 4.8** (Interface mechanism). A *cross-level interface mechanism* consists of:

- a discrepancy rule  $D : \Omega_I \times \Omega_I \rightarrow \mathbb{R}_{\geq 0}$  (the notion of mismatch);
- an admissible family of liftings (reconstructions)  $\mathcal{U} \subseteq \{u : \Omega_G \rightarrow \Omega_I\}$ . If a declared package calls such liftings *sections*, the package must additionally state the relevant macro-consistency clause (e.g. when  $\Omega_I$  is identified with the state fiber,  $u(g) \in F_g$ ; in interface-image typed cases,  $u(g) \in \Phi_I(F_g)$ ). Otherwise  $u$  is only an admissible reconstruction candidate.

*Remark 4.3* (Operational degrees of freedom must be locked). Any scalar cross-level diagnostic (e.g. a residual) is not well-posed until the operational degrees of freedom are locked. Minimally, one must lock

$$(\Phi_I, \Phi_G, D, \mathcal{U}, \Omega_{\text{reach}}, \varepsilon),$$

where  $\varepsilon$  is the declared reporting resolution (Def. 4.10).

### 4.3 Declared package $V$ as a coupling layer

A declared package  $V$  is an audit bundle because it first functions as a coupling layer: it specifies how objective structure is made reportable through finite access, admissible compression, reconstruction, discrepancy, comparability, and resolution clauses. The package therefore does not stand outside the analysis as a subjective annotation; it is part of the reportability conditions under which any structural quantity in this manuscript is typed and compared.

**Definition 4.9** (Declared package). A *declared package* is an audit bundle that locks the operational degrees of freedom under which cross-level statements are made. Concretely, we write

$$V := (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C}),$$

where:

- $\varepsilon > 0$  is the declared reporting threshold (resolution convention);
- $\mathcal{Q}$  is a declared set of audited predicates/queries (optional in purely structural sections, mandatory in audit sections);
- $\mathcal{C}$  is a declared admissible class of constrained histories or admissible trajectories (optional unless dynamics/audit questions are invoked);



- $\mathcal{I}$  is the *interface module* responsible for uniquely locking cross-level diagnostics, and must minimally include the core degrees of freedom

$$\mathcal{I} \supseteq (\Phi_I, \Phi_G, D, \mathcal{U}, \Omega_{\text{reach}}),$$

where  $(\Phi_I, \Phi_G)$  are the chosen representation maps,  $D$  is the discrepancy rule on  $\Omega_I$ ,  $\mathcal{U}$  is the admissible lifting family  $\Omega_G \rightarrow \Omega_I$ , and  $\Omega_{\text{reach}}$  is the reachable set over which worst-case statements are evaluated.

*Remark 4.4* (Typed internal refinements preserve the package schema). The tuple

$$V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$$

remains the authoritative top-level package schema throughout the paper. Optional modules may refine the internal content of an existing component without adding a new top-level slot. In particular, the fixed-target resolution module first declares a raw realization clause  $\mathcal{R}_\lambda^{\text{raw}} = (\mathcal{N}_\lambda, D^*, \hat{D}_\lambda)$ , then extends it by the certification domain, one-sided envelopes, and certificate object to obtain  $\mathcal{R}_\lambda^{\text{cert}}$ , with

$$\mathcal{I}_\lambda = \mathcal{I}^{\text{base}} \oplus \mathcal{R}_\lambda^{\text{cert}}.$$

The module refines  $\mathcal{Q}$  first by task clauses (ordered null/separation pairs and directional pairwise or partition semantics) and only then, when invoked, by a separate policy-floor clause. The scope under which certificates are reused in scans or adaptive paths, and the adaptive control law itself, remain protocol declarations associated with the package card. None replaces the package definition. Here  $\oplus$  denotes typed extension of an existing component, not a new algebraic primitive.

*Remark 4.5* (Threshold binding within a declared package). When a declared package  $V$  provides a reporting threshold  $\varepsilon$ , all level specifications  $\Lambda$  invoked inside  $V$  are understood to inherit the same  $\varepsilon$  unless an explicit multi-threshold conversion clause is declared. If thresholds differ across reports without such a conversion clause, comparisons are treated as *undefined* (optional diagnostic tag RES / resolution-mismatch) in the sense fixed by Normative Rule 2.1.

**Definition 4.10** (Resolution-aware indistinguishability). Under a declared threshold  $\varepsilon > 0$ , the symbol “ $\approx_\varepsilon$ ” denotes the threshold predicate used for resolution-aware indistinguishability. Conditional on package well-posedness and on every additional operability, comparability, and certification gate invoked by the report, a nonnegative scalar  $x \geq 0$  is evaluated by

$$x \approx_\varepsilon 0 \iff x < \varepsilon.$$

The inequality fixes the threshold semantics; it does not by itself establish full report licensing. When a discrepancy rule  $D$  is used, the same conditional convention applies to declared comparisons induced by a symmetric report rule. A directional threshold predicate such as  $E_{\lambda, \varepsilon}$  in Sec. 4.4 is not, by itself, an instance of the symmetric relation  $\approx_\varepsilon$ ; an explicit symmetric rule or certified symmetrization is required.

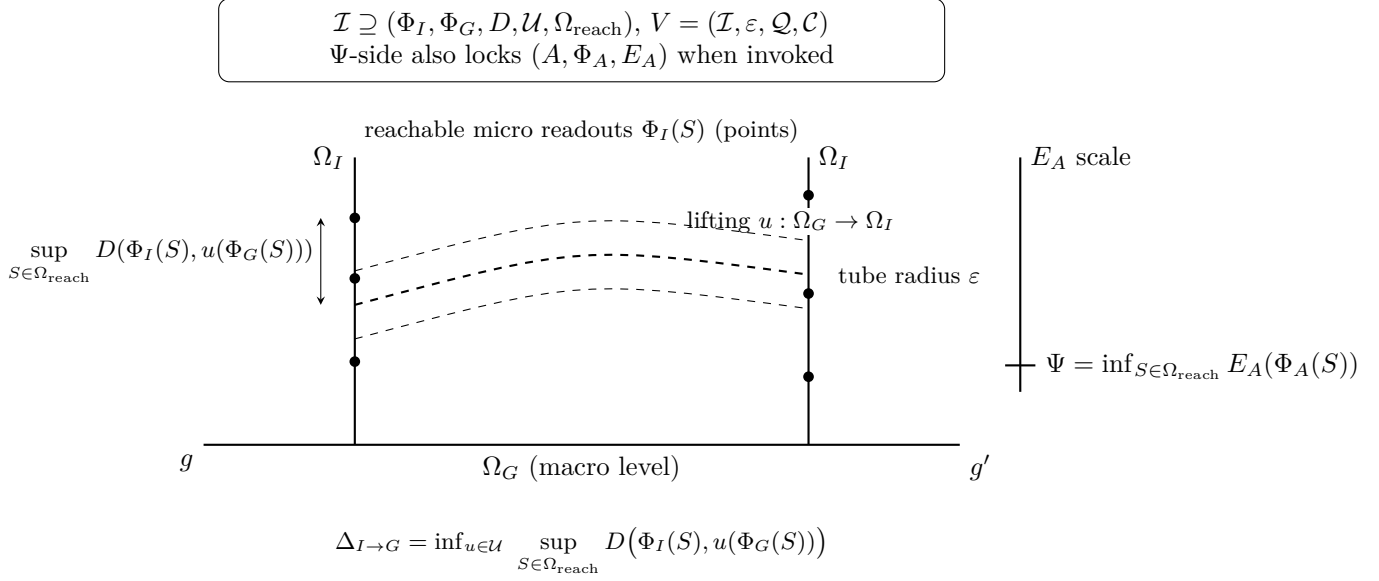


Figure 2: Fiber view of cross-level gluing under a declared interface. The macro representation  $\Phi_G$  induces fibers above  $\Omega_G$ , while reachable micro readouts  $\Phi_I(S)$  appear as points. A declared lifting  $u \in \mathcal{U}$  selects a reconstruction candidate;  $\Delta_{I \rightarrow G}$  is the minimax worst-case residual measured by the declared discrepancy rule  $D$  over  $\Omega_{\text{reach}}$ . The dashed band of radius  $\varepsilon$  visualizes the reporting convention  $\Delta_{I \rightarrow G} \approx_\varepsilon 0 \Leftrightarrow \Delta_{I \rightarrow G} < \varepsilon$  under the locked interface package  $(\Phi_I, \Phi_G, D, \mathcal{U}, \Omega_{\text{reach}}, \varepsilon)$ . The intra-level quantity  $\Psi$  (right) is shown only as a floor marker  $\inf_{S \in \Omega_{\text{reach}}} E_A(\Phi_A(S))$  for a declared perspective  $A$  with representation map  $\Phi_A$  and conflict functional  $E_A$ , and is distinct from  $\Delta_{I \rightarrow G}$ .

*Remark 4.6* (The reporting threshold is declared but may also require feasibility certification). The framework does not supply a universal, task-independent rule for choosing  $\varepsilon$ . The threshold is a package-declared reporting parameter, not a hidden constant. Declaration alone, however, is not always sufficient. When the report invokes a certified resolution or numerical-realization refinement of the interface, the completed package instance must also lie in the corresponding certificate-relative set of Sec. 4.4. Measurement resolution, numerical precision, truncation analysis, sampling certification, task loss, and policy tolerance therefore enter through typed clauses rather than as an undifferentiated list of “sources of  $\varepsilon$ .” If neither a threshold criterion nor an applicable certificate is declared, a resolution-negligibility report is not licensed. Failure under a current certificate does not by itself prove absolute impossibility under every sharper certificate or improved mechanism.

**Protocol 4.1** (Minimal reporting card for  $V$ ). Any reported value of a package-relative quantity (including later  $\Psi$  and  $\Delta_{I \rightarrow G}$ ) is regarded as well-posed and comparable only if it reports at least:

1. **Substrate and boundary:** the substrate regime  $\mathfrak{R}$  (explicitly or by reference) and the induced fine-grained domain  $(\Omega, \Omega_{\text{reach}})$ ;
2. **Designated perspectives:** the designated perspectives  $(A_I, A_G)$  and representation maps  $(\Phi_I, \Phi_G)$ ;
3. **Reachability:** the reachable set  $\Omega_{\text{reach}}$  (and any declared dynamics or admissibility constraints used to define it);
4. **Interface mechanism:** discrepancy rule  $D$  on  $\Omega_I$  and the admissible lifting family  $\mathcal{U}$ ;
5. **Reporting convention:** threshold  $\varepsilon$  and the meaning of “ $\approx_\varepsilon$ ” used in the report;

6. **Resolution-certification clause (when invoked):** the raw and certified refinement family  $\{\mathcal{R}_\lambda^{\text{raw}}, \mathcal{R}_\lambda^{\text{cert}}\}$  inside  $\mathcal{I}$ , fixed target discrepancy  $D^*$ , reported discrepancy  $\widehat{D}_\lambda$ , certification domain and envelopes  $\rho_\pm$ , ideal query margin and declared certified lower margin, ordered query refinements  $Q_0$  and/or  $Q_{\text{sep}}$ , directional pairwise or partition semantics, policy floor if any, and pointwise/simultaneous/sequential certificate scope;
7. **Audit clause (when applicable):** the audited query set  $\mathcal{Q}$  and admissible history class  $\mathcal{C}$ .

**Definition 4.11** (Package-locked reporting structure (logical structure)). Fix a target system  $\mathcal{L}$  (equivalently, a substrate regime together with its induced domain  $(\Omega, \Omega_{\text{reach}})$ ). Given a declared package  $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$ , the *reporting structure locked by*  $(\mathcal{L}, V)$  is the logical structure

$$\mathcal{V} := (\Omega_{\text{reach}}, \approx_\varepsilon, \Phi_I, \Phi_G, D, \mathcal{U}, \mathcal{Q}, \models_V),$$

where  $\models_V$  denotes evaluation/satisfaction carried out under the conventions, encodings, and admissibility constraints locked by  $V$ . All quantities reported in this manuscript (including  $\Psi$  and  $\Delta$ -type diagnostics) are to be understood as invariants/obstructions of  $\mathcal{V}$  (hence *package-relative*). In most of the manuscript we suppress  $\mathcal{V}$  and simply say “under  $V$ ”.

**Example 4.1** (Package Card:  $V_{\text{toy}}$  (XOR 2-bit), report ID: `Pkg_toy_xor`). • **ID / scope.** Target quantities:  $\Delta_{I \rightarrow G}(\mathcal{L}; V_{\text{toy}})$  and  $\Psi(\mathcal{L}, A; V_{\text{toy}})$ . Boundary: system state is 2-bit; no environment variables.

- **System object.**  $L$ : toy 2-bit system.  $\Omega = \{00, 01, 10, 11\}$ .
- **Audited domain.**  $\Omega_{\text{reach}} = \Omega$ .
- **Level specifications / perspectives.** For  $\Psi$ :  $\Lambda = \text{bit-level}$ ,  $A = \text{identity perspective}$ ,  $\Phi_A = \text{id} : \Omega \rightarrow \Omega$ . For  $\Delta_{I \rightarrow G}$ :  $\Phi_I = \text{id} : \Omega \rightarrow \Omega_I (= \Omega)$ ,  $\Phi_G(x_1 x_2) = x_1 \oplus x_2 : \Omega \rightarrow \Omega_G (= \{0, 1\})$ .
- **Conflict functional (locked for this package).**  $E_A(x_1 x_2) = \mathbf{1}\{(x_1 = x_2 = 1)\}$  (thus  $\Psi = 0$  in the infimum reading; the infimum is realized here). The alternative choice  $E_A \equiv 1$  would define a different package and is not part of  $V_{\text{toy}}$ .
- **Interface mechanism.** Discrepancy  $D$ : Hamming distance on 2-bit strings. Admissible liftings  $\mathcal{U}$ : macro-consistent section maps  $u : \{0, 1\} \rightarrow \Omega$  satisfying  $\Phi_G(u(g)) = g$  for  $g \in \{0, 1\}$ .
- **Reporting resolution / threshold.** This package locks the reporting threshold at  $\varepsilon = 2$ . Convention: “ $\Delta_{I \rightarrow G} \approx_\varepsilon 0$ ” iff “ $\Delta_{I \rightarrow G} < \varepsilon$ ” under the declared interface package.
- **Comparability.** No transport/comparability law  $\tau_{V \rightarrow V'}$  is declared; cross-package comparison is undefined (Normative Rule 2.2).
- **Admissibility constraints.** (toy)  $u$  is a macro-consistent section map  $\{0, 1\} \rightarrow \Omega$  and  $D$  is Hamming.

**Example 4.2** (Minimal end-to-end pipeline report under  $V_{\text{toy}}$ ). Consider the declared package card  $V_{\text{toy}}$  in Example 4.1. Both  $\Psi(\mathcal{L}, A; V_{\text{toy}})$  and  $\Delta_{I \rightarrow G}(\mathcal{L}; V_{\text{toy}})$  are well-defined because the required clauses  $(A, \Phi_A, E_A, \Phi_I, \Phi_G, D, \mathcal{U}, \Omega_{\text{reach}}, \varepsilon)$  are explicitly locked.

**Computing  $\Delta_{I \rightarrow G}$  (fixed-slice).** For the XOR macro, the fibers are  $\Phi_G^{-1}(0) = \{00, 11\}$  and  $\Phi_G^{-1}(1) = \{01, 10\}$ . Fix any admissible lifting  $u : \{0, 1\} \rightarrow \Omega$ , i.e. a macro-consistent section satisfying  $\Phi_G(u(g)) = g$ . On the fiber  $\Phi_G^{-1}(0)$ , macro-consistency forces  $u(0) \in \{00, 11\}$ ; the best admissible choice still has worst-case Hamming residual 2 (because  $D(00, 11) = 2$ ). Likewise, on  $\Phi_G^{-1}(1)$ , macro-consistency forces  $u(1) \in \{01, 10\}$ , and the worst-case residual is 2. Therefore

$$\sup_{S \in \Omega_{\text{reach}}} D(\Phi_I(S), u(\Phi_G(S))) = 2 \quad \text{for every admissible } u,$$

and hence

$$\Delta_{I \rightarrow G}(\mathcal{L}; V_{\text{toy}}) = \inf_{u \in \mathcal{U}} 2 = 2.$$

Because  $V_{\text{toy}}$  locks  $\varepsilon = 2$ , the strict threshold convention gives  $\Delta_{I \rightarrow G} = 2 \not\leq 2$ . The licensed output is therefore “not resolution-negligible” under  $V_{\text{toy}}$ .

**Computing  $\Psi$  and the induced H/M regime label.** The conflict functional locked by  $V_{\text{toy}}$  yields  $\Psi = 0$  in the infimum reading, and the infimum is realized in this toy domain. Under the level-locked H/M classification (Definition 5.5, with the admissible family declared as part of the level specification), this package yields an H-type regime. Replacing  $E_A$  by  $E_A \equiv 1$  would define a distinct package with an M-type report; that alternative is not silently identified with the present package.

**Undefinedness check.** If, in the package card, either the discrepancy rule  $D$  or the admissible lifting family  $\mathcal{U}$  is not declared, then  $\Delta_{I \rightarrow G}(\mathcal{L}; V_{\text{toy}})$  is **undefined** under the declared regime (under-specification; optional tag **U-spec**).

**Cross-package boundary.** Changing  $D$ ,  $\varepsilon$ , or  $\mathcal{U}$  defines a different package. Numerical comparison across such packages is licensed only by an explicit transport clause; otherwise the comparison is **undefined** (Normative Rule 2.2).

#### 4.4 Resolution-controlled refinements and certified threshold feasibility

Declaring a reporting threshold does not by itself establish that the threshold is supportable by the invoked resolution mechanism or fine enough for the declared task. This subsection records a minimal fixed-target certification result. It introduces no new top-level component of the package  $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$ . Instead, when this optional module is invoked, it adds typed subclauses inside the existing interface and query components. Its scope is deliberately narrow: a scalar target discrepancy is held fixed, a parameterized mechanism family reports estimates of that same discrepancy, and the audited task declares ordered-pair separation requirements and, when a full two-sided classifier is invoked, ordered-pair null-preservation requirements. Partition output is treated only after a symmetric threshold relation or a declared certified symmetrization is supplied. The full pairwise/partition module is given in Appendix E.

**Definition 4.12** (Fixed-target raw resolution refinement). Fix base package clauses

$$\mathcal{I}^{\text{base}}, \quad \mathcal{Q}^{\text{base}}, \quad \mathcal{C}^{\text{base}},$$

where  $\mathcal{I}^{\text{base}}$  retains the original structural interface, including the core clauses required by Definition 4.9. Let  $X_{\text{adm}}$  be a declared admissible report domain and let

$$D^* : X_{\text{adm}} \times X_{\text{adm}} \rightarrow \mathbb{R}_{\geq 0}$$

be a fixed target discrepancy. The target  $D^*$  may be the base discrepancy  $D$  itself or a declared scalar report induced by the base interface; it does not replace or redefine the underlying structural interface and need not be symmetric.

A *raw fixed-target resolution refinement* is a declared family

$$\mathfrak{R}_{\text{res}}^{\text{raw}} := \{\mathcal{R}_\lambda^{\text{raw}}\}_{\lambda \in \Lambda_{\text{res}}}, \quad \mathcal{R}_\lambda^{\text{raw}} := (\mathcal{N}_\lambda, D^*, \widehat{D}_\lambda),$$

where  $\mathcal{N}_\lambda$  is the resolution or numerical-realization mechanism and

$$\widehat{D}_\lambda : X_{\text{adm}} \times X_{\text{adm}} \rightarrow \mathbb{R}_{\geq 0}$$

is its report of the same fixed target discrepancy. The control parameter  $\lambda$  may represent, for example, a step size, mesh width, sampling rate, quadrature order, or truncation order; it is not the report threshold  $\varepsilon$ .

**Definition 4.13** (Uniform certification and certified interface refinement). Let  $\mathcal{D}_{\text{cert}} \subseteq X_{\text{adm}} \times X_{\text{adm}}$  be the declared certification domain. A two-sided certification scheme consists of one-sided envelopes  $\rho_+, \rho_- : \Lambda_{\text{res}} \rightarrow \mathbb{R}_{\geq 0}$  and a declared certificate object  $\text{Cert}$  such that, deterministically or on a declared uniform confidence event,

$$\widehat{D}_\lambda(x, y) - D^*(x, y) < \rho_+(\lambda), \quad D^*(x, y) - \widehat{D}_\lambda(x, y) < \rho_-(\lambda)$$

for all  $(x, y) \in \mathcal{D}_{\text{cert}}$ . Define the certified refinement subclause

$$\mathcal{R}_\lambda^{\text{cert}} := (\mathcal{R}_\lambda^{\text{raw}}, \mathcal{D}_{\text{cert}}, \rho_+(\lambda), \rho_-(\lambda), \text{Cert}),$$

and the refined interface

$$\mathcal{I}_\lambda := \mathcal{I}^{\text{base}} \oplus \mathcal{R}_\lambda^{\text{cert}}.$$

Thus the certificate data are part of the typed internal refinement of  $\mathcal{I}$ , not a new top-level package slot.

Under the strict-envelope convention above, an exact or non-strict error certificate is represented by adding a declared positive slack. A zero envelope is valid only when the corresponding signed report error is uniformly strictly negative; in particular, symmetric zero envelopes do not represent an exact mechanism under the strict convention.

**Definition 4.14** (True and certified query margins). Let  $Q_{\text{sep}} \subseteq \mathcal{D}_{\text{cert}}$  be a nonempty, predeclared set of ordered pairs whose declared directional distinctions must remain reportably non-negligible, with  $x \neq y$  for every  $(x, y) \in Q_{\text{sep}}$ . Define the ideal pairwise margin

$$\delta_Q^{\text{pair}} := \inf_{(x, y) \in Q_{\text{sep}}} D^*(x, y).$$

Operational use requires a declared certified lower-margin family

$$\underline{\delta}_Q^{\text{pair}} : \Lambda_{\text{res}} \rightarrow \mathbb{R}_{\geq 0}, \quad 0 \leq \underline{\delta}_Q^{\text{pair}}(\lambda) \leq \delta_Q^{\text{pair}},$$

together with the pointwise, simultaneous, or sequential scope under which it remains valid. A family-uniform certificate is the special case in which  $\underline{\delta}_Q^{\text{pair}}(\lambda) \equiv \underline{\delta}_Q^{\text{pair}}$  on the invoked mechanism family. When the ideal margin is exactly available, one may take  $\underline{\delta}_Q^{\text{pair}}(\lambda) = \delta_Q^{\text{pair}}$  for every invoked  $\lambda$ .

When a full two-sided threshold classifier is invoked, also declare a nonempty ordered-pair null set  $Q_0 \subseteq \mathcal{D}_{\text{cert}}$  satisfying

$$D^*(x, y) = 0 \quad \text{for every } (x, y) \in Q_0,$$

whose pairs must be reported negligible. Let  $\text{Sem}$  specify directional pairwise output or partition output, and let  $p \geq 0$  be any declared policy floor.

**Definition 4.15** (Task-indexed query refinements and completed packages). Let

$$\sigma \in \{\text{pair}, \text{part}\}, \quad \tau \in \{\text{sep}, \text{class}\}$$

index the report semantics and the task mode. Define the certificate-level query refinements

$$\mathcal{Q}^{\sigma, \text{sep}} := \mathcal{Q}^{\text{base}} \oplus (Q_{\text{sep}}, \text{Sem} = \sigma),$$

and, when the declared null set  $Q_0$  is present,

$$\mathcal{Q}^{\sigma, \text{class}} := \mathcal{Q}^{\text{base}} \oplus (Q_{\text{sep}}, Q_0, \text{Sem} = \sigma).$$

If a policy floor is invoked, define

$$\mathcal{Q}^{\sigma, \tau, p} := \mathcal{Q}^{\sigma, \tau} \oplus p.$$

For each  $(\lambda, \varepsilon) \in \Lambda_{\text{res}} \times \mathbb{R}_{>0}$ , the corresponding ordinary completed packages under the unchanged schema are

$$V_{\lambda, \varepsilon}^{\sigma, \tau} := (\mathcal{I}_{\lambda}, \varepsilon, \mathcal{Q}^{\sigma, \tau}, \mathcal{C}^{\text{base}}),$$

and, when policy is invoked,

$$V_{\lambda, \varepsilon}^{\sigma, \tau, p} := (\mathcal{I}_{\lambda}, \varepsilon, \mathcal{Q}^{\sigma, \tau, p}, \mathcal{C}^{\text{base}}).$$

Thus directional-pairwise versus partition semantics, separation-only versus full classification, and certificate-only versus policy-restricted use are distinct internal query refinements of the same top-level package schema. The  $\rho_+$  floor belongs only to the  $\tau = \text{class}$  mode.

**Definition 4.16** (Task-indexed package-admissible threshold sets). For each fixed mechanism parameter  $\lambda$  and task index  $(\sigma, \tau)$ , define

$$\mathcal{E}_{\text{adm}}^{\sigma, \tau}(\lambda) := \left\{ \varepsilon > 0 : V_{\lambda, \varepsilon}^{\sigma, \tau} \text{ is admissible} \right\},$$

and, when a policy clause is invoked,

$$\mathcal{E}_{\text{adm}}^{\sigma, \tau, p}(\lambda) := \left\{ \varepsilon > 0 : V_{\lambda, \varepsilon}^{\sigma, \tau, p} \text{ is admissible} \right\}.$$

These task-indexed sets collect the package-admissibility restrictions not already expressed by the algebraic certificate inequalities. The certificate inequalities below generate algebraic threshold slices; the actually licensed slices are their intersections with the corresponding  $\mathcal{E}_{\text{adm}}^{\sigma, \tau}$  sets. No endpoint criterion may silently replace that intersection unless the required algebraic slice is known to lie inside the relevant task-indexed package-admissible threshold set.

| Envelope | Error controlled                                               | Operational roles                                                                                                                                                                                                               |
|----------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\rho_+$ | over-report allowance: $\widehat{D}_{\lambda} - D^* < \rho_+$  | sets the null-preservation floor for a declared $Q_0$ ; converts an observed report into a certified lower bound $D^* > \widehat{D}_{\lambda} - \rho_+$ ; may therefore also enter the construction of a certified query margin |
| $\rho_-$ | under-report allowance: $\widehat{D}_{\lambda} < D^* - \rho_-$ | erodes the certified separation-query ceiling and converts a certified lower bound on $D^*$ into the display guarantee $\widehat{D}_{\lambda} > D^* - \rho_-$                                                                   |

The subscripts refer to the sign of the report error, not to whether the resulting bound on  $D^*$  is upper or lower. In particular,  $\rho_+$  is the envelope used when an observed positive report is converted into a lower bound on the target discrepancy.

**Definition 4.17** (Certified directional pairwise feasibility sets). Under the directional threshold convention

$$\widehat{D}_\lambda(x, y) < \varepsilon \implies (x, y) \text{ is reported negligible in the declared direction,}$$

define the algebraic slices

$$I_\lambda^{\text{sep, cert}} := \left(0, \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda)\right],$$

and, when  $Q_0$  is declared,

$$I_\lambda^{\text{class, cert}} := (0, \infty) \cap \left[\rho_+(\lambda), \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda)\right].$$

The package-conditioned certified sets are

$$\mathcal{A}_{Q_{\text{sep}}}^{\text{sep, cert}} := \left\{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{pair, sep}}(\lambda) \cap I_\lambda^{\text{sep, cert}}\right\},$$

and

$$\mathcal{A}_{Q_0, Q_{\text{sep}}}^{\text{class, cert}} := \left\{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{pair, class}}(\lambda) \cap I_\lambda^{\text{class, cert}}\right\}.$$

The first protects the declared directional distinctions; the second additionally guarantees negligible reports on the declared ordered null pairs.

The policy-restricted slices are

$$I_{\lambda, p}^{\text{sep, cert}} := I_\lambda^{\text{sep, cert}} \cap [p, \infty), \quad I_{\lambda, p}^{\text{class, cert}} := I_\lambda^{\text{class, cert}} \cap [p, \infty),$$

and the corresponding licensed sets use  $\mathcal{E}_{\text{adm}}^{\text{pair, sep}, p}(\lambda)$  and  $\mathcal{E}_{\text{adm}}^{\text{pair, class}, p}(\lambda)$ , respectively.

**Proposition 4.1** (Fixed-mechanism certified feasibility). *Fix  $\lambda$  and write*

$$\underline{b}_\lambda := \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda).$$

*The exact fixed-mechanism licensed slices are*

$$\mathcal{E}_{\text{adm}}^{\text{pair, sep}}(\lambda) \cap (0, \underline{b}_\lambda],$$

*for separation-only reporting, and*

$$\mathcal{E}_{\text{adm}}^{\text{pair, class}}(\lambda) \cap (0, \infty) \cap [\rho_+(\lambda), \underline{b}_\lambda]$$

*for two-sided classification. Policy replaces the separation and classification admissibility sets by  $\mathcal{E}_{\text{adm}}^{\text{pair, sep}, p}$  and  $\mathcal{E}_{\text{adm}}^{\text{pair, class}, p}$ , respectively, and intersects the algebraic slice with  $[p, \infty)$ . Consequently, each licensed slice is nonempty if and only if the displayed intersection is nonempty.*

**Corollary 4.1** (Endpoint criteria under admissibility-neutrality). *Suppose the relevant algebraic slice is contained in the corresponding package-admissible threshold set. Then the separation-only slice is nonempty exactly when  $\underline{b}_\lambda > 0$ , and the two-sided classification slice is nonempty exactly when*

$$\rho_+(\lambda) \leq \underline{b}_\lambda \quad \text{and} \quad \underline{b}_\lambda > 0.$$

*The policy-restricted endpoint conditions are, respectively,*

$$p \leq \underline{b}_\lambda, \quad \underline{b}_\lambda > 0,$$

*and*

$$\max\{p, \rho_+(\lambda)\} \leq \underline{b}_\lambda, \quad \underline{b}_\lambda > 0.$$

*If  $\rho_+ = \rho_- = \rho$ , the full-classification algebraic slice is nonempty exactly when*

$$2\rho(\lambda) \leq \underline{\delta}_Q^{\text{pair}}(\lambda) \quad \text{and} \quad \underline{\delta}_Q^{\text{pair}}(\lambda) > 0.$$

*These endpoint tests certify package-conditioned feasibility only under the stated admissibility containment.*

*Remark 4.7* (Ideal and certified margins). Replacing  $\underline{\delta}_Q^{\text{pair}}(\lambda)$  by the ideal margin  $\delta_Q^{\text{pair}}$  defines the corresponding ideal algebraic geometry. The operational certificate slices are conservative because  $\underline{\delta}_Q^{\text{pair}}(\lambda) \leq \delta_Q^{\text{pair}}$ . Failure of a package-conditioned certified set may arise from an empty algebraic certificate slice, from package inadmissibility on that slice, or from both; it need not imply failure of the ideal geometry.

*Remark 4.8* (Certificate-relative, package-conditioned output). The framework does not derive a universal, task-independent threshold. It derives certificate-relative, package- and query-conditioned sets once a fixed target discrepancy, a typed resolution refinement of the interface, typed task and policy refinements, and external physical, numerical, or statistical certificates have been declared. Failure to enter one of these sets does not, by itself, prove that no sharper margin certificate, sharper mechanism certificate, or improved mechanism could license the report.

*Remark 4.9* (Multiple audited queries). If one completed package uses one threshold for several audited query bundles, the configuration must lie in the intersection of their certified sets. Replacing the bundles by unions of their  $Q_{\text{sep}}$  and  $Q_0$  clauses is equivalent only when all bundles share the same  $D^*$ , raw and certified refinement family, certification domain, envelopes, margin-certification construction, certificate scope, report semantics, and policy clauses. If policy floors differ, the joint policy clause uses  $p_{\text{joint}} := \sup_i p_i$ . If directional pairwise and partition semantics or certificate scopes differ, the intersection formulation must be retained. A threshold certified for only one member of  $\mathcal{Q}$  is not thereby certified for the whole package.

*Remark 4.10* (Pairwise scope and partition boundary). Proposition 4.1 uses an ordered-pair threshold-report predicate and does not require a symmetric discrepancy. This directional predicate is not, by itself, the framework’s symmetric resolution-indistinguishability predicate  $\approx_\varepsilon$ ; it may instantiate that relation only after a symmetric rule or a declared certified symmetrization is supplied. If the deliverable must be an equivalence partition containing all threshold-level identifications, direct pairwise separation is insufficient because finite-resolution indistinguishability need not be transitive. Appendix E introduces the corresponding bottleneck-chain margin and identifies transitive closure as the precise contact point with finite-resolution repair non-uniqueness.

## 4.5 Comparability across packages

**Definition 4.18** (Comparability and transport law). Let  $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$  and  $V' = (\mathcal{I}', \varepsilon', \mathcal{Q}', \mathcal{C}')$  be two declared packages. Cross-package comparison of reported scalars is defined only if a *transport law* is explicitly declared, consisting of:

- reparameterization/identification maps between the relevant representation spaces (e.g. between  $\Omega_I$  and  $\Omega'_I$ ; between  $\Omega_G$  and  $\Omega'_G$ ), compatible with the chosen  $(\Phi_I, \Phi_G)$ ;
- a transformation rule specifying how  $D$  and  $\mathcal{U}$  are transported (pushforward/pullback) under these identifications;
- a resolution rule specifying how  $\varepsilon$  relates to  $\varepsilon'$  for reporting.

If no such transport law is declared, cross-package comparison statements are **undefined**; see Normative Rule 2.2.

## 5 Core Quantities: $\Psi$ and the Fixed-Slice Gap $\Delta_{I \rightarrow G}$

This section defines the two core quantities of the framework. Both are *package-relative*: their meanings and reported values are uniquely locked only after a declared package  $V$  has fixed the relevant interface mechanism and reporting convention (Sec. 4).



**Main-line convention.** In the slim core, the gap symbol refers to the fixed-slice diagnostic  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ . Accumulated quantities and ontological guardrails are always written explicitly as  $\Delta_{\text{acc}}(\gamma)$  and  $\Delta_R^*$  (with  $R$  suppressed when unambiguous), respectively.

## 5.1 Intra-level structural tension readout $\Psi$

**Definition 5.1** (Conflict functional (declared)). Fix a system  $\mathcal{L}$  with fine-grained state space  $\Omega$  and reachable set  $\Omega_{\text{reach}}$ . Let  $A$  be a declared perspective at a fixed level specification (Def. 4.5) with representation map  $\Phi_A : \Omega \rightarrow \Omega_A$ .

A *conflict functional* under  $A$  is a declared map

$$E_A : \Omega_A \rightarrow \mathbb{R}_{\geq 0},$$

whose interpretation is fixed as part of the declared regime (either explicitly listed in  $V$  or deterministically induced by locked clauses of  $V$ ).

**Definition 5.2** ( $\Psi$  (intra-level structural tension readout)). Fix a declared package  $V$  and a perspective  $A$  admissible at a fixed level. Assume  $V$  declares (or induces) a conflict functional  $E_A$  in the sense of Def. 5.1, and fixes the reachable set  $\Omega_{\text{reach}}$ .

The intra-level structural tension readout is defined by

$$\Psi(\mathcal{L}, A; V) := \inf_{S \in \Omega_{\text{reach}}} E_A(\Phi_A(S)).$$

(Equivalently,  $\Psi(\mathcal{L}, A; V) = \inf_{y \in \Phi_A(\Omega_{\text{reach}})} E_A(y)$ .)

**Non-vacuity clause.** The declared conflict functional  $E_A$  is admissible only when its variables, audited tension source, zero condition, discrepancy meaning, and reachable-set convention are explicitly tied to the clauses of  $V$ . Otherwise  $\Psi(\mathcal{L}, A; V)$  is **undefined** rather than freely assigned.

**Operability gate.** If the declared reachable set  $\Omega_{\text{reach}}$  is empty, numerical reporting of  $\Psi(\mathcal{L}, A; V)$  is **undefined** under Protocol 8.4 (optional tag: ADM).

*Remark 5.1* (Allowed values and interpretation). By construction,  $\Psi(\mathcal{L}, A; V) \in [0, \infty]$  and in particular  $\Psi$  may be equal to 0.

**Neutral reading.**  $\Psi(\mathcal{L}, A; V) = 0$  means that, within the declared reachability and within the chosen conflict functional, there exist reachable states whose residual conflict can be made arbitrarily small under  $A$ .  $\Psi(\mathcal{L}, A; V) > 0$  means that a strictly positive lower bound persists under the same declared regime. No cross-level claims are implied by  $\Psi$  alone.

**Regime dependence.** Changing any of the locked clauses (e.g. reachable set, admissibility, or the choice of  $E_A$ ) generally changes the operational meaning and value of  $\Psi$ . Therefore, any reported  $\Psi$  must be accompanied by the relevant package elements (Sec. 4.3, Protocol 4.1).

## 5.2 Inter-level structural gap $\Delta_{I \rightarrow G}$

**Definition 5.3** ( $\Delta_{I \rightarrow G}$  (inter-level structural gap)). Fix a system  $\mathcal{L}$  with state space  $\Omega$  and reachable set  $\Omega_{\text{reach}}$ . Let  $V$  be a declared package whose interface module  $\mathcal{I}$  (Def. 4.9) includes a designated cross-level pair

$$(\Phi_I : \Omega \rightarrow \Omega_I, \Phi_G : \Omega \rightarrow \Omega_G),$$

a discrepancy rule  $D : \Omega_I \times \Omega_I \rightarrow \mathbb{R}_{\geq 0}$ , and an admissible lifting family  $\mathcal{U} \subseteq \{u : \Omega_G \rightarrow \Omega_I\}$ .

The inter-level structural gap is defined as

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) := \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} D(\Phi_I(S), u(\Phi_G(S))).$$

This is an infimal worst-case residual over the declared admissible lifting class. If the infimum is attained, the attaining lifting may be called a best admissible reconstruction; otherwise only  $\eta$ -optimal admissible liftings are licensed. **Operability gate.** If  $\Omega_{\text{reach}} = \emptyset$  or  $\mathcal{U} = \emptyset$  under the declared regime, numerical reporting of  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  is **undefined** under Protocol 8.4 (optional tag: ADM).

**Definition 5.4** (Macro fibers and interface diameter). Under the interface pair  $(\Phi_I, \Phi_G)$  declared in  $V$ , for any macro value  $g \in \Phi_G(\Omega_{\text{reach}})$  define the *macro fiber*

$$F_g := \{S \in \Omega_{\text{reach}} : \Phi_G(S) = g\}.$$

Its interface image is  $\Phi_I(F_g) \subseteq \Omega_I$ . If the declared discrepancy rule  $D$  is a pseudometric on  $\Omega_I$ , define the *fiber diameter*

$$\text{diam}_D(g) := \sup_{S, S' \in F_g} D(\Phi_I(S), \Phi_I(S')) \in [0, \infty].$$

See Section 5.3 for a geometric reading of fibers, diameter, and lifting radius.

**Assumption 5.1** (Pseudometric discrepancy for the fiber-diameter bound). The declared discrepancy rule  $D$  is a pseudometric on  $\Omega_I$ : for all  $x, y, z \in \Omega_I$ ,

$$D(x, x) = 0, \quad D(x, y) = D(y, x), \quad D(x, z) \leq D(x, y) + D(y, z).$$

This assumption is satisfied by common metric discrepancies used in auditing (e.g. total variation or IPM distances). When  $D$  is directed, the lower-bound mechanism below is not available in this form and requires a separately declared directed or two-sided variant.

### 5.3 Geometric intuition: projection and fibers

It is helpful to read the cross-level interface as a many-to-one projection. The map  $\Phi_G : \Omega_{\text{reach}} \rightarrow \Phi_G(\Omega_{\text{reach}})$  collapses multiple micro states into a single macro report. For any  $g \in \Phi_G(\Omega_{\text{reach}})$ , the fiber

$$F_g := \{S \in \Omega_{\text{reach}} : \Phi_G(S) = g\}$$

collects the micro states that are indistinguishable at the reporting level.

A lifting  $u \in \mathcal{U}$  selects a representative  $u(g) \in \Omega_I$  for each macro value  $g$ . Given the interface pseudometric  $D$  on the internal-side images, the worst-case representative radius at  $g$  is

$$r_u(g) := \sup_{T \in F_g} D(\Phi_I(T), u(g)).$$

By Assumption 5.1, any single representative controls pairwise variation on the same fiber: for all  $S, S' \in F_g$ ,

$$D(\Phi_I(S), \Phi_I(S')) \leq D(\Phi_I(S), u(g)) + D(u(g), \Phi_I(S')) \leq 2r_u(g).$$

Taking the supremum over  $S, S' \in F_g$  yields the structural bound

$$\text{diam}_D(g) \leq 2r_u(g).$$

Consequently, whenever a fiber has nonzero interface-image diameter, no lifting can be lossless in worst case: the residual is enforced by the fiber thickness under  $D$ , rather than by estimation noise. In the diagram below, the left panel displays the interface-image fiber  $\Phi_I(F_g) \subseteq \Omega_I$ , not the state fiber  $F_g \subseteq \Omega_{\text{reach}}$  itself; this keeps  $u(g)$  and the compared points in the same type space.

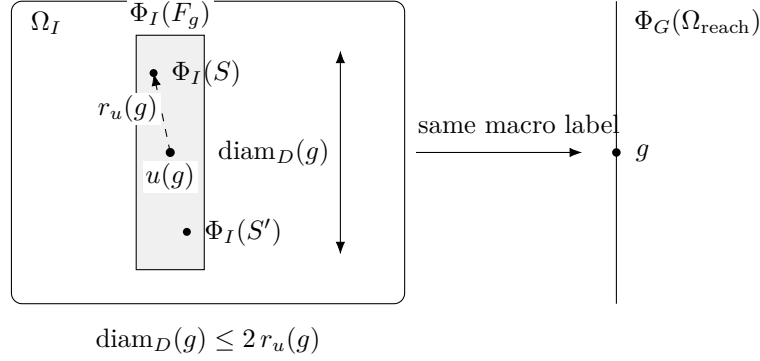


Figure 3: Projection geometry of a cross-level interface. The reporting map  $\Phi_G$  collapses micro states into a macro value  $g$ , producing a state fiber  $F_g$  whose interface image is  $\Phi_I(F_g) \subseteq \Omega_I$ . The figure displays this interface-image fiber so that  $u(g)$ ,  $\Phi_I(S)$ , and  $\Phi_I(S')$  live in the same type space. A lifting  $u$  selects a representative  $u(g)$ ; the interface-image thickness  $\text{diam}_D(g)$  controls the unavoidable worst-case representative radius  $r_u(g)$  via  $\text{diam}_D(g) \leq 2r_u(g)$  (Assumption 5.1).

Finally, the declared reporting package fixes a finite reporting resolution  $\varepsilon$ . Within a locked audit package  $V^{\text{aud}} = (\mathcal{I}^{\text{aud}}, \varepsilon, \mathcal{Q}, \mathcal{C})$ , whose audit interface  $\mathcal{I}^{\text{aud}}$  locks an endpoint operator **End** and the corresponding endpoint discrepancy, statements such as “reportable as 0” refer only to the thresholded convention at resolution  $\varepsilon$  (i.e., the relevant residual is below  $\varepsilon$  under  $V^{\text{aud}}$ ), and do not express any ontological equality across levels.

**Theorem 5.1** (Fiber-diameter lower bound for  $\Delta_{I \rightarrow G}$ ). *Assume the package clauses make  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  well-defined, with  $\Omega_{\text{reach}} \neq \emptyset$  and  $\mathcal{U} \neq \emptyset$ . Assume also Assumption 5.1. Then the package-relative inter-level gap satisfies*

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) \geq \frac{1}{2} \sup_{g \in \Phi_G(\Omega_{\text{reach}})} \text{diam}_D(g).$$

*In particular, if there exists  $g$  with  $\text{diam}_D(g) > 0$ , then  $\Delta_{I \rightarrow G}(\mathcal{L}; V) > 0$ .*

*Proof.* Fix an admissible lifting  $u \in \mathcal{U}$  and any  $g \in \Phi_G(\Omega_{\text{reach}})$ . For any  $S, S' \in F_g$ , Assumption 5.1 gives, using symmetry in the second term,

$$\begin{aligned} D(\Phi_I(S), \Phi_I(S')) &\leq D(\Phi_I(S), u(g)) + D(u(g), \Phi_I(S')) \\ &\leq 2 \sup_{T \in F_g} D(\Phi_I(T), u(g)). \end{aligned}$$

Taking the supremum over  $S, S' \in F_g$  yields

$$\begin{aligned} \text{diam}_D(g) &\leq 2 \sup_{T \in F_g} D(\Phi_I(T), u(g)) \\ &\leq 2 \sup_{T \in \Omega_{\text{reach}}} D(\Phi_I(T), u(\Phi_G(T))). \end{aligned}$$

Now take the infimum over  $u \in \mathcal{U}$  and then the supremum over  $g$ . □

**Corollary 5.1** (Resolution witness for “not  $\approx_\varepsilon 0$ ” reports). *Assume the package clauses make  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  well-defined, with  $\Omega_{\text{reach}} \neq \emptyset$  and  $\mathcal{U} \neq \emptyset$ , and assume Assumption 5.1. If there exist  $S, S' \in \Omega_{\text{reach}}$  such that  $\Phi_G(S) = \Phi_G(S')$  and  $D(\Phi_I(S), \Phi_I(S')) \geq 2\varepsilon$ , then  $\Delta_{I \rightarrow G}(\mathcal{L}; V) \geq \varepsilon$  and hence  $\Delta_{I \rightarrow G}(\mathcal{L}; V) \not\approx_\varepsilon 0$  under the reporting convention of Normative Rule 2.1.*

**Remark 5.2** (Finite-precision licensing of a  $\Delta_{I \rightarrow G}$  witness). If  $\Delta_{I \rightarrow G}(\mathcal{L}; V) = +\infty$ , then it is non-negligible at every finite threshold and no finite residual-axis estimation construction is required. Assume henceforth that

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) < \infty.$$

The finite-precision mechanisms below are typed realization refinements of the fixed package interface: they leave the structural clauses  $(\Phi_I, \Phi_G, D, \mathcal{U}, \Omega_{\text{reach}})$ , and hence the target value  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ , unchanged while certifying how that fixed target or its witnesses are numerically reported. For every completed refinement package used below, declare the identity transport on the shared residual report axis. Under this transport,

$$\tau_{\lambda, \varepsilon}^\Delta = \text{id}_{\mathbb{R}_{\geq 0}}, \quad \Delta_{I \rightarrow G}(\mathcal{L}; V_{\lambda, \varepsilon}^{\text{pair, class}}) \equiv_{\tau_{\lambda, \varepsilon}^\Delta} \Delta_{I \rightarrow G}(\mathcal{L}; V).$$

The equality of target values is therefore licensed by the shared structural projection and the declared identity transport, not inferred merely from reuse of the symbol. For this fixed system-package instance, take the residual report domain

$$X_\Delta := \{0, \Delta_{I \rightarrow G}(\mathcal{L}; V)\} \subseteq \mathbb{R}_{\geq 0}, \quad D^*(r, s) := |r - s|,$$

with

$$Q_0^\Delta := \{(0, 0)\}, \quad Q_{\text{sep}}^\Delta := \{(\Delta_{I \rightarrow G}(\mathcal{L}; V), 0)\}.$$

Its pairwise separation margin is  $\delta_Q^{\text{pair}} = \Delta_{I \rightarrow G}(\mathcal{L}; V)$ . Write

$$L := \frac{1}{2} \sup_{g \in \Phi_G(\Omega_{\text{reach}})} \text{diam}_D(g),$$

so that Theorem 5.1 gives  $L \leq \Delta_{I \rightarrow G}(\mathcal{L}; V)$ . When  $L$  is not exactly available, let  $0 \leq \underline{L} \leq L$  be any declared certified lower bound; taking  $\underline{L} = L$  recovers the exact case. Thus  $\underline{L}$  is a certified lower bound on the residual-axis separation margin, not the margin itself, and any set constructed from it is conservative.

The minimal licensing rule for a positive witness is one-sided: any certified quantity  $\underline{\Delta}_\lambda$  satisfying

$$\underline{\Delta}_\lambda \leq \Delta_{I \rightarrow G}(\mathcal{L}; V), \quad \underline{\Delta}_\lambda \geq \varepsilon,$$

licenses  $\Delta_{I \rightarrow G}(\mathcal{L}; V) \not\approx_\varepsilon 0$ . Three common sources of such a lower bound should be kept distinct.

1. **Structural-bound route.** The theorem-generated certificate itself gives  $\underline{\Delta}_\lambda := \underline{L}$ ; hence  $\varepsilon \leq \underline{L}$  is sufficient, with no additional envelope for a direct estimate of  $\Delta_{I \rightarrow G}$ .
2. **Witness-pair route.** Suppose a finite-precision mechanism reports a fiber-pair distance  $\hat{d}_\lambda$  for states  $S, S' \in \Omega_{\text{reach}}$  with  $\Phi_G(S) = \Phi_G(S')$ , and supplies a one-sided *over-report allowance*  $\eta_+(\lambda)$  such that

$$\hat{d}_\lambda - D(\Phi_I(S), \Phi_I(S')) < \eta_+(\lambda).$$

Then

$$\underline{\Delta}_\lambda^{\text{fib}} := \frac{1}{2}(\hat{d}_\lambda - \eta_+(\lambda)) \leq \Delta_{I \rightarrow G}(\mathcal{L}; V).$$

In particular,  $\hat{d}_\lambda - \eta_+(\lambda) \geq 2\varepsilon$  licenses the non-negligibility report through Corollary 5.1.

3. **Direct-estimate route.** If a mechanism reports  $\hat{\Delta}_\lambda$  with

$$\hat{D}_\lambda(\Delta_{I \rightarrow G}(\mathcal{L}; V), 0) = \hat{\Delta}_\lambda$$

and the over-report envelope of Definition 4.13 gives

$$\hat{\Delta}_\lambda - \Delta_{I \rightarrow G}(\mathcal{L}; V) < \rho_+(\lambda),$$

then  $\underline{\Delta}_\lambda^{\text{dir}} := \hat{\Delta}_\lambda - \rho_+(\lambda)$  is a certified lower bound. Hence  $\hat{\Delta}_\lambda - \rho_+(\lambda) \geq \varepsilon$  is sufficient for the positive report.

These one-sided witness rules are weaker than the full two-sided classifier of Sec. 4.4. A scalar estimate  $\hat{\Delta}_\lambda$  alone supports only the one-sided positive-witness route. The full two-sided slice additionally requires a certified residual-axis discrepancy report  $\hat{D}_\lambda$  on both the target pair  $(\Delta_{I \rightarrow G}, 0)$  and the declared null pair  $Q_0^\Delta = \{(0, 0)\}$ . If the same threshold must both merge the declared null residual and keep the fixed target residual  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  non-negligible, define the conservative algebraic full-classification slice

$$I_{\lambda, \underline{L}}^{\Delta, \text{class}} := (0, \infty) \cap [\rho_+(\lambda), \underline{L} - \rho_-(\lambda)].$$

The actually licensed slice is

$$\mathcal{E}_{\text{adm}}^{\text{pair, class}}(\lambda) \cap I_{\lambda, \underline{L}}^{\Delta, \text{class}}.$$

The algebraic slice is nonempty exactly when

$$\rho_+(\lambda) \leq \underline{L} - \rho_-(\lambda) \quad \text{and} \quad \underline{L} - \rho_-(\lambda) > 0.$$

In the symmetric case its algebraic nonemptiness is equivalent to

$$2\rho(\lambda) \leq \underline{L} \quad \text{and} \quad \underline{L} > 0.$$

These endpoint conditions imply a licensed package slice only when the algebraic slice intersects the task-indexed package-admissible threshold set. Failure of the conservative algebraic slice does not establish failure of the true algebraic slice because  $\underline{L}$  may be strictly below  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ ; failure of the licensed intersection may additionally arise from package inadmissibility.

The two factors of two appearing in this discussion express two-sided separation geometry but arise through different arguments. In Corollary 5.1, the factor follows from the triangle inequality on a structural fiber. In the symmetric classification criterion, it is the sum of the over-report and under-report allowances. The constants are analogous but are not identified. This remark licenses only fiber-witness-supported or directly certified  $\Delta_{I \rightarrow G}$  reports; it does not cover every  $\Delta_{I \rightarrow G}$  or  $\Psi$  computation.

*Remark 5.3* (Fiber witnesses as audit artifacts). A pair  $(S, S')$  with  $\Phi_G(S) = \Phi_G(S')$  and large interface discrepancy constitutes a *fiber witness*. When a report asserts that  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  is not resolution-negligible at threshold  $\varepsilon$ , including such a witness (or a reproducible procedure for generating it under  $V$ ) provides an explicit, package-locked explanation of why the “ $\approx_\varepsilon 0$ ” convention is unavailable.

**Probabilistic reporting branch.** A distribution-valued implementation of the same diagnostic, including TV/IPM reporting semantics and event-probability bounds, is recorded in Appendix B. This branch is not needed for the core definitions in this section and introduces no new primitive; it only translates package-locked residual statements into a declared distribution-level reporting vocabulary when the relevant interface clauses are supplied.

*Remark 5.4* (Probabilistic reporting when single-representative sufficiency fails). The probabilistic reading used in Appendix B should not be treated as a new core primitive of the framework. It is an interpretive consequence of the same  $V$ -locked reporting discipline. When an endpoint or compressed report is sufficient under  $V$ , the original structure can be represented by a single admissible  $V$ -level report. When single-representative sufficiency fails, the original structure is not thereby invalidated; what fails is the license to report it through one unique representative at the declared resolution.

In this situation,  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  marks the obstruction to such single-representative reporting. It does not by itself determine a probability law. A distribution-valued report is licensed only when the package additionally declares a candidate/fiber space, a fiber measure or prior, and the corresponding reporting interface. Under those clauses, if  $V$  is not enlarged by additional complement coordinates, a finer interface, or a different admissible reconstruction class, one

available replacement is a distribution-valued report. Informally, the unresolved candidates inside the  $V$ -fiber are no longer collapsed into a single representative, but are reported as a conditional distribution or mixture relative to the original structure.

One may write this intuition schematically as follows. This notation is not a new diagnostic layer; it only re-labels the currently invoked distribution-valued reporting situation, with  $e_V$  playing the role of the declared compressed report and  $\rho_{\mathfrak{S}}$  playing the role of the original-structure report object whose single-representative recovery is at issue. Let  $\mathcal{X}_V$  be the declared candidate/fiber space, let  $e_V : \mathcal{X}_V \rightarrow \mathbf{End}_V$  be the declared endpoint map, and let  $\rho_{\mathfrak{S}} : \mathcal{X}_V \rightarrow \mathbf{Rep}$  denote the report object induced by the original structure  $\mathfrak{S}$ , where  $\mathbf{Rep}$  may itself be a space of probability laws. If  $\mu_V(\cdot | e)$  is a declared conditional measure on the fiber  $e_V^{-1}(e)$ , then the corresponding probabilistic report is the pushforward

$$P_V^{\mathfrak{S}}(\cdot | e) := (\rho_{\mathfrak{S}})_{\#} \mu_V(\cdot | e) \in \mathcal{P}(\mathbf{Rep}).$$

Bayesian conditioning may then be read as the familiar update of this fiber measure after a  $V$ -declared evidence event. This reading explains why Bayesian forms become natural once a distribution-valued reporting clause and a declared evidence event are supplied, but it does not make Bayesian probability part of the structural core.

#### 5.4 Optional H/M regime labels induced by $\Psi$

The  $H/M$  vocabulary is an optional shorthand for a fixed-level  $\Psi$ -report. The naming is mnemonic:  $H$  refers to a homogeneous-cluster closure regime, where some admissible perspective removes the declared intra-level tension;  $M$  refers to a mixed heterogeneous package, where heterogeneous contributions remain mixed under every admissible perspective. The formal test is only the  $\Psi$ -criterion below.

**Definition 5.5** (Optional H/M label at a fixed level). Fix a level specification  $\Lambda$ , an admissible family  $\mathcal{A}_{\Lambda}$ , and a declared package  $V$  under which  $\Psi(\mathcal{L}, A; V)$  is well-defined for each  $A \in \mathcal{A}_{\Lambda}$ .

- $H$ -type at  $\Lambda$ : there exists  $A \in \mathcal{A}_{\Lambda}$  such that  $\Psi(\mathcal{L}, A; V) = 0$  in the infimum reading.
- $M$ -type at  $\Lambda$ :  $\Psi(\mathcal{L}, A; V) > 0$  for every  $A \in \mathcal{A}_{\Lambda}$ .

At finite reporting resolution, replace 0 by the declared threshold convention:  $\Psi(\mathcal{L}, A; V) < \varepsilon$  for the  $H$ -type report and  $\Psi(\mathcal{L}, A; V) \geq \varepsilon$  for the  $M$ -type report. If the level, admissible family, conflict functional, reachable set, or threshold is not declared, the label is **undefined**. The label is level-locked and must not be identified with  $\Delta_{I \rightarrow G}$  or  $\Delta_{\text{acc}}$ .

## 6 Structural Fracture Law and Diagnostic Translation

### 6.1 Structural fracture law (canonical statement)

**Structural fracture law.** When a logical system simultaneously contains opposing elements, and these elements produce feedback on the system, if one attempts to connect, across levels, the system's whole (macroscopic determinacy) and the system's internal factors (microscopic behavioral logic) by a single continuous-logic principle within a single perspective, then a judgment discontinuity is structurally unavoidable. In short:

opposing elements + system self-feedback + cross-level continuous logic  $\Rightarrow$  judgment discontinuity.

### 6.2 Diagnostic translation into $\Psi$ - $\Delta_{I \rightarrow G}$ language

The fracture law is used here as orientation, not as a derivation engine. Under a locked package, it is translated into three reportable diagnostics.

**Cross-level gluing.** A continuity-biased gluing claim asks for an admissible lift  $u \in \mathcal{U}$  such that

$$\sup_{S \in \Omega_{\text{reach}}} D(\Phi_I(S), u(\Phi_G(S))) < \varepsilon.$$

The corresponding diagnostic object is

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} D(\Phi_I(S), u(\Phi_G(S))).$$

If this value is not resolution-negligible under the declared package, the single-valued cross-level report is not licensed at that resolution.

**Intra-level oppositionality.** At a fixed level, opposing contributions are read through the conflict functional used to define  $\Psi(\mathcal{L}, A; V)$ . If no admissible perspective at the level drives the readout below the declared threshold, the situation receives the optional  $M$ -type label of Definition 5.5.

**Audit/fiber formulation.** For audited decidability or extendability claims, the same pressure appears as fiber non-constancy: two admissible histories share the declared endpoint/readout but differ on a declared query. This gives a concrete witness that the queried judgment is not single-valued under the locked viewpoint. Any closure-restoring response requires a separate complement or extension package; it is not built into the present core.

### 6.3 Meta-gap $\Delta^*$ (cross-level non-identity postulate)

**Definition 6.1** ( $\Delta_R^*$  (meta-gap schema)). The meta-gap  $\Delta_R^*$  is a semantic non-identity marker attached to a genuine cross-level relation  $R$ . It records that a genuine cross-level identification is not an ontological equality. It is not an operational inf-sup diagnostic and is not itself compared with  $\varepsilon$ ; only a separately declared package-level realization may enter such a comparison. The framework adopts the postulate schema

$$(\text{genuine cross-level relation } R) \implies \Delta_R^* > 0.$$

Here “ $> 0$ ” is schematic notation for semantic non-identity, not a numerical order comparison on an operational carrier. When  $R$  is unambiguous from context, we may write  $\Delta^*$  for  $\Delta_R^*$ .

*Remark 6.1* (Visibility is resolution-dependent; ontology is not). The meta-gap  $\Delta_R^*$  itself is neither assigned an operational magnitude nor compared with  $\varepsilon$ . If a package separately declares an operational realization  $r_V(R) \in \mathbb{R}_{\geq 0}$  of the relation, that realized residual may satisfy  $r_V(R) < \varepsilon$  and therefore be reported as resolution-negligible, while  $\Delta_R^* > 0$  remains unchanged as a semantic non-identity postulate. Accordingly, *diagnostic* threshold comparisons are carried out using the package-relative gap  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  (Sec. 5.2), not  $\Delta_R^*$ .

### 6.4 From static fracture to declared dynamics

The structural fracture law supplies the static orientation for the framework: opposing elements, system self-feedback, and a single cross-level continuous-logic principle mark the regime in which a cross-level judgment discontinuity is expected under the locked viewpoint. The first move of the present framework is to translate that discontinuity into a package-conditioned residual report: the obstruction is no longer left as an informal statement, but is read through declared interfaces, admissible reconstructions, discrepancy rules, and resolution clauses.

The second move is dynamical. When the same package-locked obstruction persists along a declared evolution, refinement, hierarchy, or scale parameter, the framework needs a minimal way to report persistence without importing a domain-specific dynamics. The role of the next

section is exactly this: it supplies the bookkeeping interface that turns a running local readout  $\Psi_{\text{run}}(\gamma)$  into an accumulated quantity  $\Delta_{\text{acc}}(\gamma)$  under declared transport across  $\gamma$ .

The dynamical interface does not add a new fracture law. It supplies the minimal reporting structure required when the same static obstruction is followed along a declared evolution parameter.

## 7 Minimal Dynamical Interface

### Module summary.

**Scope.** Provide the minimal core interface by which the static fracture diagnostic is extended along a declared evolution parameter  $\gamma$ , without adding new core primitives.

**Dependencies.** A declared evolution parameter  $\gamma$ , a declared evolution path  $\mathfrak{P}_\gamma$ , and an explicit transport/comparability clause across  $\gamma$ .

**Outputs.** A reproducible accumulated readout  $\Delta_{\text{acc}}(\gamma)$  and its running driver  $\Psi_{\text{run}}(\gamma)$ , together with resolution-aware reporting of claims such as  $\Delta_{\text{acc}}(\gamma_1) \approx_\varepsilon \Delta_{\text{acc}}(\gamma_0)$ .

**Non-invocation rule.** If the required evolution/transport clauses are not declared, any accumulated-gap claim must be reported as undefined under the reporting protocol. The interface is core to the framework only after such a declared evolution axis is supplied.

### 7.1 Declared evolution parameter and accumulated quantity

This section is the point at which the static fracture diagnostic becomes a minimal dynamics of declared residual readout.

**Source and status of the dynamical interface.** The relation  $d\Delta_{\text{acc}}/d\gamma := \Psi_{\text{run}}$  is introduced as a definition-level reporting interface. It is motivated by repeated cases in which local unresolved tension persists along a declared parameter and contributes to an accumulated residual readout. It is not asserted as a universal conservation law, and it does not identify  $\Delta_{\text{acc}}$  with every slice-local residual carrier by default.

**Definition 7.1** (Declared evolution path and accumulated structural quantity). Let  $\gamma \in [\gamma_0, \gamma_1]$  be a declared evolution parameter. A *declared evolution path* is a  $\gamma$ -indexed family

$$\mathfrak{P}_\gamma := (\mathcal{L}_\gamma, A_\gamma, V_\gamma),$$

such that for each  $\gamma$ :

- $\mathcal{L}_\gamma$  is the system object at stage  $\gamma$  (e.g. an effective description under refinement/coarse-graining/RG [7, 14, 15, 10, 13], if such a reading is intended);
- $A_\gamma$  is a declared perspective at the corresponding fixed level specification;
- $V_\gamma$  is a declared package whose locked clauses make  $\Psi(\mathcal{L}_\gamma, A_\gamma; V_\gamma)$  well-posed (Sec. 5).

In addition, the path must include an explicit transport/comparability law across  $\gamma$  in the sense of Def. 4.18, so that quantities reported at different  $\gamma$  are interpreted within a single declared convention.

Define the *running intra-level readout* along the path by

$$\Psi_{\text{run}}(\gamma) := \Psi(\mathcal{L}_\gamma, A_\gamma; V_\gamma).$$

The *accumulated structural quantity* along the evolution parameter is then defined by

$$\Delta_{\text{acc}}(\gamma) := \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^{\gamma} \Psi_{\text{run}}(s) ds,$$



whenever the integral is well-defined. If  $\Psi_{\text{run}} \in L^1_{\text{loc}}$  on the declared interval, then  $\Delta_{\text{acc}}$  is absolutely continuous and, in differential form,

$$\frac{d}{d\gamma}\Delta_{\text{acc}}(\gamma) = \Psi_{\text{run}}(\gamma) \quad \text{for almost every } \gamma,$$

with the initial condition  $\Delta_{\text{acc}}(\gamma_0)$  declared.

When resolution-aware comparisons of accumulated values are invoked, the path declaration must also lock a symmetric discrepancy on the transported accumulated carrier. Under the default scalar rule

$$d_{\text{acc}}(a, b) := |a - b|, \quad a \approx_\varepsilon b \iff d_{\text{acc}}(a, b) < \varepsilon,$$

where the displayed use of  $\approx_\varepsilon$  is restricted to this declared accumulated-carrier context.

**Parameter-measure convention.** The accumulated readout is defined relative to the declared  $\gamma$ -axis and measure convention. Under reparameterization,  $\Psi_{\text{run}}$  must be transported as a density; otherwise cross-axis comparisons of  $\Delta_{\text{acc}}$  are **undefined**.

**Separation from the fixed-slice inter-level diagnostic.**  $\Delta_{\text{acc}}(\gamma)$  is an accumulated bookkeeping quantity attached to the declared evolution path  $\mathfrak{P}_\gamma$ . It is distinct from the fixed-slice inter-level gap  $\Delta_{I \rightarrow G}(\mathcal{L}_\gamma; V_\gamma)$  of Def. 5.3, which is defined at each  $\gamma$  by an inf-sup reconstruction residual under the locked interface module of  $V_\gamma$ .

| Symbol                                     | Role in this manuscript                                                                                                                   | Gating / notes                                                                                                                                                                                                  |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ | Fixed-slice inter-level diagnostic (inf-sup reconstruction residual) under a locked interface module.                                     | Requires $(\Phi_I, \Phi_G, D, \mathcal{U}, \Omega_{\text{reach}}, \varepsilon)$ to be declared; $\approx_\varepsilon$ semantics fixed by Normative Rule 2.1.                                                    |
| $\Delta_{\text{acc}}(\gamma)$              | Accumulated bookkeeping quantity along a declared evolution parameter $\gamma$ (integral / discrete accumulation interface).              | Requires a declared evolution path $\mathfrak{P}_\gamma$ and an explicit transport/comparability clause across $\gamma$ . Not identified with $\Delta_{I \rightarrow G}$ unless an explicit bridge is declared. |
| $\Delta_R^*$                               | Meta-gap schema (cross-level non-identity postulate) guarding against ontological equality readings across genuine cross-level relations. | Semantic guardrail, not an operational inf-sup diagnostic; never used as a numerical threshold criterion.                                                                                                       |

Table 2: Disambiguation of  $\Delta$ -roles used in this paper. Numerical reporting is always package-relative; ontological statements use  $\Delta_R^*$  (with  $R$  suppressed when unambiguous) only as a semantic axiom (Sec. 6.3).

*Remark 7.1* (Domain of validity and reporting convention). The interface in Def. 7.1 is operational and path-dependent: it requires a declared evolution parameter, a declared path  $\mathfrak{P}_\gamma$ , and an explicit transport law across  $\gamma$ .

**Discrete variant.** If  $\gamma$  indexes a discrete sequence (e.g.  $\gamma = n \in \mathbb{N}$ ), one may use the declared discrete accumulation rule

$$\Delta_{\text{acc}}(n) = \Delta_{\text{acc}}(n_0) + \sum_{k=n_0}^{n-1} \Psi_{\text{run}}(k),$$

under the same comparability/transport requirements.

**Resolution.** Any statement of the form “ $\Delta_{\text{acc}}(\gamma) \approx_{\varepsilon} 0$ ” (if used) is evaluated under the declared accumulated-carrier discrepancy; under the default rule above it means  $|\Delta_{\text{acc}}(\gamma)| < \varepsilon$  and does not assert an ontological identity. Geometric closures and self-similar solutions are deferred to appendices (or later chapters) to keep the main line slim.

*Remark 7.2* (Role-level analogies deferred). Analogies with familiar dynamical lexicons such as force, motion, energy, action, driver, or accumulated response are not part of the slim core. They may be used only as optional role-level alignments under an explicit package and are deferred to the extended dynamics module when needed. The main text uses only the minimal accumulation interface of Def. 7.1.

## 8 Well-posed Reporting and Licensed Comparisons

This section states the minimal declaration conditions needed for reproducible reporting, and specifies when cross-report comparisons are licensed. In this framework,  $\Psi$  and  $\Delta_{I \rightarrow G}$  (and any other  $\Delta$ -variant when explicitly invoked) are not treated as context-free invariants: their meanings are uniquely locked only after the relevant clauses of a declared package have been stated. When the required clauses are not declared (or when cross-report comparisons invoke undeclared transport/domain/resolution conversions), the corresponding numerical statement is *undefined relative to the declared regime*. For compactness, we optionally attach short diagnostic status tags (Appendix A), but the core semantics is simply partiality (well-posed vs. undefined) and licensed vs. unlicensed comparisons.

### 8.1 Package card template (fillable)

The minimal reporting card (Protocol 4.1) specifies what must be *declared*. For long-term reuse, it is often convenient to record these clauses in a single “package card” block with a short identifier. The following template is a fillable form; it introduces no new primitives.

**Package Card Template (minimal).** Fill the right column; omit rows that are not invoked.

| Field                                                                                                                                                                                                                                     | Declaration (fill) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| ID / scope                                                                                                                                                                                                                                | _____              |
| Target system $\mathcal{L}$ and boundary                                                                                                                                                                                                  | _____              |
| Audited domain / reachability $\Omega_{\text{reach}}$ (and how fixed)                                                                                                                                                                     | _____              |
| Level specification(s) and perspectives $(A_I, A_G) / (\Phi_I, \Phi_G)$                                                                                                                                                                   | _____              |
| Conflict functional for $\Psi$ : $E_A$ (and hidden clauses, if any)                                                                                                                                                                       | _____              |
| Discrepancy rule for $\Delta_{I \rightarrow G}$ : $D$ (and induced interface sub-clauses)                                                                                                                                                 | _____              |
| Admissible liftings $\mathcal{U}$ (and admissibility constraints)                                                                                                                                                                         | _____              |
| Reporting threshold $\varepsilon$ (and meaning of $\approx_\varepsilon$ ; see Normative Rule 2.1)                                                                                                                                         | _____              |
| Resolution refinement, if invoked: raw/certified clauses $\mathcal{R}_\lambda^{\text{raw}}, \mathcal{R}_\lambda^{\text{cert}}$ , completed task package and optional policy refinement, and $D^\star, \hat{D}_\lambda, \rho_\pm(\lambda)$ | _____              |
| Resolution queries / semantics / certificate scope: ordered $Q_0$ and/or $Q_{\text{sep}}$ ; ideal and certified query margins; directional pairwise or partition; pointwise, simultaneous, or sequential                                  | _____              |
| Audit clause (only if invoked): $\mathcal{Q}, \mathcal{C}$                                                                                                                                                                                | _____              |
| Probabilistic/Bayesian clause, if invoked                                                                                                                                                                                                 | _____              |
| Comparability / transport (only if cross-package comparisons are invoked)                                                                                                                                                                 | _____              |
| Invoked optional modules (e.g. audit, dynamics, closure)                                                                                                                                                                                  | _____              |

If the probabilistic/Bayesian clause is invoked, the report must declare the candidate/fiber space  $\mathcal{X}_V$ , the fiber measure or prior, the evidence event, the likelihood/readout kernel, and the normalization domain.

## 8.2 What must be declared to make $\Psi$ and $\Delta_{I \rightarrow G}$ well-defined

**Protocol 8.1** (Minimal declaration requirements). Any report of  $\Psi(\mathcal{L}, A; V)$  or  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  is considered well-defined only if the following elements are explicitly declared (or are deterministically induced by locked clauses that are themselves declared):

1. **Target system and boundary.** State the target system object  $\mathcal{L}$  and the boundary of the discussion: what is treated as inside the system versus environment/background.
2. **Reachability / audited domain.** Declare the reachable set  $\Omega_{\text{reach}} \subseteq \Omega$  (Def. 4.3) or declare the rule that determines it (dynamics, admissibility constraints, initial set, and any conditioning).
3. **Level specification and perspectives.** Declare the fixed level specification(s) used (Def. 4.5) and the perspective(s) used to compute the quantity:
  - for  $\Psi$ : the perspective  $A$  (or admissible family at that level) and the associated representation map  $\Phi_A$ ;
  - for  $\Delta_{I \rightarrow G}$ : the designated cross-level pair  $(A_I, A_G)$  and the induced maps  $(\Phi_I, \Phi_G)$  (Def. 4.7).
4. **Conflict functional for  $\Psi$ .** Declare the conflict functional  $E_A$  (Def. 5.1), including any weighting, normalization, or hidden clauses (e.g. interface aggregation rules) that affect its value.
5. **Interface mechanism for  $\Delta_{I \rightarrow G}$ .** Declare the discrepancy rule  $D$  on the individual-level representation space and the admissible lifting family  $\mathcal{U}$  (Def. 4.8 and Def. 5.3). If  $D$  is induced by a measurement/observation interface family together with an aggregation protocol, these sub-clauses must be declared as part of  $D$ .
6. **Reporting threshold and convention.** Declare the reporting threshold  $\varepsilon$  and the semantics of resolution-aware reporting, including the meaning of “ $\approx_\varepsilon$ ” (Normative Rule 2.1). If a certified resolution refinement is invoked, additionally declare the base interface; the raw and certified typed refinements  $\mathcal{R}_\lambda^{\text{raw}}$  and  $\mathcal{R}_\lambda^{\text{cert}}$ ; the completed task package and, when invoked, its separate policy refinement; the fixed target discrepancy and reported discrepancy; certification domain and envelopes; the ideal query margin and its certified lower bound; the ordered null/separation query clauses; directional pairwise or partition semantics; and the certificate scope required by Sec. 4.4. Declaration of  $\varepsilon$  does not by itself certify membership in the relevant set.
7. **Admissibility constraints and non-degeneracy gates.** Declare any constraints defining admissible families (e.g. exclusion of degenerate encodings, bounded complexity of liftings, fixed architecture class, fixed regularity assumptions, or restricted parameter ranges). If an intrinsic bound is claimed by taking an infimum over a family, the family itself must be declared (cf. Defs. 4.8, 5.2, and 5.3).

**Operability (non-emptiness).** The declared audited domain  $\Omega_{\text{reach}}$  and any declared admissible family invoked by the relevant inf/sup definition (in particular the lifting family  $\mathcal{U}$  when reporting  $\Delta_{I \rightarrow G}$ ) must be nonempty. If  $\Omega_{\text{reach}} = \emptyset$  or an invoked admissible family is empty, numerical reporting under the declared regime is **undefined** (optional tag: ADM) rather than a surrogate value induced by empty-set conventions.

If any of the above is absent, the symbols “ $\Psi$ ” and “ $\Delta_{I \rightarrow G}$ ” denote under-specified *families* of quantities rather than a single well-posed value. In that case, numerical reporting under the declared regime is **undefined**; the appropriate report is an explicit statement of under-specification (optionally tagged as U-spec, Appendix A) rather than an unscoped numerical surrogate.

### 8.3 Resolution robustness and locked adaptation

When a fixed-target resolution refinement is invoked, the certified sets of Sec. 4.4 determine which ordinary completed task packages, and which further policy-refined packages when a policy floor is invoked, are licensed under the declared report semantics. Reporting across more than one configuration requires an additional comparability and certification layer.

**Definition 8.1** (Resolution-control robustness). Fix a report semantics and a corresponding certified configuration set  $\mathcal{A}$ . Suppose the family declares either:

1. a common report space  $\mathcal{Y}$  whose identity maps are explicitly adopted as the transports across the invoked configurations; or
2. separately declared transports into a common report space  $\mathcal{Y}$ .

Let

$$R : \mathcal{A} \rightarrow \mathcal{Y}$$

be the transported diagnostic output. The report is:

- *configuration-robust* if  $R$  is constant on all of  $\mathcal{A}$ ;
- *component-robust* if  $R$  is constant on a declared connected or ordered component of  $\mathcal{A}$ ;
- *configuration-sensitive* otherwise.

A numerical report may replace literal equality by a separately declared equivalence in  $\mathcal{Y}$ , but the reporting threshold  $\varepsilon$  may not silently be reused for a second semantic role. A common codomain alone is not a transport declaration.

**Protocol 8.2** (Certification scope for scans and adaptive paths). A certificate must match the way the completed resolution packages are used.

1. A report at one predeclared configuration  $(\lambda, \varepsilon)$  may use a pointwise certificate.
2. A finite scan over several predeclared configurations requires a simultaneous or multiplicity-corrected certificate over the scanned set.
3. A continuous scan requires a uniform certificate over its declared configuration domain.
4. An adaptive path whose next configuration or stopping time depends on previous observations requires a sequentially valid or anytime-valid certificate under the declared control and stopping rules.

Pointwise confidence guarantees may not be reinterpreted as a joint guarantee for a scan or adaptive path.

**Definition 8.2** (Adaptive resolution meta-protocol). An adaptive resolution procedure is governed by the meta-protocol

$$\bar{V} := (\mathfrak{R}_{\text{res}}^{\text{cert}}, \mathcal{Q}^{\sigma, \tau}, [p], \hat{e}, \text{Cert}, \top, \text{Ctrl}, \text{Accept}, \text{Reject}, \text{Stop}),$$

where

$$\mathfrak{R}_{\text{res}}^{\text{cert}} := \{\mathcal{R}_{\lambda}^{\text{cert}}\}_{\lambda \in \Lambda_{\text{res}}}$$

is the declared certified interface-refinement family,  $\mathcal{Q}^{\sigma,\tau}$  contains the task-indexed query clauses and report semantics,  $[p]$  is the optional separate policy refinement,  $\hat{e}$  is the declared error estimator,  $\text{Cert}$  is valid for the adaptive use,  $\mathsf{T}$  declares the transports between visited package reports (including any explicitly adopted identity transports),  $\text{Ctrl}$  is the update law, and the remaining entries are the acceptance, rejection, and stopping rules. Writing the full configuration as

$$c_n := (\lambda_n, \varepsilon_n),$$

the controller may evolve by

$$c_{n+1} = \text{Ctrl}(c_n, \hat{e}_n),$$

with every visited  $c_n$  required to lie in the applicable package-conditioned certified set and, when policy is invoked, in its policy-restricted subset. For partition semantics, the sequential certificate must also cover every visited threshold-edge domain.

The object  $\bar{V}$  is a higher-level protocol governing a declared path through ordinary packages

$$V_{\lambda_0, \varepsilon_0}^{\sigma, \tau}, V_{\lambda_1, \varepsilon_1}^{\sigma, \tau}, \dots$$

or through their policy-refined counterparts. It does not replace or redefine the top-level package schema of Definition 4.9. Fixed-threshold adaptation is the special case  $\varepsilon_n \equiv \varepsilon_0$ .

**Protocol 8.3** (Lock the package path, not necessarily a constant value). Adaptive variation defines a declared path through completed packages. It is therefore package variation, but it is not *undeclared* viewpoint drift when the certified refinement family, task clauses, optional policy clause, report semantics, margin certificates, sequential mechanism certificate, transports, update law, acceptance/rejection rules, and stopping rule are locked before the substantive diagnostic. What is locked is  $\bar{V}$  and the admissible package path, not necessarily constant numerical values of  $\lambda$  or  $\varepsilon$ . A post-hoc change of any of these clauses in response to the observed conclusion defines a new meta-protocol and must be reported as such.

*Remark 8.1* (Sensitive reports). If  $R$  is configuration-sensitive, a report should provide a certified scan, certified transition boundaries between stable regions, or a declared selection objective. Certified feasibility alone does not select a unique completed package.

## 8.4 Partiality and licensed comparisons (when numerical reporting is undefined)

The framework distinguishes two outcomes: (i) a numerical report (when the quantity is uniquely locked and well-defined under a declared package and domain), and (ii) an explicit *undefinedness* or *non-licensing* statement (when the attempted report or comparison is not meaningful under the declared clauses). The latter is a semantic classification, not a value judgment.

**Definition 8.3** (Undefinedness / non-licensing modes (minimal set)). The following modes cover the minimal reasons used in this paper for treating a numerical report or comparison as undefined under the declared regime:

1. **Under-specification:** one or more required clauses of Protocol 8.1 are missing, so the quantity is not uniquely locked.
2. **Admissibility violation:** the attempted computation invokes an element (perspective, discrepancy, lifting, encoding, or interface sub-clause) that lies outside the declared admissible family; the report is undefined relative to the declared regime.
3. **Domain mismatch:** the audited domain (reachable set, history class, or query set) differs across reports without an explicit reconciliation rule; comparisons or aggregations across domains are undefined.

4. **Non-comparability (missing transport):** two values are produced under packages  $V, V'$  without a declared transport/comparability law in the sense of Def. 4.18; equality/inequality comparisons are not licensed.
5. **Resolution mismatch:** reporting thresholds or audit resolutions differ without a declared conversion rule; cross-report comparisons are not licensed under the declared resolution-aware semantics.

| Status | Meaning under the declared regime                                                                                                                                                                                    | Minimal witness (optional)                                           |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| ok     | Quantity is well-posed and the numerical report is licensed (package clauses locked, admissible instantiation, domain matches; and transport/resolution clauses declared if a cross-report comparison is performed). | Package card (Protocol 4.1) + reproduction notes.                    |
| U-spec | Under-specification: at least one clause needed to lock the quantity or the comparison is missing.                                                                                                                   | List missing declaration items (Protocol 8.1).                       |
| ADM    | Admissibility violation: an invoked component lies outside the declared admissible family.                                                                                                                           | Identify the offending component and the declared admissible family. |
| DOM    | Domain mismatch: reachable/audited domains differ across reports with no declared reconciliation rule.                                                                                                               | Specify both domains and the missing reconciliation clause.          |
| NC     | Non-comparability: cross-package comparison is requested without a declared transport law.                                                                                                                           | Specify $(V, V')$ and the missing transport/comparability clause.    |
| RES    | Resolution mismatch: cross-report comparison is requested with different $\varepsilon$ (or audit resolutions) and no declared conversion rule.                                                                       | Specify the resolutions and the missing conversion rule.             |

Table 3: Status-style reporting (optional). These labels are short names for the undefinedness / non-licensing modes (Def. 8.3) and inherit the domain-gating protocol (Protocol 8.4).

**Protocol 8.4** (Licensing rule for reporting and cross-report comparison). When a numerical report (or a cross-report comparison) is requested, the following checks determine whether the request is licensed under the declared regime:

1. **Locking check:** verify that the package clauses required by Protocol 8.1 have been declared. If not, the request is **undefined** by under-specification (optional tag: **U-spec**, Appendix A).
2. **Admissibility check:** verify that instantiated components lie in the declared admissible family. If not, the request is **undefined** by admissibility violation (optional tag: **ADM**, Appendix A).
3. **Domain check:** verify that the audited domain matches the declared one, or that an explicit reconciliation rule is given. If not, the request is **undefined** by domain mismatch (optional tag: **DOM**, Appendix A).

4. **Operability check:** verify that the declared audited domain is nonempty and that any admissible family invoked by the requested definition is nonempty on that domain (in particular,  $\mathcal{U} \neq \emptyset$  when reporting  $\Delta_{I \rightarrow G}$ ). If not, the request is **undefined** by admissibility/degeneracy (optional tag: ADM, Appendix A).
5. **Resolution-feasibility check (when a certified mechanism family is invoked):** verify that the realized completed package  $V_{\lambda, \varepsilon}$  belongs to the invoked certificate-relative set and that the certificate scope matches the use (fixed, scanned, or adaptive). If not, the object-level threshold report is not licensed; report current-certificate infeasibility, policy-induced infeasibility, chain-separation obstruction, or certificate-scope failure as applicable (Appendix E).
6. **Comparability check (cross-report only):** when comparing across packages, verify that a transport law is declared (Def. 4.18). If not, the comparison is not licensed (optional tag: NC, Appendix A).
7. **Resolution check (cross-report only):** verify that thresholds are identical or related by a declared conversion rule. If not, the comparison is not licensed (optional tag: RES, Appendix A).
8. **Otherwise:** report the numerical value together with the minimal package card (Protocol 4.1) and any additional clauses needed for reproducibility.

*Remark 8.2* (Neutrality of undefinedness diagnoses). An undefinedness/non-licensing statement indicates that a numerical report or comparison is not meaningful under the *declared* package clauses, domain, and resolution semantics. It does not evaluate the system under study; it specifies which declaration or reconciliation clause would be required to make the requested numerical statement well-posed.

## 8.5 Canonical audited example: Markov coarse-graining and fiber non-constancy

This subsection records a short, non-toy instantiation showing how the protocol-level outputs (ok vs. undefined) arise from an explicit witness.

**Setup (finite Markov chain under a locked endpoint).** Let  $X$  be a finite state space with Markov kernel  $K(x, \cdot)$  [9]. Fix a declared coarse map  $\phi : X \rightarrow Y$  and take the endpoint operator for one-step histories  $h = (x_0, x_1)$  to be  $\text{End}(h) := \phi(x_0) \in Y$ . Lock a discrete endpoint discrepancy and declare any  $\varepsilon < 1$ , so endpoint indistinguishability reduces to equality of macro labels.

**Audited predictive queries.** For each event  $A \subseteq Y$ , consider the one-step macro predictive query

$$q_A(h) := \mathbb{P}(\phi(X_1) \in A \mid X_0 = x_0). \quad (1)$$

This is a natural audited family when the intended statement treats macro-level predictions as the reportable content.

**Witness for non-single-valuedness.** If there exist micro states  $x, x' \in X$  with  $\phi(x) = \phi(x')$  such that the induced macro predictive laws differ, i.e.  $\bar{K}_x(\cdot) \neq \bar{K}_{x'}(\cdot)$ , then there exists an event  $A$  with  $q_A(h_1) \neq q_A(h_2)$  for histories  $h_1, h_2$  sharing the same endpoint label. Hence any endpoint-based report that requires these audited predictive queries to be single-valued on endpoint fibers is not licensed under the declared package.



**Protocol consequence.** The witness shows the mechanism: the coarse endpoint identifies microstates inside one macro fiber, while the audited predictive content varies within that fiber. Therefore any claim requiring single-valued endpoint-based prediction is not licensed under the locked package. With a declared discrepancy this becomes a non-negligibility witness; without the required clauses it is reported as **undefined** rather than replaced by an endpoint-matching surrogate.

## 9 Discussion

The philosophical point of the coupling layer  $V$  is not that structural gaps are failures of description, nor that reality contains a primitive viewpoint-independent discontinuity. A successful formalism may fit its target structure with high precision inside a declared regime; quantum mechanics is a paradigmatic example of such a deep alignment between formal description and objective behavior. The boundary appears when a valid descriptive structure is extended across levels, scales, or reporting interfaces while being asked to preserve structural content without loss. In this sense, the gap is neither merely linguistic nor a standalone ontological discontinuity. It is a report, conditioned by  $V$ , of an objective cross-level structural relation: the package does not create the underlying structure, but it determines the form in which cross-level residuals, admissible equivalences, and non-licensing states become reportable.

Under this coupling,  $\Psi$  and  $\Delta_{I \rightarrow G}$  separate two different questions.  $\Psi$  asks whether residual tension remains inside a declared level and perspective.  $\Delta_{I \rightarrow G}$  asks whether lower-level structure can be compressed into an upper-level report and reconstructed within the declared admissible class. Their separation prevents within-level tension, cross-level residuality, approximation, and undefinedness from collapsing into a single informal notion of error.

The remaining protocol discipline follows from this distinction. When the package clauses are sufficient, the framework reports a typed residual, a near-zero convention, or a non-negligibility witness. When a certified fixed-target resolution refinement is invoked, the declaration layer is preceded by a certificate-relative gate: the mechanism error and the typed query clauses must leave a nonempty positive-threshold set. When the required clauses or certificates are absent, the corresponding report is undefined or not licensed. Macro-closure, residual complements, and closure restoration remain downstream mechanisms because they require additional future-readout and enlarged-fiber declarations.

## 10 Conclusion

The framework’s main output is not a universal scalar formula for every obstruction. It is a disciplined report space: under a declared package  $V$  and finite reporting resolution  $\varepsilon$ , cross-level sufficiency, residuality, approximation, comparability, and undefinedness become typed and reproducible outputs. Within that report space,  $V$  functions as the coupling layer between descriptive access and objective structure;  $\Psi$  records intra-level residual tension;  $\Delta_{I \rightarrow G}$  records cross-level residuality under declared compression and reconstruction; and  $\Delta_{\text{acc}}$  provides a definition-level accumulated readout when a declared evolution axis is supplied. The resolution-feasibility layer adds one further distinction: a threshold can be declared without being feasible, and when a certified fixed-target resolution refinement is invoked its licensed values are constrained jointly by the mechanism certificate and the declared query.

The framework therefore treats  $V$ –structure coupling as the semantic basis of cross-level reportability.

## A Diagnostic status tags and optional witnesses

### Module summary.

**Scope.** Provide optional short tags and optional minimal witnesses for the undefinedness / non-licensing modes used in the main text.

**Dependencies.** None beyond the core reporting protocol (Sec. 8) and Def. 8.3.

**Outputs.** A lightweight status vocabulary (ok, U-spec, ADM, DOM, NC, RES) and suggested minimal witness fields.

**Non-invocation rule.** If omitted, the manuscript may report undefinedness in full prose; the core semantics is unchanged.

This appendix provides an *optional* compact labeling scheme for the undefinedness/non-licensing modes used in the main text (Def. 8.3). These tags are not part of the core semantics; they serve as short names for common regimes of under-specification or unlicensed comparison.

**Status-based presentation (optional).** A report may be presented in the lightweight form

$$\text{Report} := (\text{status}, \text{payload}, \text{witness}),$$

where  $\text{status} \in \{\text{ok}, \text{U-spec}, \text{ADM}, \text{DOM}, \text{NC}, \text{RES}\}$ ,  $\text{payload}$  is the numerical value when **ok** (and omitted otherwise), and  $\text{witness}$  is an optional minimal record that makes the relevant missing clause or mismatch explicit.

**Definition A.1** (Diagnostic status tags). The diagnostic tags used in this paper are:

1. **U-spec (under-specification):** one or more clauses required to lock the quantity (Protocol 8.1) are missing.
2. **ADM (admissibility violation):** the attempted computation invokes an element outside the declared admissible family.
3. **DOM (domain mismatch):** the audited domain differs across reports without an explicit reconciliation rule.
4. **NC (non-comparability):** two reports are produced under packages  $V, V'$  without a declared transport/comparability law (Def. 4.18); equality/inequality comparisons are not licensed.
5. **RES (resolution mismatch):** thresholds/resolutions differ without a declared conversion rule; cross-report comparisons are not licensed under the declared resolution-aware semantics.

*Remark A.1* (Scope of the short status vocabulary). The short tags above cover the core undefinedness and cross-package mismatch modes. They are not intended to exhaust the certificate-relative non-licensing states introduced by Appendix E, such as current-certificate infeasibility, policy-induced infeasibility, insufficient certificate scope, or chain-separation obstruction. Those states are reported in full typed prose unless a later application explicitly declares an extended status vocabulary.

*Remark A.2* (Minimal witnesses (optional)). When a diagnostic tag is used, the following witness data are sufficient (but not required) to make the mode explicit:

- **U-spec:** list the missing declaration items relative to Protocol 8.1 (or to the minimal package card, Protocol 4.1).

- ADM: specify the offending component (e.g.  $A$ ,  $D$ ,  $u \in \mathcal{U}$ , interface sub-clause) and the declared admissible family.
- DOM: specify the two audited domains and the missing reconciliation clause.
- NC: specify the pair of packages  $(V, V')$  and the missing transport/comparability declaration.
- RES: specify the compared thresholds/resolutions and the missing conversion rule.

**Worked example (relative-entropy endpoint–path gaps).** An explicit instantiation of a divergence-based diagnostic under the reporting protocol, including a status-style report and an operational witness for a reported gap, is given in [18].

## B Extended probabilistic instantiation of $\Delta_{I \rightarrow G}$

### Module summary.

**Scope.** Clarify how the package-locked residual  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  functions as a certificate-level structural limit and how distribution-valued interface clauses translate it into TV/IPM reporting readouts without replacing the slim-core witness mechanisms.

**Dependencies.** A declared package  $V$  fixing  $\Omega_{\text{reach}}$ , admissible liftings  $\mathcal{U}$ , a discrepancy rule  $D$ , and a reporting resolution  $\varepsilon$ . For the probabilistic branch, additionally a declared observation space  $(\mathcal{Y}, \Sigma)$  and distribution-valued interface clauses; for the Bayesian reading, a declared candidate/fiber space  $\mathcal{X}_V$ , fiber measure or prior, evidence event, likelihood/readout kernel, and normalization domain.

**Outputs.** (i) Certificate lower bounds via fiber witnesses / fiber diameter and via audit single-valuedness; (ii) probabilistic reporting readouts: uniform event-gap control under TV and uniform audit predicate / advantage-error control under an IPM; (iii) an optional Bayesian intuition for conditionalized distribution-valued reports.

**Non-invocation rule.** If the interface is not distribution-valued (or the required clauses are absent), the probabilistic branch is not invoked; the slim core remains unaffected. If the Bayesian clauses are absent, the Bayesian reading is simply not licensed.

### B.1 Minimax residual as a certificate-level structural limit

The slim core treats  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  as a package-locked minimax residual:

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} D(\Phi_I(S), u(\Phi_G(S))),$$

where  $\Omega_{\text{reach}} = \Omega_{\text{reach}}(V)$ ,  $\mathcal{U} = \mathcal{U}(V)$ , and  $D = D_V$  are fixed by the declared package. The primary role of  $\Delta_{I \rightarrow G}$  in this branch is certificate-level: explicit witnesses can force resolution-non-negligibility under the authoritative reporting semantics (Normative Rule 2.1).

A canonical certificate is provided by the fiber-diameter mechanism. Under the pseudometric assumption on  $D$ , Theorem 5.1 gives the witness lower bound

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) \geq \frac{1}{2} \sup_{g \in \Phi_G(\Omega_{\text{reach}})} \text{diam}_D(g),$$

and Corollary 5.1 upgrades this to a resolution-aware non-negligibility certificate: if there exist  $S, S' \in \Omega_{\text{reach}}$  with  $\Phi_G(S) = \Phi_G(S')$  and  $D(\Phi_I(S), \Phi_I(S')) \geq 2\varepsilon$ , then  $\Delta_{I \rightarrow G}(\mathcal{L}; V) \geq \varepsilon$  and hence  $\Delta_{I \rightarrow G}(\mathcal{L}; V) \not\approx_\varepsilon 0$  under the declared reporting convention. In this sense, a fiber witness (Remark 5.3) functions as an audit artifact: it explains, in package-locked terms, why the “ $\approx_\varepsilon 0$ ” convention is unavailable at the stated resolution.

## B.2 Alignment with audit semantics: single-valued decidability and fiber non-constancy

When a claim is explicitly about decidability/extendability under a locked audit package  $V^{\text{aud}} = (\mathcal{I}^{\text{aud}}, \varepsilon, \mathcal{Q}, \mathcal{C})$ , whose interface  $\mathcal{I}^{\text{aud}}$  locks the endpoint operator **End** and the corresponding endpoint discrepancy, the relevant local predicate is single-valuedness on endpoint fibers: for every declared query  $q \in \mathcal{Q}$ , admissible histories that share the same endpoint/readout at resolution  $\varepsilon$  must satisfy  $q(h_1) = q(h_2)$ . If some  $q$  varies within an endpoint fiber at the locked audit resolution, the audited claim is not licensed as a single-valued endpoint report. Any restoration then requires a separately declared extension/complement workflow rather than an implicit repair.

This audit reading is consistent with the slim-core witness logic above: a non-closure witness supplies an explicit obstruction to single-valued audit decidability under the declared viewpoint, and the correct output is either an extension workflow or a non-licensing report. Throughout, “reportable as 0” is used only as a resolution-aware reporting convention at finite  $\varepsilon$ , not as an ontological identity.

## B.3 Probabilistic instantiation as reporting interface (TV/IPM)

**Setup (declared in  $V$ ).** Fix an observation space  $(\mathcal{Y}, \Sigma)$ . Under the declared interface, each audited state  $S \in \Omega_{\text{reach}}$  induces an outcome distribution  $P_S := \Phi_I(S) \in \mathcal{P}(\mathcal{Y})$ . Given the coarse description  $\Phi_G(S)$  and an admissible lifting  $u \in \mathcal{U}$ , the reconstructed outcome distribution is  $Q_S^u := u(\Phi_G(S)) \in \mathcal{P}(\mathcal{Y})$ . The diagnostic retains its minimax form:

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} D(P_S, Q_S^u),$$

with  $D$  declared as a discrepancy on distributions.

*Remark B.1* (interpretation boundary; no sampling semantics). In this branch,  $P_S$  and  $Q_S^u$  are declared distribution-valued readouts. TV/IPM consequences are deterministic reporting bounds for the minimax residual, not sampling-based statistical inference. Estimation, significance, or average-case semantics requires an additional package clause.

**(i) Event-probability gaps via total variation.** If  $V$  declares total variation distance  $D_{\text{TV}}(P, Q) := \sup_{B \in \Sigma} |P(B) - Q(B)|$  [3], then

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} \sup_{B \in \Sigma} |P_S(B) - Q_S^u(B)|.$$

Accordingly, a report  $\Delta_{I \rightarrow G}(\mathcal{L}; V) < \varepsilon$  licenses the existence of an admissible  $u_\varepsilon \in \mathcal{U}$  such that

$$\sup_{S \in \Omega_{\text{reach}}} \sup_{B \in \Sigma} |P_S(B) - Q_S^{u_\varepsilon}(B)| < \varepsilon,$$

i.e. all event-probability gaps are  $\varepsilon$ -bounded at the declared reporting resolution.

**(ii) Audit predicates and advantage/error control via an IPM.** Let  $V$  declare an admissible audit predicate family  $\mathcal{T} \subseteq \{f : \mathcal{Y} \rightarrow [0, 1]\}$ , and define the induced IPM [8, 3]

$$d_{\mathcal{T}}(P, Q) := \sup_{f \in \mathcal{T}} |\mathbb{E}_P[f] - \mathbb{E}_Q[f]|.$$

If  $V$  sets  $D := d_{\mathcal{T}}$ , then

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} d_{\mathcal{T}}(P_S, Q_S^u),$$

and a report  $\Delta_{I \rightarrow G}(\mathcal{L}; V) < \varepsilon$  licenses the existence of  $u_\varepsilon \in \mathcal{U}$  such that

$$\sup_{S \in \Omega_{\text{reach}}} \sup_{f \in \mathcal{T}} |\mathbb{E}_{P_S}[f] - \mathbb{E}_{Q_S^{u_\varepsilon}}[f]| < \varepsilon.$$

To translate  $d_{\mathcal{T}}$  into an audit advantage/error readout, consider the equal-prior one-shot decision task induced by the declared predicate family  $\mathcal{T}$  [2] with  $Y \sim P$  under  $H = 0$  and  $Y \sim Q$  under  $H = 1$ . This is a decision-theoretic interpretation of the IPM under the declared test family, not an additional statistical testing protocol. Any  $f \in \mathcal{T}$  induces a randomized decision rule that outputs 1 with probability  $f(Y)$ . Its equal-prior error probability is

$$P_e(f; P, Q) = \frac{1}{2} (\mathbb{E}_P[f] + \mathbb{E}_Q[1 - f]) = \frac{1}{2} (1 + \mathbb{E}_P[f] - \mathbb{E}_Q[f]).$$

Define the achievable audit advantage within  $\mathcal{T}$  as  $\text{Adv}_{\mathcal{T}}(P, Q) := \frac{1}{2} - \inf_{f \in \mathcal{T}} P_e(f; P, Q)$ . Then

$$\text{Adv}_{\mathcal{T}}(P, Q) \leq \frac{1}{2} d_{\mathcal{T}}(P, Q),$$

with equality if  $\mathcal{T}$  is closed under complement ( $f \in \mathcal{T} \Rightarrow 1 - f \in \mathcal{T}$ ). Under the declared instantiation  $D = d_{\mathcal{T}}$ , a report  $\Delta_{I \rightarrow G}(\mathcal{L}; V) < \varepsilon$  therefore bounds the worst-case audit advantage (after admissible reconstruction) by  $\varepsilon/2$  at the declared resolution, and yields the corresponding equal-prior audit error lower bound.

**Interpretation.** The probabilistic branch is a reporting interface: it converts package-locked residual statements into declared distribution-level readouts (event-gap bounds; predicate-gap and advantage/error bounds) once the relevant clauses are declared. When a fiber witness or audit non-closure witness is available, the certificate lower bounds above already lock the corresponding non-licensing or extension conclusion within the same package.

## B.4 Bayesian intuition under declared evidence

Bayesian conditionalization is not used as a primitive in the framework. It is an optional interpretation available only when the probabilistic branch has already declared the needed clauses: a candidate/fiber space  $\mathcal{X}_V$ , a prior or fiber measure, an evidence event, a likelihood or readout kernel, and a normalization domain. This schematic notation presupposes the existence of the relevant conditional measure; otherwise the probabilistic report is **undefined**.

Let  $\mathcal{X}_V$  be the declared candidate/fiber space, let  $e_V : \mathcal{X}_V \rightarrow \text{End}_V$  be the declared endpoint map, let  $e \in \text{End}_V$  be the reported endpoint, and let  $\mu_V(\cdot \mid e)$  denote a declared conditional measure on the fiber  $e_V^{-1}(e)$ . If  $B$  is a  $V$ -declared evidence event with likelihood/readout kernel  $L_V(B \mid x)$  and the normalizing denominator is positive and finite, the updated fiber measure is schematically

$$d\mu_V(x \mid e, B) = \frac{L_V(B \mid x) d\mu_V(x \mid e)}{\int_{e_V^{-1}(e)} L_V(B \mid x') d\mu_V(x' \mid e)}.$$

For a report object  $\rho_{\mathfrak{S}} : \mathcal{X}_V \rightarrow \text{Rep}$  induced by the original structure  $\mathfrak{S}$ , the updated distribution-valued report is then

$$\mathbf{P}_V^{\mathfrak{S}}(\cdot \mid e, B) := (\rho_{\mathfrak{S}})_{\#} \mu_V(\cdot \mid e, B).$$

This subsection is only a Bayesian intuition for conditionalized distribution-valued reporting. It introduces no new gap object and does not change the meanings of  $\Psi$ ,  $\Delta_{I \rightarrow G}$ , or  $\Delta_{\text{acc}}$ . If the prior/fiber measure, evidence event, likelihood/readout kernel, or normalization condition is not declared, the Bayesian reading is **undefined** under the same non-licensing semantics used throughout the framework.

## C Extended $\Psi$ - $\Delta_{I \rightarrow G}$ Dynamics (short optional module)

### Module summary.

**Scope.** Record a short optional dynamical appendix for variant bookkeeping and simple accumulated-readout templates.

**Dependencies.** A declared evolution axis  $\gamma$ , a transport/comparability clause across  $\gamma$ , and any additional carrier assumptions used by the chosen template.

**Outputs.** Variant-safe statements about  $\Delta_{\text{acc}}(\gamma)$ , together with optional proportional and conservation templates.

**Non-invocation rule.** If the required declarations are not supplied, this module is not invoked.

This appendix is a compact support module. It does not introduce a new dynamical theory; it only records how the minimal accumulated interface may be reused without confusing fixed-slice, accumulated, and meta-gap objects.

### C.1 Notation and $\Delta$ -variants

- **Fixed-slice gap:**  $\Delta_{I \rightarrow G}(\mathcal{L}_\gamma; V_\gamma)$  is the compression–reconstruction residual defined by a declared interface module.
- **Accumulated readout:**  $\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^{\gamma} \Psi_{\text{run}}(s) ds$  is meaningful only under a declared evolution path and transport law.
- **Meta-gap:**  $\Delta_R^*$  is the cross-level non-identity marker of Definition 6.1; it is not a numerical substitute for either of the above.

**Lemma C.1** (Variant separation). *The three symbols  $\Delta_{I \rightarrow G}$ ,  $\Delta_{\text{acc}}$ , and  $\Delta_R^*$  name distinct objects. No equality, bound, or limit relation among them is licensed unless an explicit package-level bridge is declared.*

**Lemma C.2** (Accumulation interface). *If  $\Psi_{\text{run}} \in L_{\text{loc}}^1$  on the declared interval, then*

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^{\gamma} \Psi_{\text{run}}(s) ds$$

*defines an absolutely continuous accumulated readout and*

$$\frac{d}{d\gamma} \Delta_{\text{acc}}(\gamma) = \Psi_{\text{run}}(\gamma)$$

*holds almost everywhere. If  $\Psi_{\text{run}} \geq 0$ , then  $\Delta_{\text{acc}}$  is non-decreasing on the declared interval.*

**Remark C.1** (Discrete index). For a discrete hierarchy or grid index  $k$ , the corresponding interface is

$$\Delta_{\text{acc},k+1} - \Delta_{\text{acc},k} := \Psi_k,$$

with the same package and resolution conventions.

### C.2 Optional proportional and conservation templates

**Lemma C.3** (Optional proportional closure). *Assume  $\Delta_{\text{acc}}$  is absolutely continuous and the optional ansatz*

$$\Psi_{\text{run}}(\gamma) = b \Delta_{\text{acc}}(\gamma) \quad \text{a.e.}$$

*holds on a declared interval. Since  $\Psi_{\text{run}} \geq 0$ , the sign of  $b$  must be compatible with the sign of the accumulated carrier on that interval. On any region where  $\Delta_{\text{acc}} \neq 0$ ,*

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) e^{b(\gamma - \gamma_0)}.$$

*This is a modeling ansatz, not a core law.*

**Lemma C.4** (Regime-relative conservation test). *Fix a declared interval  $[\gamma_0, \gamma_1]$ , a transport law across  $\gamma$ , an initial value, the default accumulated-carrier discrepancy  $d_{\text{acc}}(a, b) = |a - b|$ , and a threshold  $\varepsilon$ . Suppose an audit gives an estimate  $I$  of  $\int_{\gamma_0}^{\gamma_1} \Psi_{\text{run}}(s) ds$  with error bound  $E \geq 0$ :*

$$\left| I - \int_{\gamma_0}^{\gamma_1} \Psi_{\text{run}}(s) ds \right| \leq E.$$

*If  $|I| + E < \varepsilon$ , then the report*

$$\Delta_{\text{acc}}(\gamma_1) \approx_{\varepsilon} \Delta_{\text{acc}}(\gamma_0)$$

*is licensed. Otherwise this certificate alone does not license conservation at threshold  $\varepsilon$ .*

**Remark C.2** (Required fields). Any use of the conservation template reports the evolution path, transport law, accumulation rule, initial value, estimate/error bound, and threshold. Missing fields make the conservation statement **undefined**.

## D Logarithmic Spiral Closure and Self-Similar Geometry

### Module summary.

**Scope.** Provide a short optional visualization for the accumulated interface  $(\gamma, \Delta_{\text{acc}}(\gamma))$  under a declared planar embedding.

**Dependencies.** An accumulated quantity  $\Delta_{\text{acc}}(\gamma)$ , embedding constants, and an optional proportional-closure ansatz.

**Outputs.** A logarithmic-spiral visualization when the proportional closure and positivity gates hold.

**Non-invocation rule.** If the embedding or proportional closure is not declared, no spiral claim is invoked.

This appendix records an *optional* geometric visualization for the accumulated interface introduced in Section 7. It turns the scalar pair  $(\gamma, \Delta_{\text{acc}}(\gamma))$  into a declared planar curve and isolates the proportional-closure case in which the embedding becomes a logarithmic spiral.

### D.1 Objects and scope of this appendix

Fix a declared evolution parameter  $\gamma$  and an accumulated quantity  $\Delta_{\text{acc}}(\gamma)$  as in Section 7, together with an intra-level rate  $\Psi_{\text{run}}(\gamma)$  satisfying the interface

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^{\gamma} \Psi_{\text{run}}(s) ds, \quad \frac{d}{d\gamma} \Delta_{\text{acc}}(\gamma) = \Psi_{\text{run}}(\gamma) \quad \text{a.e. when } \Psi_{\text{run}} \in L_{\text{loc}}^1 \quad (2)$$

on any interval where the integral is well-defined. In this appendix,  $\Delta_{\text{acc}}(\gamma)$  is an *accumulated readout* along  $\gamma$  and is not identified with any fixed-slice inter-level gap  $\Delta_{I \rightarrow G}(\mathcal{L}; V)$  unless such an identification is explicitly declared elsewhere.

### D.2 Declared planar embedding (auxiliary)

**Definition D.1** (Planar embedding of  $(\gamma, \Delta_{\text{acc}})$ ). Declare constants  $a_{\theta} \neq 0$ ,  $a_r > 0$ , and an offset  $\theta_0 \in \mathbb{R}$ . On any interval where  $a_r \Delta_{\text{acc}}(\gamma) > 0$ , define a planar curve in polar coordinates  $(r, \theta)$  by

$$\theta(\gamma) = a_{\theta} \gamma + \theta_0, \quad r(\gamma) = a_r \Delta_{\text{acc}}(\gamma), \quad (3)$$

and interpret the geometric trajectory as  $\gamma \mapsto (r(\gamma), \theta(\gamma))$ . The positivity condition is part of the embedding declaration; if it fails on a region, the polar-spiral report is not licensed there.

**Remark D.1** (Status of the embedding). The map (3) is an auxiliary visualization/closure device. It does not assert any claim about the physical phase-space geometry of  $\mathcal{L}$ . All geometric statements below are statements about the embedded curve defined by the declared map.

### D.3 Logarithmic-spiral closure as a local proportionality ansatz

**Definition D.2** (Proportional closure ansatz). On a declared  $\gamma$ -interval  $I$ , assume  $\Delta_{\text{acc}}$  is absolutely continuous and that the following *optional* closure relation holds almost everywhere:

$$\Psi_{\text{run}}(\gamma) = b \cdot \Delta_{\text{acc}}(\gamma) \quad \text{for a.e. } \gamma \in I, \quad (4)$$

where  $b \in \mathbb{R}$  is a declared constant (a regime parameter). Because  $\Psi_{\text{run}} \geq 0$ , this ansatz carries the same sign-compatibility gate as in Lemma C.3: on any positively embedded interval with  $\Delta_{\text{acc}}(\gamma) > 0$ , it requires  $b \geq 0$ . If a signed accumulated carrier is intended, that carrier must be declared separately.

Combining (2) and (4) yields

$$\frac{d}{d\gamma} \Delta_{\text{acc}}(\gamma) = b \Delta_{\text{acc}}(\gamma) \quad \text{for almost every } \gamma \in I, \quad (5)$$

hence, after choosing an absolutely continuous representative, on any subregion where  $\Delta_{\text{acc}}(\gamma) \neq 0$ ,

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) \exp(b(\gamma - \gamma_0)). \quad (6)$$

Under the embedding (3), this becomes a spiral law.

**Lemma D.1** (Logarithmic spiral under proportional closure). *Assume Definitions D.1 and D.2 on an interval  $I$ , and suppose  $a_r \Delta_{\text{acc}}(\gamma) > 0$  on  $I$ . Then the embedded curve satisfies*

$$\frac{dr}{d\theta} = \kappa r, \quad \kappa := \frac{b}{a_\theta}, \quad (7)$$

hence

$$r(\theta) = r(\theta_{\text{ref}}) \exp(\kappa(\theta - \theta_{\text{ref}})). \quad (8)$$

If  $\kappa \neq 0$ , the trajectory is a non-degenerate logarithmic spiral in the  $(r, \theta)$  plane. If  $\kappa = 0$ , it is the degenerate constant-radius case.

*Remark D.2* (Inward vs. outward winding). If  $\kappa > 0$ , then  $r(\theta)$  is increasing in  $\theta$  on the interval. If  $\kappa < 0$ , then  $r(\theta)$  is decreasing in  $\theta$  on the interval and approaches 0 as  $\theta \rightarrow +\infty$  along that regime (if the interval extends to  $+\infty$  in  $\theta$ ). If  $\kappa = 0$ , the embedded curve has constant radius. These are geometric properties of the embedded curve under the declared closure.

### D.4 Equiangular property and a characterization

A classical geometric characterization of logarithmic spirals is equiangularity: the angle between the tangent direction and the radius vector is constant along the curve.

**Lemma D.2** (Equiangular relation in polar form). *Let a  $C^1$  polar curve be given by  $r = r(\theta)$  with  $r(\theta) > 0$  on an interval and  $dr/d\theta \neq 0$  at the point under consideration. For the unsigned geometric angle  $\varphi(\theta) \in (0, \pi/2)$  between the tangent direction and the radius vector,*

$$|\tan \varphi(\theta)| = \left| \frac{r(\theta)}{dr/d\theta} \right|. \quad (9)$$

Equivalently, the signed formula  $\tan \varphi_{\text{or}} = r/(dr/d\theta)$  may be used when an oriented-angle convention is explicitly declared.

**Corollary D.1** (Log spiral implies equiangularity). *If  $r(\theta) = r_0 e^{\kappa(\theta - \theta_0)}$  with  $\kappa \neq 0$ , then  $dr/d\theta = \kappa r$  and (9) yields  $|\tan \varphi(\theta)| = 1/|\kappa|$ , hence  $\varphi(\theta)$  is constant under the unsigned angle convention.*

**Corollary D.2** (Equiangularity implies log spiral). *If a signed angle convention is declared and  $\varphi_{\text{or}}(\theta) \equiv \varphi_0$  is constant with  $\tan \varphi_0 \neq 0$ , then the signed relation gives  $dr/d\theta = \kappa r$  with  $\kappa = 1/\tan \varphi_0$ , hence  $r(\theta) = r_0 e^{\kappa(\theta - \theta_0)}$ . For an unsigned-angle statement, the same conclusion requires a fixed orientation/sign gate for  $dr/d\theta$  on the declared interval.*



## D.5 Self-similarity constraints and a functional characterization

The spiral in Lemma D.1 can also be related to “rotation + scaling” self-similarity constraints. A single declared scaling step is not, by itself, enough to force a pure logarithmic spiral: it permits a shift-invariant modulation on the declared interval. A pure logarithmic spiral follows only after a no-modulation condition or a stronger continuous local one-parameter scaling law is declared.

**Definition D.3** (Rotation–scaling self-similarity (single-step shape invariance)). A polar curve  $r = r(\theta)$  is said to satisfy a single-step rotation–scaling law if there exist constants  $(\phi, \lambda)$  with  $\phi \neq 0$  and  $\lambda > 0$  such that for all  $\theta$  for which both sides are in the declared interval,

$$r(\theta + \phi) = \lambda r(\theta). \quad (10)$$

**Lemma D.3** (Single-step self-similarity and the no-modulation gate). *Assume  $r(\theta) > 0$  and  $r$  is continuous on a declared interval stable under the shift  $\theta \mapsto \theta + \phi$ , and suppose (10) holds for fixed  $(\phi, \lambda)$ . Put*

$$\kappa := \frac{\log \lambda}{\phi}, \quad p(\theta) := e^{-\kappa\theta} r(\theta).$$

*Then*

$$p(\theta + \phi) = p(\theta), \quad r(\theta) = e^{\kappa\theta} p(\theta).$$

*Thus the single-step law licenses an exponential factor plus a  $\phi$ -shift-invariant modulation on the declared interval. A pure logarithmic spiral follows only if the package declares the no-modulation condition that  $p$  is constant on the interval, or if it strengthens the symmetry to a continuous local one-parameter scaling law*

$$r(\theta + s) = e^{\kappa s} r(\theta)$$

*for all sufficiently small admissible  $s$ . Under either strengthened declaration,  $dr/d\theta = \kappa r$  and the curve is a logarithmic spiral.*

**Remark D.3** (Role of symmetry declarations). The single-step constraint alone does not eliminate shift-invariant modulation; the no-modulation or local one-parameter scaling clause is an additional declaration.

## D.6 Discrete evolution index (hierarchy depth) version

When the evolution parameter is a discrete index (e.g. a hierarchy depth  $k \in \mathbb{N}$ ), the same structure appears in difference form.

**Definition D.4** (Discrete proportional closure (difference form)). Let  $\Delta_{\text{acc},k}$  be a discrete accumulated readout and define  $\Psi_k := \Delta_{\text{acc},k+1} - \Delta_{\text{acc},k}$ . Assume an optional proportionality

$$\Psi_k = b \Delta_{\text{acc},k}, \quad (11)$$

with declared constant  $b$  (regime parameter). On any positive accumulated regime where  $\Delta_{\text{acc},k} > 0$  and  $\Psi_k \geq 0$ , the closure requires  $b \geq 0$ .

Then  $\Delta_{\text{acc},k+1} = (1+b)\Delta_{\text{acc},k}$ , hence  $\Delta_{\text{acc},k} = \Delta_{\text{acc},0}(1+b)^k$  on any regime where the closure is asserted. Under the embedding  $\theta_k = a_\theta k + \theta_0$  and  $r_k = a_r \Delta_{\text{acc},k}$ , this yields a discrete sampling of a non-degenerate logarithmic spiral when  $b \neq 0$ ; when  $b = 0$ , it yields the degenerate constant-radius sampling.

## D.7 Reproducible reporting clauses for spiral closure

Any use of the spiral closure in later case studies is regarded as well-posed only if it reports the following declarations explicitly:

1. the  $\Delta$ -variant used (*accumulated*  $\Delta_{acc}$ , not a fixed-slice  $\Delta_{I \rightarrow G}$  unless such an identification is explicitly declared);
2. the evolution parameter domain (continuous  $\gamma$  or discrete  $k$ ) and the reference point;
3. the closure ansatz (e.g.  $\Psi_k = b \Delta_{acc,k}$ ) and the regime interval on which it is asserted;
4. the embedding constants  $(a_\theta, a_r, \theta_0)$  used to map the trajectory into polar coordinates.

*Remark D.4* (Relation to the core framework). Appendix D introduces no new core definition. It only records one optional closure path: the accumulation interface plus proportional closure plus a declared planar embedding  $\Rightarrow$  logarithmic spiral geometry for the accumulated readout. If the proportional closure is not declared (or not supported by an instantiation), none of the spiral claims are invoked.

## E Resolution Feasibility and Control: Pairwise and Partition Semantics

### Module summary.

**Scope.** Provide the full fixed-target scalar resolution-feasibility module underlying Sec. 4.4, including separation-only and two-sided classification tasks under pairwise and partition output semantics.

**Dependencies.** The unchanged package schema  $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$ ; typed fixed-target refinements of the existing interface and query components; certified one-sided error envelopes; and a positive report threshold.

**Outputs.** Certificate-relative pairwise and partition feasibility sets; chain-separation conditions; certificate/policy localization; robustness, selection, and adaptive-control clauses.

**Non-invocation rule.** If the resolution parameter changes the semantic target object rather than estimating a fixed target discrepancy, this appendix is not invoked without an additional transport construction.

### E.1 Purpose and restricted scope

This appendix addresses a prior licensing question: given typed resolution refinements of an existing declared package and declared threshold-reporting requirements, when does any positive reporting threshold remain certified? The threshold is confined by a certified lower bound on the query margin, the certified error of the mechanism, any invoked null-preservation demand, and any additional policy floor. Certificate-only and policy-restricted sets are kept distinct.

The construction preserves the top-level package tuple and is restricted to a fixed scalar target discrepancy and to mechanisms that estimate or numerically realize that same target. The additional objects below are internal refinements of  $\mathcal{I}$  and  $\mathcal{Q}$ , not new top-level slots. It treats two output semantics:

1. *directional pairwise semantics*, in which each ordered queried pair is judged directly by a thresholded discrepancy and no symmetry is presumed;
2. *partition semantics*, in which a symmetric threshold relation, or a declared certified symmetrization, is closed transitively into equivalence classes.

It also separates two task types:

1. *separation-only licensing*, which requires declared ordered positive pairs to remain reportably non-negligible;
2. *two-sided classification*, which additionally requires declared ordered null pairs to be reported negligible by the same threshold.

Closure, reconstructability, future-readout constancy, and scale families that change the target object itself require different demand functionals and are not reduced to the present module.

## E.2 Target discrepancy, interface family, and certification

**Definition E.1** (Fixed target discrepancy). Let  $X_{\text{adm}}$  be a declared admissible domain. A fixed target discrepancy is a declared map

$$D^* : X_{\text{adm}} \times X_{\text{adm}} \rightarrow \mathbb{R}_{\geq 0}$$

that does not depend on the resolution-control parameter or on the report threshold. It may be a metric, pseudometric, declared symmetrization of a divergence, or another scalar discrepancy whose type is fixed by the enclosing package.

**Definition E.2** (Raw and certified resolution refinements of an existing interface). Fix base clauses  $\mathcal{I}^{\text{base}}$ ,  $\mathcal{Q}^{\text{base}}$ , and  $\mathcal{C}^{\text{base}}$  under Definition 4.9. Let  $\lambda \in \Lambda_{\text{res}}$  parameterize raw realization subclauses

$$\mathfrak{R}_{\text{res}}^{\text{raw}} := \{\mathcal{R}_{\lambda}^{\text{raw}}\}_{\lambda \in \Lambda_{\text{res}}}, \quad \mathcal{R}_{\lambda}^{\text{raw}} := (\mathcal{N}_{\lambda}, D^*, \hat{D}_{\lambda}),$$

where  $\mathcal{N}_{\lambda}$  is the declared measurement, resolution, or numerical-realization mechanism and

$$\hat{D}_{\lambda} : X_{\text{adm}} \times X_{\text{adm}} \rightarrow \mathbb{R}_{\geq 0}$$

reports the same fixed target discrepancy. The base structural interface is retained, and  $\lambda$  is not the report threshold  $\varepsilon$ .

Let  $\mathcal{D} \subseteq X_{\text{adm}} \times X_{\text{adm}}$  be the declared certification domain. For each  $\lambda$ , let  $\rho_+(\lambda), \rho_-(\lambda) \geq 0$  satisfy, deterministically or on a declared confidence event,

$$\hat{D}_{\lambda}(x, y) - D^*(x, y) < \rho_+(\lambda), \tag{12}$$

$$D^*(x, y) - \hat{D}_{\lambda}(x, y) < \rho_-(\lambda) \tag{13}$$

for every  $(x, y) \in \mathcal{D}$ . If a non-strict bound or exact equality is the available result, a declared positive slack may be added to obtain the strict envelope used by the threshold rule. A zero envelope is valid only when the corresponding signed report error is uniformly strictly negative; symmetric zero envelopes do not represent an exact mechanism under this strict convention.

With  $\text{Cert}$  denoting the declared certificate object, define

$$\mathcal{R}_{\lambda}^{\text{cert}} := (\mathcal{R}_{\lambda}^{\text{raw}}, \mathcal{D}, \rho_+(\lambda), \rho_-(\lambda), \text{Cert}), \quad \mathcal{I}_{\lambda} := \mathcal{I}^{\text{base}} \oplus \mathcal{R}_{\lambda}^{\text{cert}}.$$

The symbol  $\oplus$  denotes typed internal extension of  $\mathcal{I}$ , not a new top-level package operation.

| Envelope | Error controlled                                           | Operational roles                                                                                                                                                                                          |
|----------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\rho_+$ | over-report allowance: $\hat{D}_{\lambda} - D^* < \rho_+$  | null-preservation floor for a declared $Q_0$ ; converts an observed report into the certified lower bound $D^* > \hat{D}_{\lambda} - \rho_+$ ; may also enter the construction of a certified query margin |
| $\rho_-$ | under-report allowance: $D^* - \hat{D}_{\lambda} < \rho_-$ | certified separation-ceiling erosion; converts a certified lower bound on $D^*$ into the display guarantee $\hat{D}_{\lambda} > D^* - \rho_-$                                                              |

The subscripts encode the sign of the report error. They do not name the direction of the resulting bound on  $D^*$ . This role map is used throughout the appendix.

**Remark E.1** (Exogenous certification). The physical, numerical, or statistical arguments that produce the mechanism envelopes and the certified query-margin bounds are inputs to this module. Instrument specifications, conditioning analysis, rounding-error bounds, concentration inequalities, and independent calibration may contribute to those certificates; none is automatically identical to the final certified quantity. The framework operates downstream of the declared certificates.

**Definition E.3** (Pointwise, simultaneous, and sequential certificates). How the declared certificate is used is a protocol-level scope clause. A certificate for one predeclared completed configuration  $(\lambda, \varepsilon)$  is *pointwise in the configuration*. For a declared scan set  $\mathcal{G}_0 \subseteq \Lambda_{\text{res}} \times \mathbb{R}_{>0}$ , a simultaneous certificate satisfies

$$\mathbb{P}\left(\bigcap_{(\lambda, \varepsilon) \in \mathcal{G}_0} \mathcal{C}(\lambda)\right) \geq 1 - \alpha,$$

where  $\mathcal{C}(\lambda)$  is the event on which (12)–(13) and every certified query-margin lower bound used at that configuration are valid. A finite scan may use a declared familywise correction; an infinite scan requires a uniform certificate. For an adaptively chosen sequence of completed configurations  $c_1, c_2, \dots$ , a sequentially valid certificate satisfies

$$\mathbb{P}(\mathcal{C}_n \text{ holds at every visited step } n) \geq 1 - \alpha$$

under the declared data-dependent stopping rule. Pointwise guarantees cannot be reused as a joint guarantee for a scan or adaptive path; time-uniform confidence sequences provide one standard sequential construction [6].

**Protocol E.1** (Certification scope must match use). A single predeclared configuration may use a pointwise certificate. A predetermined scan requires simultaneous certification. An adaptive controller requires sequentially valid certification under the declared controller and stopping rule.

### E.3 Threshold semantics and typed query clauses

**Definition E.4** (Direct threshold relation). For  $\varepsilon > 0$ , define the ordered relation

$$x E_{\lambda, \varepsilon} y \quad :\Longleftrightarrow \quad \widehat{D}_\lambda(x, y) < \varepsilon.$$

Under directional pairwise semantics, an ordered pair in  $E_{\lambda, \varepsilon}$  is reported negligible in that direction. Symmetry is not presumed, and the relation need not be transitive. Accordingly,  $E_{\lambda, \varepsilon}$  is a directional threshold-report predicate, not the framework’s symmetric resolution-indistinguishability predicate  $\approx_\varepsilon$ . It may instantiate  $\approx_\varepsilon$  only after a symmetric rule or a declared certified symmetrization is supplied. The word *merge* is reserved below for a symmetric relation that is closed into a partition.

**Definition E.5** (True and certified pairwise query margins). Let

$$Q_{\text{sep}} \subseteq \mathcal{D}, \quad Q_{\text{sep}} \neq \emptyset,$$

be declared before the main diagnostic, with every ordered pair required to remain reportably non-negligible in its declared direction and with  $x \neq y$  for every  $(x, y) \in Q_{\text{sep}}$ . Define the ideal margin

$$\delta_Q^{\text{pair}} := \inf_{(x, y) \in Q_{\text{sep}}} D^*(x, y).$$

Operational certification requires a declared lower-margin family

$$\delta_Q^{\text{pair}} : \Lambda_{\text{res}} \rightarrow \mathbb{R}_{\geq 0}, \quad 0 \leq \delta_Q^{\text{pair}}(\lambda) \leq \delta_Q^{\text{pair}},$$

together with the pointwise, simultaneous, or sequential scope under which it remains valid. A family-uniform lower bound is the special case in which  $\delta_Q^{\text{pair}}(\lambda)$  is constant on the invoked mechanism family. If the ideal margin is exactly available, one may take  $\delta_Q^{\text{pair}}(\lambda) = \delta_Q^{\text{pair}}$  for every invoked  $\lambda$ .

When a full two-sided classifier is invoked, additionally declare a nonempty ordered-pair set

$$Q_0 \subseteq \mathcal{D}$$

whose pairs must be reported negligible, with

$$D^*(x, y) = 0 \quad \text{for every } (x, y) \in Q_0.$$

The  $Q_0$  clause is not implied by  $Q_{\text{sep}}$  and must not be introduced silently.

**Definition E.6** (Policy floor). A reporting policy may impose  $p \geq 0$ , forbidding thresholds below  $p$  even when the certificate-only task clauses would permit them. Policy does not certify capability and is therefore introduced as a further query refinement rather than built into the pre-policy task package.

**Definition E.7** (Task-indexed query refinements and completed packages). Let

$$\sigma \in \{\text{pair}, \text{part}\}, \quad \tau \in \{\text{sep}, \text{class}\}$$

index report semantics and task mode. Define

$$\mathcal{Q}^{\sigma, \text{sep}} := \mathcal{Q}^{\text{base}} \oplus (Q_{\text{sep}}, \text{Sem} = \sigma),$$

and, when  $Q_0$  is declared,

$$\mathcal{Q}^{\sigma, \text{class}} := \mathcal{Q}^{\text{base}} \oplus (Q_{\text{sep}}, Q_0, \text{Sem} = \sigma).$$

If policy is invoked, define

$$\mathcal{Q}^{\sigma, \tau, p} := \mathcal{Q}^{\sigma, \tau} \oplus p.$$

For each  $(\lambda, \varepsilon) \in \Lambda_{\text{res}} \times \mathbb{R}_{>0}$ , define

$$V_{\lambda, \varepsilon}^{\sigma, \tau} := (\mathcal{I}_{\lambda}, \varepsilon, \mathcal{Q}^{\sigma, \tau}, \mathcal{C}^{\text{base}}),$$

and, when the policy clause is invoked,

$$V_{\lambda, \varepsilon}^{\sigma, \tau, p} := (\mathcal{I}_{\lambda}, \varepsilon, \mathcal{Q}^{\sigma, \tau, p}, \mathcal{C}^{\text{base}}).$$

All four combinations are instances of the original package schema and remain distinct because adding  $Q_0$  or changing pairwise to partition semantics may change package admissibility.

**Definition E.8** (Task-indexed package-admissible threshold sets). For each fixed  $\lambda$  and task index  $(\sigma, \tau)$ , define

$$\mathcal{E}_{\text{adm}}^{\sigma, \tau}(\lambda) := \{\varepsilon > 0 : V_{\lambda, \varepsilon}^{\sigma, \tau} \text{ is admissible}\},$$

and, when policy is invoked,

$$\mathcal{E}_{\text{adm}}^{\sigma, \tau, p}(\lambda) := \{\varepsilon > 0 : V_{\lambda, \varepsilon}^{\sigma, \tau, p} \text{ is admissible}\}.$$

The certificate inequalities generate algebraic threshold slices. The licensed configuration slices are their intersections with the corresponding task-indexed package-admissible threshold sets. Endpoint inequalities alone are sufficient only when the required algebraic slice is known to lie inside the relevant  $\mathcal{E}_{\text{adm}}^{\sigma, \tau}$  set.

**Definition E.9** (Collision and vanishing uniform margin). A *representation collision* occurs when some  $(x, y) \in Q_{\text{sep}}$  satisfies  $D^*(x, y) = 0$ . A *vanishing ideal margin* occurs when every required pair has positive discrepancy but  $\delta_Q^{\text{pair}} = 0$ . A certified lower bound may also vanish even when the ideal margin is positive; that is certificate-relative conservatism, not a representation collision.

On the certification event, every required-distinct ordered pair satisfies

$$\hat{D}_\lambda(x, y) > D^*(x, y) - \rho_-(\lambda) \geq \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda).$$

For each declared null ordered pair, the over-report envelope gives

$$\hat{D}_\lambda(x, y) < \rho_+(\lambda).$$

Thus  $\rho_-$  supplies the upper threshold constraint for a declared certified margin. The  $\rho_+$  envelope enters the threshold interval as a lower endpoint only when null preservation is required; separately, it may be used upstream to construct a certified lower bound from observed discrepancy reports.

#### E.4 Pairwise certified feasibility

**Definition E.10** (Directional pairwise certified sets). For fixed  $\lambda$ , define the algebraic slices

$$I_\lambda^{\text{pair,sep,cert}} := \left(0, \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda)\right],$$

and, when  $Q_0$  is declared,

$$I_\lambda^{\text{pair,class,cert}} := (0, \infty) \cap \left[\rho_+(\lambda), \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda)\right].$$

The package-conditioned sets are

$$\mathcal{A}_{Q_{\text{sep}}}^{\text{pair,sep,cert}} := \{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{pair,sep}}(\lambda) \cap I_\lambda^{\text{pair,sep,cert}}\},$$

and

$$\mathcal{A}_{Q_0, Q_{\text{sep}}}^{\text{pair,class,cert}} := \{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{pair,class}}(\lambda) \cap I_\lambda^{\text{pair,class,cert}}\}.$$

Policy intersects the algebraic slice with  $[p, \infty)$  and uses  $\mathcal{E}_{\text{adm}}^{\text{pair,sep},p}(\lambda)$  or  $\mathcal{E}_{\text{adm}}^{\text{pair,class},p}(\lambda)$ , respectively.

For fixed  $\lambda$ , write

$$\underline{b}_\lambda^{\text{pair}} := \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda).$$

**Proposition E.1** (Directional pairwise fixed-mechanism slices). *For fixed  $\lambda$ , the exact licensed slices are*

$$\mathcal{E}_{\text{adm}}^{\text{pair,sep}}(\lambda) \cap (0, \underline{b}_\lambda^{\text{pair}}]$$

*for separation-only reporting and*

$$\mathcal{E}_{\text{adm}}^{\text{pair,class}}(\lambda) \cap (0, \infty) \cap [\rho_+(\lambda), \underline{b}_\lambda^{\text{pair}}]$$

*for full classification. Policy uses  $\mathcal{E}_{\text{adm}}^{\text{pair,sep},p}(\lambda)$  or  $\mathcal{E}_{\text{adm}}^{\text{pair,class},p}(\lambda)$ , respectively, and additionally intersects with  $[p, \infty)$ . Each licensed slice is nonempty exactly when the corresponding displayed intersection is nonempty.*

**Corollary E.1** (Pairwise endpoint criteria under admissibility-neutrality). *Assume the relevant algebraic slice is contained in the corresponding package-admissible threshold set. Then the separation-only slice is nonempty exactly when  $\underline{b}_\lambda^{\text{pair}} > 0$ , and the full-classification slice is nonempty exactly when*

$$\rho_+(\lambda) \leq \underline{b}_\lambda^{\text{pair}} \quad \text{and} \quad \underline{b}_\lambda^{\text{pair}} > 0.$$

*Policy replaces the lower endpoint by  $p$  or  $\max\{p, \rho_+(\lambda)\}$ , respectively. If  $\rho_+ = \rho_- = \rho$ , the full-classification algebraic slice is nonempty exactly when*

$$2\rho(\lambda) \leq \underline{\delta}_Q^{\text{pair}}(\lambda) \quad \text{and} \quad \underline{\delta}_Q^{\text{pair}}(\lambda) > 0.$$

*These endpoint criteria certify package-conditioned feasibility only under the stated admissibility containment.*

*Remark E.2* (Ideal and certified margins). Replacing  $\underline{\delta}_Q^{\text{pair}}(\lambda)$  by the true margin  $\delta_Q^{\text{pair}}$  gives the corresponding ideal algebraic geometry. The operational certificate slices are conservative. Failure of a package-conditioned certified set may reflect a weak margin certificate, package inadmissibility on the algebraic slice, or both, even when the ideal margin is positive.

*Remark E.3* (Classical adjacency). The geometry  $\rho_+ + \rho_- \leq \delta$  is the standard requirement that two-sided uncertainty allowances fit inside a required separation margin. Closely related forms appear in margin, packing, and two-point minimax arguments [11]. No novelty is claimed for the inequality in isolation. Its role here is to locate that geometry inside a declared package with explicitly typed null and separation queries and a declared certified margin.

*Remark E.4* (Multiple queries and family geometry). If one package uses one threshold for several audited query bundles, its completed configuration must lie in the intersection of their certified sets. Replacing those bundles by unions of their  $Q_0$  and  $Q_{\text{sep}}$  clauses is equivalent only when they share the same target discrepancy, raw and certified refinement family, certification domain, envelopes, margin-certification construction, certificate scope, report semantics, and policy clauses. If policy floors differ, use  $p_{\text{joint}} := \sup_i p_i$ . When semantics or certificate scopes differ, retain the intersection formulation.

For fixed  $\lambda$ , a threshold slice may be empty, a singleton, or a nondegenerate interval. The family-level set need not be connected: discrete mechanism families may yield isolated configurations or several components. The word *margin* is therefore reserved for an ordered slice or component, not for every multi-point family-level set.

## E.5 Partition semantics and chain separation

Assume for this subsection that the direct threshold relation is symmetric, either because  $D^*$  and  $\hat{D}_\lambda$  are symmetric or because a symmetrization was explicitly declared and certified.

**Definition E.11** (Threshold graph and generated partition). Let  $G_{\lambda,\varepsilon}$  be the undirected graph on  $X_{\text{adm}}$  with edge  $\{x, y\}$  when  $\hat{D}_\lambda(x, y) < \varepsilon$ . The threshold partition  $\Pi_{\lambda,\varepsilon}$  is the partition into connected components of  $G_{\lambda,\varepsilon}$ , equivalently the equivalence relation generated by  $E_{\lambda,\varepsilon}$ .

**Definition E.12** (Chain-edge certification condition). Let

$$\mathcal{D}_{\text{edge}}(\lambda, \varepsilon) := \{(x, y) \in X_{\text{adm}}^2 : \hat{D}_\lambda(x, y) < \varepsilon\}.$$

Write  $\text{EdgeCert}(\lambda, \varepsilon)$  when

$$\mathcal{D}_{\text{edge}}(\lambda, \varepsilon) \subseteq \mathcal{D},$$

and the certified under-report envelope is valid jointly on that edge domain. The stronger condition  $\mathcal{D} = X_{\text{adm}}^2$  is sufficient. For a scan, **EdgeCert** must hold simultaneously on every visited edge domain; for an adaptive path it must hold under the declared sequential certificate.

**Definition E.13** (Bottleneck chain distance). Set

$$B^*(x, x) := 0.$$

For  $x \neq y$ , define

$$B^*(x, y) := \inf_{\substack{n \geq 1 \\ x=x_0, \dots, x_n=y}} \max_{1 \leq i \leq n} D^*(x_{i-1}, x_i),$$

where all chain vertices lie in  $X_{\text{adm}}$ . Equivalently, one may allow the length-zero chain and assign it bottleneck 0. This is the smallest target-level edge threshold at which  $x$  and  $y$  can be connected through a chain. The construction is the standard bottleneck geometry underlying single-linkage threshold connectivity [4].

**Definition E.14** (True and certified partition query margins). Define the ideal bottleneck margin

$$\delta_Q^{\text{chain}} := \inf_{(x,y) \in Q_{\text{sep}}} B^*(x, y).$$

Because the one-edge chain is admissible,

$$\delta_Q^{\text{chain}} \leq \delta_Q^{\text{pair}}.$$

Operational use requires a declared certified lower-margin family

$$\underline{\delta}_Q^{\text{chain}} : \Lambda_{\text{res}} \rightarrow \mathbb{R}_{\geq 0}, \quad 0 \leq \underline{\delta}_Q^{\text{chain}}(\lambda) \leq \delta_Q^{\text{chain}},$$

with a certificate scope matching the scan or adaptive use. When pairwise and partition certificate slices are compared, choose compatible lower bounds satisfying

$$\underline{\delta}_Q^{\text{chain}}(\lambda) \leq \underline{\delta}_Q^{\text{pair}}(\lambda)$$

for every invoked  $\lambda$ . Exact margins may be used when available; family-uniform bounds are the constant special case.

**Proposition E.2** (Certified chain-separation sufficiency). *Assume  $\text{EdgeCert}(\lambda, \varepsilon)$ . If*

$$\varepsilon \leq \underline{\delta}_Q^{\text{chain}}(\lambda) - \rho_-(\lambda),$$

*then no pair in  $Q_{\text{sep}}$  lies in the same block of  $\Pi_{\lambda, \varepsilon}$ .*

*Proof.* Suppose a required pair  $(x, y)$  is connected by a reported threshold chain  $x = x_0, \dots, x_n = y$ . For every edge,

$$\widehat{D}_\lambda(x_{i-1}, x_i) < \varepsilon.$$

By the certified under-report envelope,

$$D^*(x_{i-1}, x_i) < \widehat{D}_\lambda(x_{i-1}, x_i) + \rho_-(\lambda) < \varepsilon + \rho_-(\lambda) \leq \underline{\delta}_Q^{\text{chain}}(\lambda) \leq \delta_Q^{\text{chain}}.$$

Hence the chain has target bottleneck strictly below  $\delta_Q^{\text{chain}}$ , contradicting the definition of the query margin.  $\square$

**Definition E.15** (Partition certificate-only and policy-restricted sets). For fixed  $\lambda$ , define

$$I_\lambda^{\text{part, sep, cert}} := \left(0, \underline{\delta}_Q^{\text{chain}}(\lambda) - \rho_-(\lambda)\right],$$

and, when  $Q_0$  is declared,

$$I_\lambda^{\text{part, class, cert}} := (0, \infty) \cap \left[\rho_+(\lambda), \underline{\delta}_Q^{\text{chain}}(\lambda) - \rho_-(\lambda)\right].$$



The package-conditioned partition sets are

$$\mathcal{A}_{Q_{\text{sep}}}^{\text{part,sep,cert}} := \left\{ (\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{part,sep}}(\lambda) \cap I_{\lambda}^{\text{part,sep,cert}}, \quad \text{EdgeCert}(\lambda, \varepsilon) \right\},$$

and, when  $Q_0$  is declared,

$$\mathcal{A}_{Q_0, Q_{\text{sep}}}^{\text{part,class,cert}} := \left\{ (\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{part,class}}(\lambda) \cap I_{\lambda}^{\text{part,class,cert}}, \quad \text{EdgeCert}(\lambda, \varepsilon) \right\}.$$

Policy uses  $\mathcal{E}_{\text{adm}}^{\text{part,sep},p}(\lambda)$  or  $\mathcal{E}_{\text{adm}}^{\text{part,class},p}(\lambda)$ , respectively, intersects the algebraic slice with  $[p, \infty)$ , and retains the same edge-certification condition.

With compatible certified margins, the algebraic inclusions

$$I_{\lambda}^{\text{part,sep,cert}} \subseteq I_{\lambda}^{\text{pair,sep,cert}}, \quad I_{\lambda}^{\text{part,class,cert}} \subseteq I_{\lambda}^{\text{pair,class,cert}}$$

hold. The corresponding package-conditioned set inclusions require, in addition, that partition-admissible configurations are pairwise-admissible under the declared task transports and that the partition edge certificate is available. No inclusion of full package-conditioned sets is inferred from the margin inequality alone.

**Corollary E.2** (Symmetric partition endpoint criteria). *Assume  $\rho_+ = \rho_- = \rho$ , the relevant algebraic slice lies inside the corresponding package-admissible threshold set, and  $\text{EdgeCert}(\lambda, \varepsilon)$  holds throughout that slice. Then the full partition-classification slice is nonempty exactly when*

$$2\rho(\lambda) \leq \underline{\delta}_Q^{\text{chain}}(\lambda) \quad \text{and} \quad \underline{\delta}_Q^{\text{chain}}(\lambda) > 0.$$

*The partition separation-only slice is algebraically nonempty exactly when*

$$\underline{\delta}_Q^{\text{chain}}(\lambda) - \rho(\lambda) > 0.$$

*Policy uses the corresponding lower endpoint  $p$  or  $\max\{p, \rho(\lambda)\}$ . These endpoint conditions are sufficient for package-conditioned feasibility only under the stated admissibility and edge-certification assumptions.*

**Proposition E.3** (Pairwise feasibility need not imply partition feasibility). *It is possible that*

$$\delta_Q^{\text{pair}} > 0 \quad \text{and} \quad \delta_Q^{\text{chain}} = 0.$$

*Consequently positive pairwise-certified thresholds may exist while every valid certified lower bound on the partition margin is zero, so no positive partition-certified threshold exists.*

*Proof.* Take  $X_{\text{adm}} = [0, 1]$  with  $D^*(x, y) = |x - y|$  and  $Q_{\text{sep}} = \{(0, 1)\}$ . Then  $\delta_Q^{\text{pair}} = 1$ . For every  $n$ , the chain  $0, 1/n, 2/n, \dots, 1$  has bottleneck  $1/n$ , so  $B^*(0, 1) = 0$  and  $\delta_Q^{\text{chain}} = 0$ .  $\square$

**Remark E.5** (Practical scope of partition semantics). Positive partition-level feasibility requires a positive ideal bottleneck margin  $\delta_Q^{\text{chain}} > 0$  and, operationally at an invoked mechanism, a positive certified lower margin  $\underline{\delta}_Q^{\text{chain}}(\lambda) > 0$ . This holds, for example, in finite metric domains with distinct required pairs, in uniformly gapped domains, and in other settings where the required pairs are separated by a positive bottleneck gap. If  $D^*$  is a metric and the relevant component of  $X_{\text{adm}}$  is path-connected in the topology induced by  $D^*$ , every two points in that component admit chains with arbitrarily small maximal edge; hence  $B^*(x, y) = 0$  for pairs in that component and  $\delta_Q^{\text{chain}} = 0$  whenever the query lies within it.

At the opposite boundary, if  $D^*$  is an ultrametric, then its strong triangle inequality propagates along every finite chain and gives

$$D^*(x, y) \leq \max_i D^*(x_{i-1}, x_i).$$

The one-edge chain supplies the reverse inequality after taking the infimum, so

$$B^*(x, y) = D^*(x, y), \quad \delta_Q^{\text{chain}} = \delta_Q^{\text{pair}}.$$

Provided the required full edge-domain certificate is available, pairwise and partition feasibility then coincide geometrically. The margin incurs no additional loss, although the certification burden may remain stronger under partition semantics because the under-report envelope must hold on every pair that may serve as a chain edge. Ultrametric hierarchies and finite or uniformly gapped tree structures are therefore natural hosts of positive partition-level margins; unrestricted path-connected tree metrics are not.

*Remark E.6* (Contact with finite-resolution repair non-uniqueness). The directional pairwise feasibility result and the Universal Closure-Restoring Complement theorem are not identified. Their precise contact is the non-transitivity of finite-resolution indistinguishability. In the present module, transitive closure can merge required-distinct pairs through threshold chains. In the universal-complement setting, the related obstruction appears as failure of finite-resolution adequacy to remain closed under joins, permitting incomparable maximal adequate repairs with no coarsest element [19]. A theorem equating the two constructions is not claimed.

## E.6 Numerical step size as a fixed-target instance

A standard numerical differentiation or discretization method controlled by  $\lambda = h > 0$  may admit a symmetric error envelope of the schematic form

$$\rho(h) = C_{\text{tr}} h^r + C_{\text{ro}} \frac{u}{h^s}, \quad r, s > 0,$$

where the first term is truncation or discretization error and the second is finite-precision amplification [5]. Both terms belong to the mechanism certificate; neither is the query demand.

For a pairwise separation-only query, the positive thresholds satisfy

$$0 < \varepsilon \leq \underline{\delta}_Q^{\text{pair}}(h) - \rho(h),$$

so the algebraic slice is nonempty exactly when  $\underline{\delta}_Q^{\text{pair}}(h) - \rho(h) > 0$ ; the package-conditioned slice additionally requires intersection with  $\mathcal{E}_{\text{adm}}^{\text{pair, sep}}(h)$ . For a full two-sided classifier with a declared null set, the thresholds satisfy

$$\rho(h) \leq \varepsilon \leq \underline{\delta}_Q^{\text{pair}}(h) - \rho(h),$$

and the algebraic slice is nonempty exactly when

$$2\rho(h) \leq \underline{\delta}_Q^{\text{pair}}(h) \quad \text{and} \quad \underline{\delta}_Q^{\text{pair}}(h) > 0.$$

The package-conditioned full-classification slice additionally requires intersection with  $\mathcal{E}_{\text{adm}}^{\text{pair, class}}(h)$ . For partition output, replace  $\underline{\delta}_Q^{\text{pair}}(h)$  by  $\underline{\delta}_Q^{\text{chain}}(h)$  and require the chain-edge certificate.

The U-shaped error curve is the geometry of the mechanism envelope  $\rho(h)$ : decreasing  $h$  reduces truncation error but eventually increases rounding amplification. When the minimum is attained, an error-minimizing step

$$h_{\text{err}} \in \arg \min_{h>0} \rho(h)$$

does not by itself determine the report threshold. Mechanism-error minimization, threshold selection within a fixed- $h$  slice, and joint resource–error optimization are distinct problems.

## E.7 Certificate-relative infeasibility and localization

**Definition E.16** (Algebraic, package-conditioned, and policy-induced non-licensing). Fix one of the four invoked semantics/task combinations. For a given mechanism or family:

- an *algebraic certificate obstruction* occurs when the relevant certificate inequality slice is empty;
- *package-conditioned non-licensing* occurs when the algebraic slice is nonempty but its intersection with the relevant package-admissible threshold set (and, for partition semantics, the edge-certification condition) is empty;
- *policy-induced non-licensing* occurs when a pre-policy licensed set is nonempty but the corresponding policy-refined set is empty.

The union of the first two cases is the certificate-relative failure of the completed package family to license a threshold, but their causes must be reported separately.

Certificate-relative failure certifies only that the declared interface family, current mechanism and margin certificates, fixed target discrepancy, report semantics, package admissibility clauses, and task clauses are jointly unable to license a positive threshold. It does not prove that no sharper certificate or improved mechanism could do so. Nor does it identify a missing coordinate, prove closure failure, establish complement demand, or select a repair. Possible responses include extending the mechanism family, improving conditioning or calibration, defining a new discrepancy or representation package, changing report semantics, modifying the query, or reporting the query as unservable under the current certified family.

**Proposition E.4** (Positive homogeneity of the algebraic feasibility inequalities). *Let  $c > 0$  and apply the common unit rescaling*

$$D^*, \hat{D}_\lambda, \delta_Q^{\text{pair}}(\lambda), \delta_Q^{\text{chain}}(\lambda), \rho_\pm, \varepsilon, p \mapsto cD^*, c\hat{D}_\lambda, c\delta_Q^{\text{pair}}(\lambda), c\delta_Q^{\text{chain}}(\lambda), c\rho_\pm, c\varepsilon, cp.$$

*Then the algebraic pairwise and partition feasibility inequalities are invariant under the coordinate map  $(\lambda, \varepsilon) \mapsto (\lambda, c\varepsilon)$ .*

*Proof.* Every algebraic endpoint and defining inequality is multiplied by the same positive constant.  $\square$

**Remark E.7** (When full certified-set membership is scale-covariant). Full package-conditioned membership is preserved only when the package-admissibility clauses, certificate object, query-margin certificate, policy clauses, report spaces, and required transports are themselves carried by the declared unit transport. Positive homogeneity of the inequalities alone does not license cross-scale package membership. Thus a mere numerical change of units cannot repair an algebraic obstruction, while a claim about the full certified set additionally requires transport of the surrounding package clauses.

**Remark E.8** (No automatic residual-complement bridge). Neither an algebraic certificate obstruction nor package-conditioned non-licensing implies complement demand. Residual-complement claims require additional fiber, evolution, future-readout, and activation conditions. The present output may be used as evidence only after those independent clauses are verified.

## E.8 Robustness, selection, and adaptive control

Before comparing outputs across completed configurations, the family must declare either a common report space  $\mathcal{Y}$  whose identity maps are explicitly adopted as the transports, or separately declared transports into  $\mathcal{Y}$ . A common codomain alone is not a transport declaration.

**Definition E.17** (Configuration robustness). Fix an invoked certified configuration set  $\mathcal{A}$  and let

$$R : \mathcal{A} \rightarrow \mathcal{Y}.$$

The report is configuration-robust if  $R$  is constant on all of  $\mathcal{A}$ , component-robust if it is constant on a declared ordered or connected component, and configuration-sensitive otherwise.

A robustness scan is licensed only under a certificate valid jointly over the scanned set; adaptive scans require sequential validity. If  $R$  is sensitive, the report should provide a certified scan, transition boundaries between stable regions, or a declared selector.

Selection of one configuration is substantive only when the certified set has more than one element, the deliverable requires a single configuration, and either the report or operational cost varies over the set. For fixed  $\lambda$ , possible declared selectors include the lowest certified threshold when that minimum is attained, the highest task-sufficient threshold when attained, or a loss-minimizing threshold. If an endpoint is only an infimum or supremum, a selected interior value and its gap from the unattained bound must be reported. Across mechanisms, threshold order does not automatically order  $\lambda$ ; a family-level selector must declare either a two-stage mechanism/threshold criterion or a joint objective  $J(\lambda, \varepsilon)$ .

**Definition E.18** (Adaptive resolution meta-protocol). An adaptive protocol is declared by

$$\bar{V} := (\mathfrak{R}_{\text{res}}^{\text{cert}}, \mathcal{Q}^{\sigma, \tau}, [p], \hat{e}, \text{Cert}, \mathsf{T}, \text{Ctrl}, \text{Accept}, \text{Reject}, \text{Stop}),$$

where

$$\mathfrak{R}_{\text{res}}^{\text{cert}} := \{\mathcal{R}_{\lambda}^{\text{cert}}\}_{\lambda \in \Lambda_{\text{res}}}.$$

Here  $[p]$  is the optional separate policy refinement, and  $\mathsf{T}$  declares the transports between visited package reports, including any explicitly adopted identity transports. Writing  $c_n = (\lambda_n, \varepsilon_n)$ , the update rule is

$$c_{n+1} = \text{Ctrl}(c_n, \hat{e}_n),$$

and every visited configuration must lie in the applicable package-conditioned certified set and, when policy is invoked, in its policy-restricted subset. Under partition semantics, the sequential certificate must also cover every visited threshold-edge domain. The object  $\bar{V}$  governs a path through ordinary task packages  $V_{\lambda_n, \varepsilon_n}^{\sigma, \tau}$  or their policy-refined counterparts and does not redefine the top-level package tuple. Fixed-threshold adaptation is the special case  $\varepsilon_n \equiv \varepsilon_0$ .

**Protocol E.2** (Locked adaptive package path). Adaptive variation is a declared path through completed packages. It is not undeclared package drift when the certified interface refinements, task clauses, optional policy clause, report semantics, margin certificates, estimator, sequential mechanism certificate, transports, update law, acceptance/rejection rules, and stopping rule are declared before the substantive diagnostic. What is locked is  $\bar{V}$  and the admissible path, not necessarily constant values of  $\lambda$  or  $\varepsilon$ . Post-hoc modification in response to the substantive result defines a new meta-protocol.

## E.9 Reporting states and non-claims

| Condition                                                        | Licensed report                                                                                                                        |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Certificate or certification domain undeclared                   | Object-level threshold claim is not licensed.                                                                                          |
| Certified query-margin lower bound undeclared                    | Operational certified set is unavailable; only ideal geometry may be stated.                                                           |
| Pointwise certificate reused for a scan                          | Joint scan claim is not licensed.                                                                                                      |
| Adaptive path without sequential validity                        | Adaptive certificate claim is not licensed.                                                                                            |
| Algebraic certificate slice empty                                | Current mechanism or margin certificates leave no algebraically admissible positive threshold.                                         |
| Package-conditioned set empty but algebraic slice nonempty       | Package admissibility, transport, or partition edge-certification clauses block the otherwise algebraically feasible configuration.    |
| Full classification set empty but separation-only set nonempty   | Directional positive distinctions can be protected, but the same threshold is not certified as a two-sided null/separation classifier. |
| Directional pairwise certified but partition certified set empty | Chain-separation / transitive-closure obstruction.                                                                                     |
| Pre-policy licensed set nonempty but policy-restricted set empty | Policy-induced non-licensing.                                                                                                          |
| Exactly one licensed configuration                               | Unique configuration relative to family, task clauses, certificates, policy clauses, and report semantics.                             |
| Multiple configurations, robust report                           | Report the certified set; point selection is optional.                                                                                 |
| Multiple configurations, sensitive report                        | Report a certified scan, stable-region boundaries, or a declared selector.                                                             |
| Adaptive control                                                 | Report the meta-package, sequential certificate, and realized configuration path.                                                      |
| Post-hoc protocol change                                         | New package; comparison requires declared transport.                                                                                   |

Table 4: Certificate-relative resolution-feasibility reporting states for the fixed-target module.

This appendix does not claim a universal threshold, a derivation of the external error certificate, an automatic bridge to residual complements, or a complete theory of coarse-graining families in which the semantic target changes with  $\lambda$ . Its contribution is the typed location of standard robust-separation geometry inside the declared-package framework, together with the distinction between separation-only and full classification tasks, the distinction between directional pairwise and partition output semantics, and the associated certification/control requirements.

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